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1 Introduction

The LS Corporation, has faced challenges in handling its multiple projects due to its limited flexibility system. Due to the challenges, it has increase inefficiency in the workflow and communication gap among team members. For tackling the issues, new strategy is planned to apply that implements a centralized project management systems. With implementation of the new system will expect to overcome challenge enhancing system workflow.

In the proposed idea, users are able to perform multiple projects and tasks collaterally. Each task will be designed to segregate into multiple subtasks, allowing for breakdown of work ensuring organized completion of task without delays. Users are able to provide feedback encouraging for improvements and build collaboration. For avoiding any issues in project, it is broken down into multiple milestones enhancing project management. So, the corporation has aimed to provide a clear map on progress tracking and adapt a systematic approach for project completion. The implementation of this centralized project management will allow corporation to properly address the current raise challenges.

2 Aims and Objective

2.1 Aims

The main aim of this coursework is to design, develop and implement a web-based application that can be efficient and enhances project management through data centralization, communication improvements and providing task tracking.

2.2 Objectives

- Proper analyzing of case study for identifying specific need for project management.
- Create full normalized database schema for representing relationships between user, projects, tasks, subtasks and milestones.
- Develop ERD using Oracle SQL Developer Data Modeler for proper cardinality and relation insights.
- Database to be generated based on ERD.
- Develop a web-based application using ASP.NET with C#.
- Design user-friendly interface for web application that helps for navigation.
- Conduct testing performing CRUD operations for checking its functionalities.
- Proper documentation for usability and maintenance.

3 Entity-Relation Model

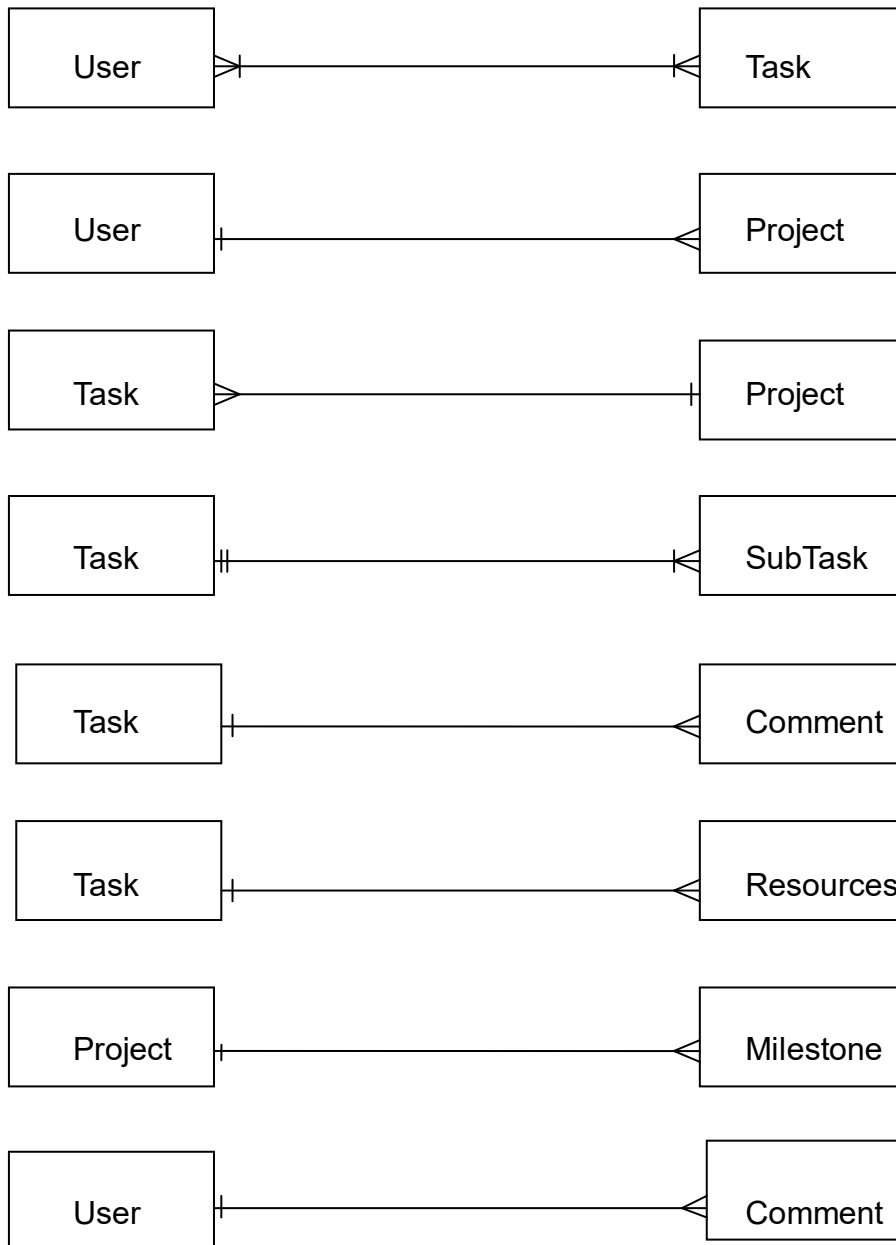
Business Rule:

1. A user can be part of multiple projects and a project can have multiple tasks.
2. A user can be assigned to multiple tasks and a task can have multiple users.
3. Each task can have multiple sub-tasks but each sub-task has only one single task linked to it.
4. Users are able to comment multiple times for a single task.
5. A task can require multiple resources.

Table 1 Entities and Attributes

Entities	Attributes
User	□ User_ID □ User_Name □ User_Email □ User_Contact □ User_Role(eg: Developer, Designer, Project Manager)
Project	□ Project_ID □ Project_Start_Date □ Project_Due_Date □ Project_Status
Task	□ Task_ID □ Task_Name

	□ Start_Date □ Due_Date □ Status □ Project_ID
Sub-Task	□ SubTask_ID □ SubTask_Name
Comment	□ Comment_ID □ Comment_Date □ User_ID □ Task_ID
Resources	□ Resource_ID □ Resource_Name □ Resource_Type □ Task_ID

Cardinality

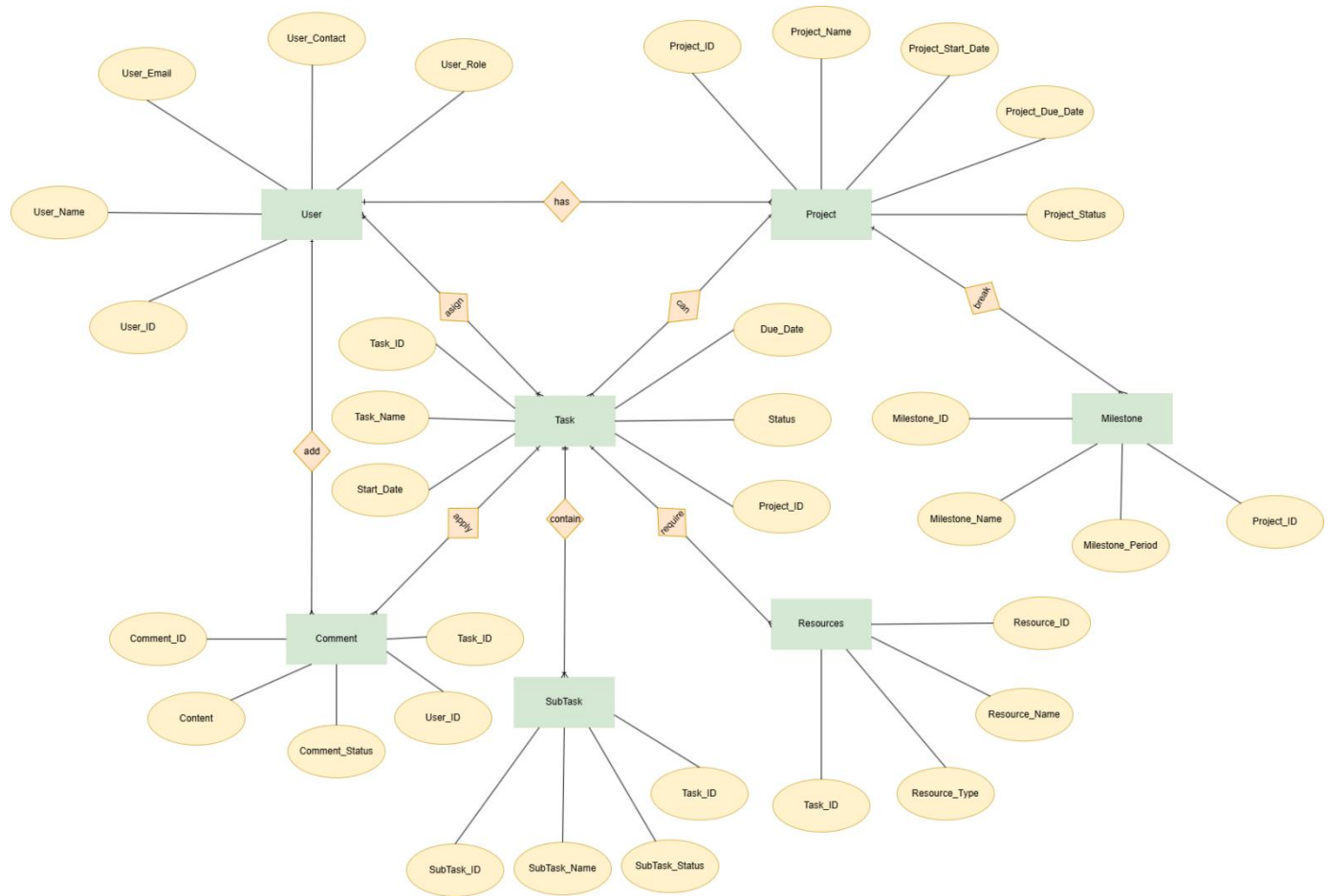


Figure 1 Initial ERD

The Entity-Relationship Model is designed to illustrate relationships build among identified entities for project management system. The User entity contains attributes such as User_ID, User_Name, User_Email, User_Contact, and User_Role which provide detail insights about individual participating in projects. The Project entity includes attributes such as Project_ID, Project_Name, Project_Start_Date, Project_Due_Date and Project_Status, along with a foreign key referencing User entity. The Task entity contains attributes like Task_ID, Task_Name, Start_Date, Due_Date, Status, which form relationship between both project and user entities. The SubTask entity contains attributes like SubTask_ID, SubTask_Name, and SubTask_Status, referencing Task_ID as foreign key form Task table. The Milestone entity is defined into Milestone_ID, Milestone_Name, and Milestone_Period, while referencing the project

entity. The comment contains Comment_ID, Content, and Comment_Status, along with referencing user and task entities. Lastly, the Resources entity is defined with attributes like Resource_ID, Resource_Name and Resource_Type

The relationship established by user and project is one-to-many relationship, while user and comment has also one-to-many relationship. Each Project has one-to-many relationship between tasks. The User and Task entities have many-to-many relationship, forming composite key. Task is divided into SubTasks creating one-to-many relationships. Each Task has one-to-many relation for both Comment and Resources. A project has multiple milestones, forming one-to-many relationship. The User entity also form link with comment creating one-to-many relationship.

4 Normalization

Normalization is a design technique of database which focuses on structuring data to minimize redundancy and dependency. It helps to remove data like insertion, update, and deletion anomalies. It makes sure to divide larger data tables into small table forms. The main objective of normalization is to eliminate redundancy, ensuring data consistency and maintain data integrity. The normalization is structured in a series of normal forms.

- Unnormalized Form (UNF)

UNF is the raw data that is presented without any normalization applied.

- First Normal Form (1NF)

The 1NF ensure each data in a column is unique while eliminating repeating groups.

- Second Normal Form (2NF)

The 2NF is achieved when a table is in 1NF and all non-key attributes are full functional dependent on primary key. It eliminates partial dependencies, where a non-key attribute is dependent on composite key.

- Third Normal Form (3NF)

The 3NF is achieved when a table is in 2NF and contains no transitive dependencies, where non-key attribute is not dependent with each other.

SQL consist of different keys like primary key, composite key and foreign key. Primary key is a unique identifier that must be unique and cannot be null. Composite key is also a primary key that has two or more columns, which is useful in scenarios where no single column can uniquely identify itself. Foreign key is present in one table that uniquely identifies another table making relation between them.

Normalization consists of three functional dependencies: Partial functional dependency, Full-functional dependency and Transitive functional dependency. The partial functional dependency occurs when an attribute in table depends on part of a composite primary key. Full functional dependency occurs when an attribute is fully dependent on the primary key. Transitive functional dependency occurs when a non-key attribute depends on another non-key attribute.

The normalization helps to reduce any data redundancy making sure to improve data consistency. With proper structuring of data through normalization make it efficient for querying. The use of normalization reduces insertion, update and deletion anomalies. On the other hand, with more number of tables can lead to complexity in understanding the database. With more tables having relations need more storage and slower the query performance. If the normalization is not done properly with accurate relation, it leads to complexity. (Anon., 2024)

Assumption: It is assumed each user can be in multiple projects while a project inherits multiple users, and also each project can contain multiple tasks.

Unnormalized Form (UNF)

User (User_ID, User_Name, User_Email, User_contact, User_Role, {Project_ID, Project_Name, Project_Start_Date, Project_Due_Date, Project_Status, {Task_ID, Task_Name, Start_Date, Due_Date, Status}}})

In this UNF, it is presented in a single record that combines multiple entities, including users, projects, and tasks. It is in a nested group form where repeating values are indicated in parentheses and repeating groups are enclosed within curly braces.

First Normal Form (1NF):

User-1 (User_ID, User_Name, User_Email, User_contact, User_Role)

Project-1 (Project_ID, Project_Name, Project_Start_Date, Project_Due_Date, Project_Status, User_ID*)

Task-1 (Task_ID, Task_Name, Start_Date, Due_Date, Status, User_ID*, Project_ID*)

In the transformation of UNF to 1NF, each of the entity is separated into distinct tables for clear representation of data. User, Project, and Task entities are separately defined while maintaining relationship through use of foreign keys. However, partial dependencies may exist, so further normalization is required for data integrity.

Second Normal Form (2NF)

Checking partial functional dependency from Project-1:

Project-1 (Project_ID, Project_Name, Project_Start_Date, Project_Due_Date, Project_Status, User_ID*)

Project_ID, User_ID $\longrightarrow$

Project $\longrightarrow$ Project_Name, Project_Start_Date, Project_Due_Date, Project_Status

Checking partial functional dependency from Task-1:

Task-1 (Task_ID, Task_Name, Start_Date, Due_Date, Status, User_ID*, Project_ID*)

Task_ID, User_ID, Project_ID $\longrightarrow$

Task $\longrightarrow$ Task_Name, Start_Date, Due_Date, Status

Final 2NF:

User-2 (User_ID, User_Name, User_Email, User_contact, User_Role)

Project-2 (Project_ID, Project_Name, Project_Start_Date, Project_Due_Date, Project_Status)

UserProject-2(Project_ID*, User_ID*)

Task-2 (Task_ID*, User_ID*, Project_ID*)

Task_Details-2 (Task_ID, Task_Name, Start_Date, Due_Date, Status)

For achieving 2NF, partial functional dependencies are checked for every independent table. It is noted to check partial dependency in Task-1 and Project-1, which leads to derive four separate tables: Project-2, UserProject-2, Task-2 and Task_Details-2. This helps to ensure all non-key attributes are fully functional dependent on primary key, enhancing data organization and reducing redundancy. However, it is acknowledged that transitive dependencies among the tables, indicating the need for further normalization.

Third Normal Form (3NF)

No transitive dependency is present among independent tables, indicating all non-key attributes are fully functional dependent on their primary keys.

Final 3NFs

User-3 (User_ID, User_Name, User_Email, User_contact, User_Role)

Project-3 (Project_ID, Project_Name, Project_Start_Date, Project_Due_Date, Project_Status)

UserProject-3(Project_ID*, User_ID*)

Task-3 (Task_ID*, User_ID*, Project_ID*)

Task_Details-3 (Task_ID, Task_Name, Start_Date, Due_Date, Status)

In the 3NF, any transitive dependency occurred is removed from the tables. It is confirmed, all the non-key attributes are fully functional dependent to their primary key while ensuring no indirect relationships occur among attributes. So, it forms distinct tables: User, Project, and Tasks, ensuring clear defined relationship. It helped in reducing redundancy enhancing data integrity.

5 Final ER-Model

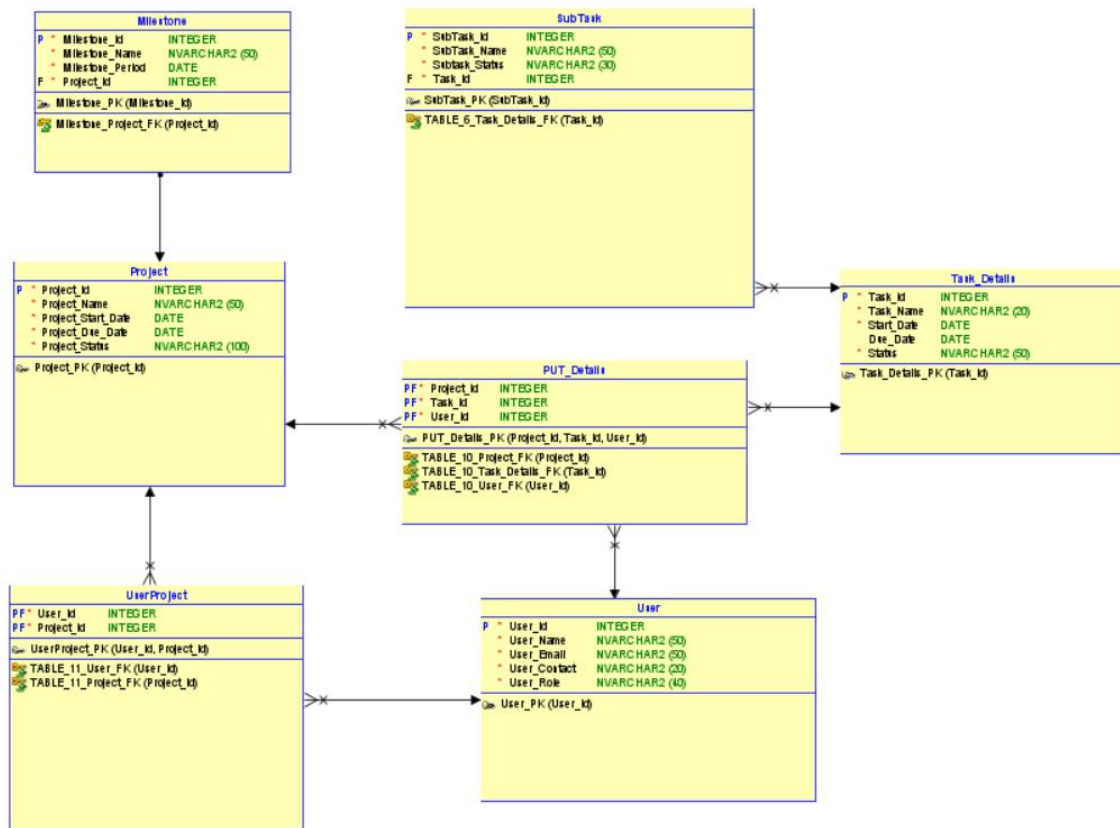


Figure 2 Final ER-Model

In the final ERD, relationships among different entities are shown, with each entity representing their own respective attributes. The User entity is linked with Task entity, indicating users are responsible for tasks. The Task_Details entity has connection between User through a bridge entity known as Task. The Project entity builds relationship with Milestone, as for tracking the progress of project. Resources are required for performing tasks, which is linked by Task_Details. The Comment entity collects feedback from both User and Task_Details entities. The SUBtask entity are connected to Task_Details entity, as a larger task contains multiple subtasks. Overall, the final erd provided us a comprehensive overview of the relationship between entities.

6 Data Dictionary

Data dictionary is a crucial element of DBMS, which serves as a centralized repository for the metadata, including data types, relationships and constraints. It helps to ensure data consistency, accuracy and efficiency. (Sudhanshu, 2025)

1. User

Table 2 Data Dictionary of User table

Column Name	Data Type	Size	Constraints	Reference Table	Example Data
USER_ID	INTEGER	20	Primary Key		1
USER_NAME	NVARCHAR	50			Ningwa
USER_EMAIL	NVARCHAR	50			ningwa@gmail.com
USER_CONTACT	NUMBER	20			9876543212
USER_ROLE	NVARCHAR	40			Designer

2. Project

Table 3 Data Dictionary of Project table

Column Name	Data Type	Size	Constraints	Reference Table	Example Data
PROJECT_ID	INTEGER	15	Primary Key		2
PROJECT_NAME	NVARCHAR	50			Project Alpha
PROJECT_START_DATE	DATE				01-10-25
PROJECT_DUE_DATE	DATE				02-20-25
PROJECT_STATUS	NVARCHAR	100			A

3. Milestone

Table 4 Data Dictionary of Milestone table

Column Name	Data Type	Size	Constraints	Reference Table	Example Data
MILESTONE_ID	INTEGER	20	Primary Key		16
MILESTONE_NAME	NVARCHAR	50			Requirement Gathering
MILESTONE_PERIOD	DATE				10-04-25
PROJECT_ID	INTEGER	20	Foreign Key	Project	2

4. Task Details

Table 5 Data Dictionary of Task Details table

Column Name	Data Type	Size	Constraints	Reference Table	Example Data
TASK_ID	INTEGER	15	Primary Key		3
TASK_NAME	NVARCHAR	20			Feasibility Study
START_DATE	DATE				11-02-2025
DUE_DATE	DATE				04-16-2025
STATUS	NVARCHAR	50			In Progress

5. Subtask

Table 6 Data Dictionary of Subtask table

Column Name	Data Type	Size	Constraints	Reference Table	Example Data
SUBTASK_ID	INTEGER	20	Primary Key		06
SUBTASK_NAME	NVARCHAR	50			Prepare Deployment Plan.
SUBTASK_STATUS	NVARCHAR	30			Not Started.
TASK_ID	INTEGER	15	Foreign Key	Task	06

6. UserProject

Column Name	Data Type	Size	Constraints	Reference Table	Example Data
USER_ID	INTEGER	15	Primary Key	User	03
PROJECT_ID	INTEGER	15	Primary Key	Project	05

7. PUT_Details

Table 7 Data Dictionary of Task table

Column Name	Data Type	Size	Constraints	Reference Table	Example Data
TASK_ID	INTEGER	20	Primary Key	Task_Details	07
USER_ID	INTEGER	15	Primary Key	User	02
PROJECT_ID	INTEGER	15	Primary Key	Project	05

7 Database

A new user is created as “Users” while allowing privileges for resources. Total of six tables is created by the result acquired from Case study initial ERD and normalized table. User table contains the information about users including identifiers, names, emails, contact and roles. The project table involves details of projects such as project ID, names, status, start and dues dates along with user references. A Task_Details table contains detailed information about tasks including task ID, names, dates, status and references to projects. A milestone contains Milestone ID, names and periods linking each milestone to a project. Subtasks are related to break down of larger task to multiple tasks. The new bridge entity is get for connecting users to specific tasks in Task_Details table along with projects. Several data is inserted into the created tables for manipulation.

Query for Table Creation

```
CREATE TABLE Milestone
```

```
(
    Milestone_Id    INTEGER NOT NULL ,
    Milestone_Name  NVARCHAR2 (50) NOT NULL ,
    Milestone_Period DATE  NOT NULL ,
    Project_Id      INTEGER NOT NULL
)
;
```

```
ALTER TABLE Milestone
```

```
ADD CONSTRAINT Milestone_PK PRIMARY KEY ( Milestone_Id );
```

```
CREATE TABLE Project
```

```
(
    Project_Id      INTEGER NOT NULL ,
    Project_Name    NVARCHAR2 (50) NOT NULL ,
    Project_Start_Date DATE  NOT NULL ,
```

```
Project_Due_Date DATE NOT NULL ,  
Project_Status NVARCHAR2 (100) NOT NULL  
)  
;
```

```
ALTER TABLE Project  
ADD CONSTRAINT Project_PK PRIMARY KEY ( Project_Id );
```

```
CREATE TABLE PUT_Details  
(  
Project_Id INTEGER NOT NULL ,  
Task_Id INTEGER NOT NULL ,  
User_Id INTEGER NOT NULL  
)  
;
```

```
ALTER TABLE PUT_Details  
ADD CONSTRAINT PUT_Details_PK PRIMARY KEY ( Project_Id, Task_Id, User_Id );
```

```
CREATE TABLE SubTask  
(  
SubTask_Id INTEGER NOT NULL ,  
SubTask_Name NVARCHAR2 (50) NOT NULL ,  
Subtask_Status NVARCHAR2 (30) NOT NULL ,  
Task_Id INTEGER NOT NULL  
)  
;
```

```
ALTER TABLE SubTask  
ADD CONSTRAINT SubTask_PK PRIMARY KEY ( SubTask_Id );
```

```
CREATE TABLE Task_Details
```

```
(
  Task_Id  INTEGER NOT NULL ,
  Task_Name NVARCHAR2 (20) NOT NULL ,
  Start_Date DATE NOT NULL ,
  Due_Date  DATE ,
  Status   NVARCHAR2 (50) NOT NULL
)
;
```

```
ALTER TABLE Task_Details
```

```
ADD CONSTRAINT Task_Details_PK PRIMARY KEY ( Task_Id );
```

```
CREATE TABLE "User"
```

```
(
  User_Id  INTEGER NOT NULL ,
  User_Name NVARCHAR2 (50) NOT NULL ,
  User_Email NVARCHAR2 (50) NOT NULL ,
  User_Contact NVARCHAR2 (20) NOT NULL ,
  User_Role NVARCHAR2 (40) NOT NULL
)
;
```

```
ALTER TABLE "User"
```

```
ADD CONSTRAINT User_PK PRIMARY KEY ( User_Id );
```

```
CREATE TABLE UserProject
```

```
(
  User_Id  INTEGER NOT NULL ,
  Project_Id INTEGER NOT NULL
)
```

;

```
ALTER TABLE UserProject
  ADD CONSTRAINT UserProject_PK
  PRIMARY KEY ( User_Id, Project_Id );
```

```
ALTER TABLE Milestone
  ADD CONSTRAINT Milestone_Project_FK FOREIGN KEY
  (
    Project_Id
  )
  REFERENCES Project
  (
    Project_Id
  )
  ON DELETE CASCADE
;
```

```
ALTER TABLE PUT_Details
  ADD CONSTRAINT TABLE_10_Project_FK FOREIGN KEY
  (
    Project_Id
  )
  REFERENCES Project
  (
    Project_Id
  )
  ON DELETE CASCADE
;
```

```
ALTER TABLE PUT_Details
```

```
ADD CONSTRAINT TABLE_10_Task_Details_FK FOREIGN KEY
(
    Task_Id
)
REFERENCES Task_Details
(
    Task_Id
)
ON DELETE CASCADE
;
```

```
ALTER TABLE PUT_Details
ADD CONSTRAINT TABLE_10_User_FK FOREIGN KEY
(
    User_Id
)
REFERENCES "User"
(
    User_Id
)
ON DELETE CASCADE
;
```

```
ALTER TABLE UserProject
ADD CONSTRAINT TABLE_11_Project_FK FOREIGN KEY
(
    Project_Id
)
REFERENCES Project
(
    Project_Id
)
```

```
)  
ON DELETE CASCADE  
;
```

```
ALTER TABLE UserProject  
ADD CONSTRAINT TABLE_11_User_FK FOREIGN KEY  
(  
    User_Id  
)  
REFERENCES "User"  
(  
    User_Id  
)  
ON DELETE CASCADE  
;
```

```
ALTER TABLE SubTask  
ADD CONSTRAINT TABLE_6_Task_Details_FK FOREIGN KEY  
(  
    Task_Id  
)  
REFERENCES Task_Details  
(  
    Task_Id  
)  
ON DELETE CASCADE  
;
```

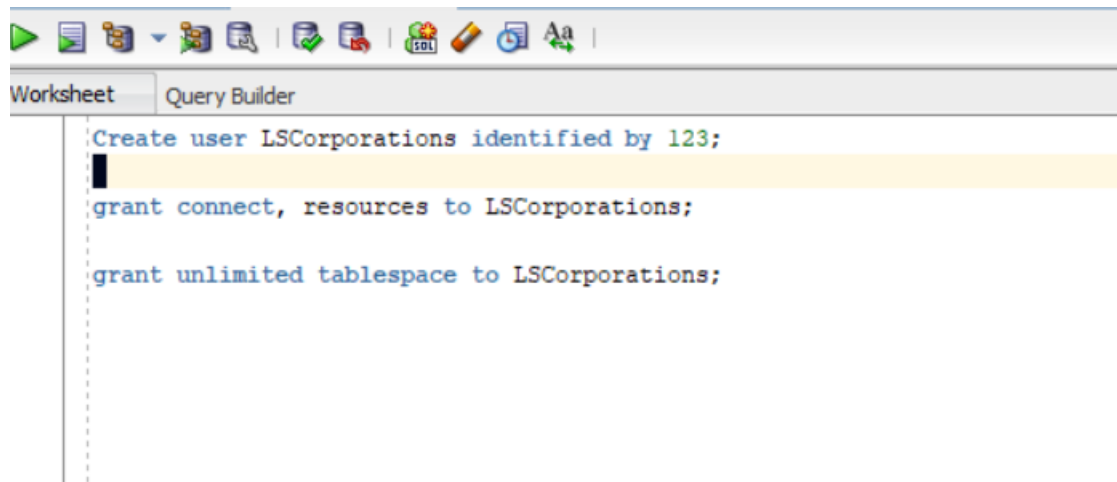



Figure 3 Create User

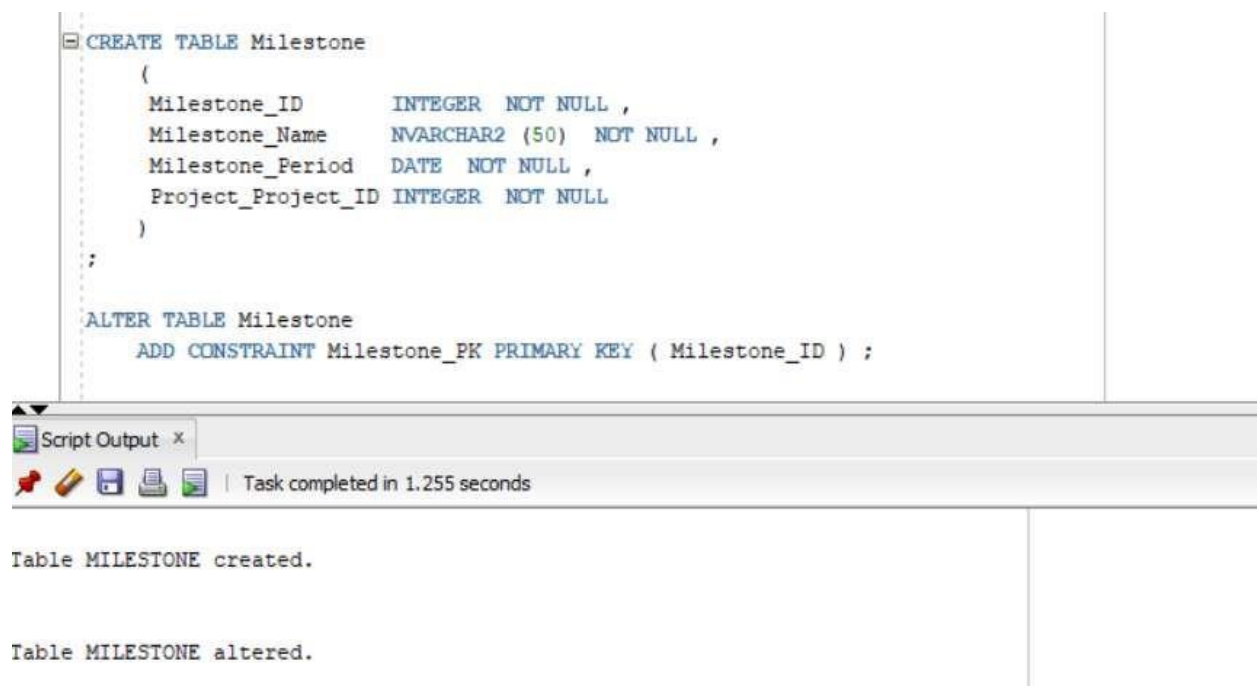


Figure 4 Creating Milestone Table

```
CREATE TABLE Project
(
    Project_ID          INTEGER NOT NULL ,
    Project_Name        NVARCHAR2 (50) NOT NULL ,
    Project_Start_Date  DATE NOT NULL ,
    Project_Due_Date    DATE NOT NULL ,
    Project_Status      NVARCHAR2 (1) NOT NULL ,
    User_User_ID        INTEGER NOT NULL
)
;

ALTER TABLE Project
ADD CONSTRAINT Project_PK PRIMARY KEY ( Project_ID ) ;
```

Script Output x

Task completed in 1.255 seconds

Table PROJECT created.

Table PROJECT altered.

Figure 5 Creating Project Table

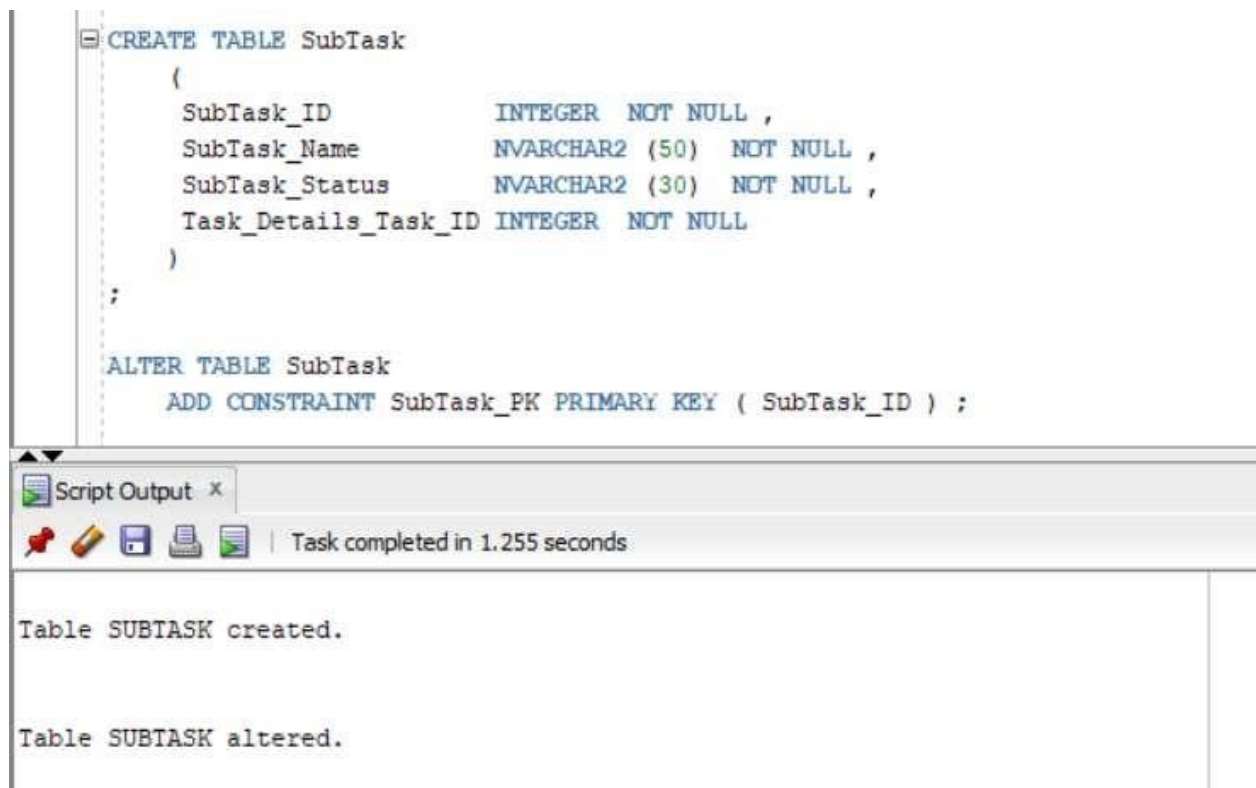
```
CREATE TABLE UserProject
(
    User_Id    INTEGER NOT NULL ,
    Project_Id INTEGER NOT NULL
)
;
```

Script Output x

Task completed in 0.086 seconds

Table USERPROJECT created.

Figure 6 Creating UserProject Table

*Figure 7 Creating UserProject Table**Figure 8 Creating SubTask Table*

```
CREATE TABLE Task_Details
(
    Task_ID          INTEGER NOT NULL ,
    Task_Name        NVARCHAR2 (50) NOT NULL ,
    Start_Date       DATE NOT NULL ,
    Due_Date         DATE NOT NULL ,
    Status           NVARCHAR2 (40)
)

;

ALTER TABLE Task_Details
ADD CONSTRAINT Task_Details_PK PRIMARY KEY ( Task_ID ) ;
```

Script Output x

Task completed in 1.255 seconds

Table TASK_DETAILS created.

Table TASK_DETAILS altered.

Figure 9 Creating Task_Details Table

```
CREATE TABLE PUT_Details
(
    Project_Id INTEGER NOT NULL ,
    Task_Id    INTEGER NOT NULL ,
    User_Id    INTEGER NOT NULL
)

;
```

Script Output x

Task completed in 0.062 seconds

Table PUT_DETAILS created.

Figure 10 Creating PUT_Details Table

```
CREATE TABLE "User"
(
    User_ID      INTEGER NOT NULL ,
    User_Name    NVARCHAR2 (50) NOT NULL ,
    User_Email   NVARCHAR2 (50) NOT NULL ,
    User_Contact NUMBER ,
    User_Role    NVARCHAR2 (40) NOT NULL
)
;

ALTER TABLE "User"
    ADD CONSTRAINT User_PK PRIMARY KEY ( User_ID ) ;
```

Script Output x

Task completed in 1.255 seconds

Table "User" created.

Table "User" altered.

Figure 11 Creating User Table

```
ALTER TABLE UserProject
    ADD CONSTRAINT TABLE_11_Project_FK FOREIGN KEY
    (
        Project_Id
    )
    REFERENCES Project
    (
        Project_Id
    )
    ON DELETE CASCADE
;

ALTER TABLE UserProject
    ADD CONSTRAINT TABLE_11_User_FK FOREIGN KEY
    (
        User_Id
    )
    REFERENCES "User"
    (
        User_Id
    )
    ON DELETE CASCADE
;
```

Figure 12 Altering UserProject Table

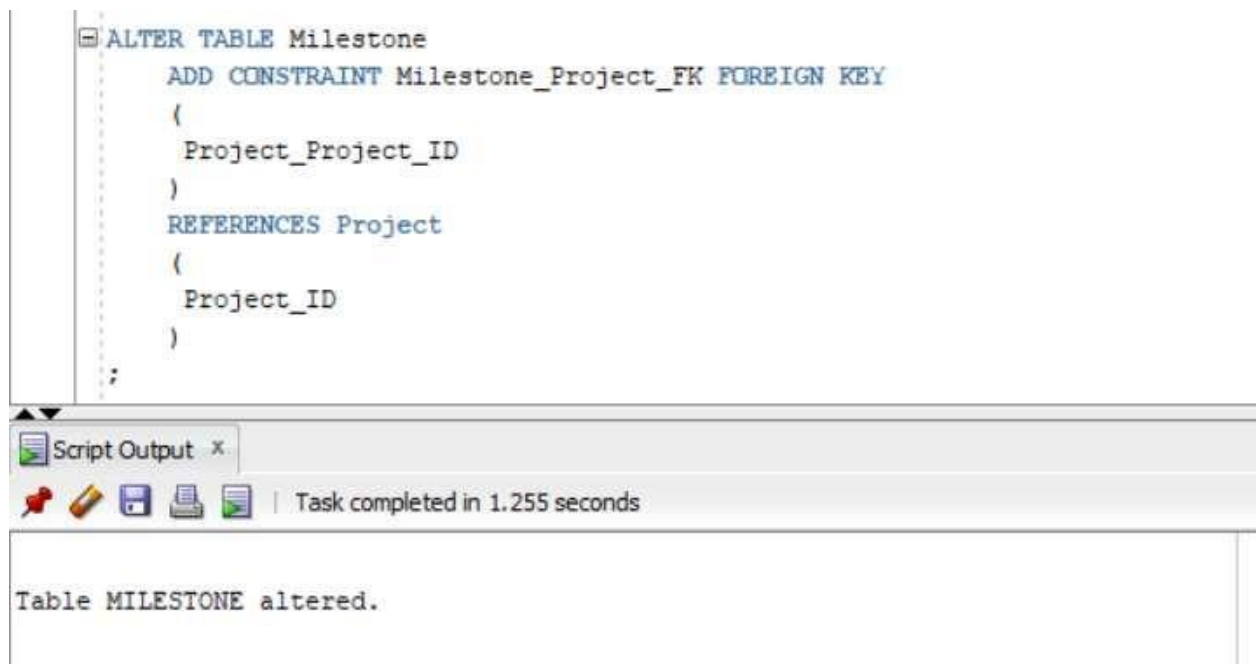


Figure 13 Altering Milestone Table



Figure 14 Altering Project Table



Figure 15 Altering SubTask Table



Figure 16 Altering Task_Details Table

```
ALTER TABLE PUT_Details
  ADD CONSTRAINT TABLE_10_User_FK FOREIGN KEY
  (
    User_Id
  )
  REFERENCES "User"
  (
    User_Id
  )
  ON DELETE CASCADE
;
```

Figure 17 Altering PUT_Details Table

```
ALTER TABLE PUT_Details
  ADD CONSTRAINT TABLE_10_Project_FK FOREIGN KEY
  (
    Project_Id
  )
  REFERENCES Project
  (
    Project_Id
  )
  ON DELETE CASCADE
;

ALTER TABLE PUT_Details
  ADD CONSTRAINT TABLE_10_Task_Details_FK FOREIGN KEY
  (
    Task_Id
  )
  REFERENCES Task_Details
  (
    Task_Id
  )
  ON DELETE CASCADE
;
```

Figure 18 Altering PUT_Details Table


```

INSERT INTO "User" (User_ID, User_Name, User_Email, User_Contact, User_Role) VALUES (1, 'Alice', 'alice@gmail.com', 9812345678, 'Developer');
INSERT INTO "User" (User_ID, User_Name, User_Email, User_Contact, User_Role) VALUES (2, 'Bibisha', 'bibisha@gmail.com', 9822345678, 'Designer');
INSERT INTO "User" (User_ID, User_Name, User_Email, User_Contact, User_Role) VALUES (3, 'Charlie', 'charlie@gmail.com', 9812345678, 'Project Manager');
INSERT INTO "User" (User_ID, User_Name, User_Email, User_Contact, User_Role) VALUES (4, 'Sugan', 'sugan@gmail.com', 9812342378, 'Developer');
INSERT INTO "User" (User_ID, User_Name, User_Email, User_Contact, User_Role) VALUES (5, 'Sujan', 'sujan@gmail.com', 9892345678, 'Designer');
INSERT INTO "User" (User_ID, User_Name, User_Email, User_Contact, User_Role) VALUES (6, 'Anjila', 'anjila@gmail.com', 9712345678, 'Developer');
INSERT INTO "User" (User_ID, User_Name, User_Email, User_Contact, User_Role) VALUES (7, 'Bob', 'bob@gmail.com', 9812345684, 'Designer');
INSERT INTO "User" (User_ID, User_Name, User_Email, User_Contact, User_Role) VALUES (8, 'Ran', 'ran@gmail.com', 9711345678, 'Project Manager');
INSERT INTO "User" (User_ID, User_Name, User_Email, User_Contact, User_Role) VALUES (9, 'Diana', 'diana@gmail.com', 9711345677, 'Developer');
INSERT INTO "User" (User_ID, User_Name, User_Email, User_Contact, User_Role) VALUES (10, 'Neha', 'neha@gmail.com', 9711345688, 'Tester');
INSERT INTO "User" (User_ID, User_Name, User_Email, User_Contact, User_Role) VALUES (11, 'Numa', 'numa@gmail.com', 9711555678, 'Developer');
INSERT INTO "User" (User_ID, User_Name, User_Email, User_Contact, User_Role) VALUES (12, 'Krish', 'krish@gmail.com', 9701345678, 'Designer');
INSERT INTO "User" (User_ID, User_Name, User_Email, User_Contact, User_Role) VALUES (13, 'Evan', 'evan@gmail.com', 9711345666, 'Tester');
INSERT INTO "User" (User_ID, User_Name, User_Email, User_Contact, User_Role) VALUES (14, 'Aisha', 'aisha@gmail.com', 9811345664, 'Developer');
INSERT INTO "User" (User_ID, User_Name, User_Email, User_Contact, User_Role) VALUES (15, 'Binu', 'binu@gmail.com', 9711335666, 'Designer');
INSERT INTO "User" (User_ID, User_Name, User_Email, User_Contact, User_Role) VALUES (16, 'Akriti', 'akriti@gmail.com', 9811845666, 'Developer');
INSERT INTO "User" (User_ID, User_Name, User_Email, User_Contact, User_Role) VALUES (17, 'Deepson', 'deepson@gmail.com', 9781345666, 'Designer');
INSERT INTO "User" (User_ID, User_Name, User_Email, User_Contact, User_Role) VALUES (18, 'Smritika', 'smritika@gmail.com', 9711345654, 'Tester');
INSERT INTO "User" (User_ID, User_Name, User_Email, User_Contact, User_Role) VALUES (19, 'Anna', 'anna@gmail.com', 9711346066, 'Developer');
INSERT INTO "User" (User_ID, User_Name, User_Email, User_Contact, User_Role) VALUES (20, 'Laxman', 'laxman@gmail.com', 9710045666, 'Designer');
COMMIT;

```

Script Output: Query Result

Task completed in 0.247 seconds

```

1 row inserted.
1 row inserted.
Commit complete.

```

Figure 19 Inserting values in User table

USER_ID	USER_NAME	USER_EMAIL	USER_CONTACT	USER_ROLE
1	Alice	alice@gmail.com	9812345678	Developer
2	Bibisha	bibisha@gmail.com	9822345678	Designer
3	Charlie	charlie@gmail.com	9812345678	Project Manager
4	Sugan	sugan@gmail.com	9812342378	Developer
5	Sujan	sujan@gmail.com	9892345678	Designer
6	Anjila	anjila@gmail.com	9712345678	Developer
7	Bob	bob@gmail.com	9812345684	Designer
8	Ran	ran@gmail.com	9711345678	Project Manager
9	Diana	diana@gmail.com	9711345677	Developer
10	Neha	neha@gmail.com	9711345688	Tester
11	Numa	numa@gmail.com	9711555678	Developer
12	Krish	krish@gmail.com	9701345678	Designer
13	Evan	evan@gmail.com	9711345666	Tester
14	Aisha	aisha@gmail.com	9811345664	Developer
15	Binu	binu@gmail.com	9711335666	Designer
16	Akriti	akriti@gmail.com	9811845666	Developer
17	Deepson	deepson@gmail.com	9781345666	Designer
18	Smritika	smritika@gmail.com	9711345654	Tester
19	Anna	anna@gmail.com	9711346066	Developer
20	Laxman	laxman@gmail.com	9710045666	Designer

Figure 20 Displaying User table

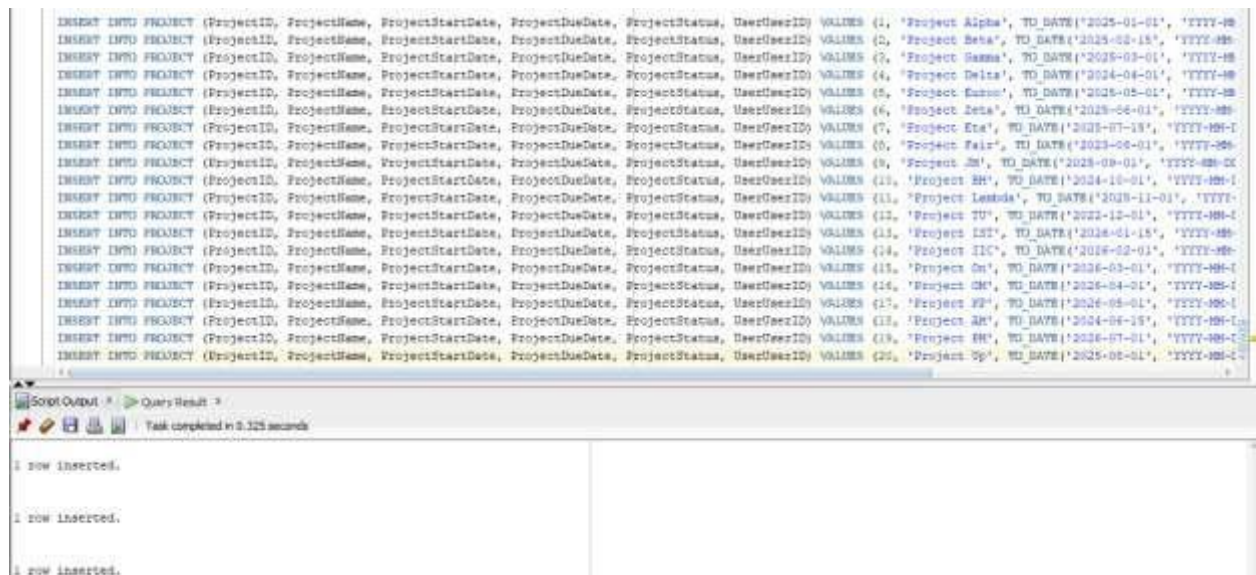


Figure 21 Inserting values in Project table

PROJECT_ID	PROJECT_NAME	PROJECT_START_DATE	PROJECT_DUE_DATE	PROJECT_STATUS	USER_USER_ID
1	Project Alpha	01-JAN-25	31-DEC-25	A	3
2	Project Beta	15-FEB-25	30-NOV-25	A	7
3	Project Gamma	01-MAR-25	15-OCT-25	B	12
4	Project Delta	01-APR-24	30-SEP-24	C	1
5	Project Epsilon	01-MAY-25	31-AUG-25	A	20
6	Project Zeta	01-JUN-25	31-JUL-25	A	15
7	Project Eta	15-JUL-25	15-DEC-25	B	9
8	Project Theta	01-AUG-25	01-OCT-25	C	4
9	Project Iota	01-SEP-25	30-NOV-25	A	10
10	Project Kappa	01-OCT-24	15-DEC-24	A	14
11	Project Lambda	01-NOV-25	15-JAN-26	B	6
12	Project Mu	01-DEC-22	28-FEB-23	C	8
13	Project Nu	15-JAN-26	15-MAR-26	A	19
14	Project Xi	01-FEB-26	01-MAY-26	A	2
15	Project Omicron	01-MAR-26	30-JUN-26	B	11
16	Project Pi	01-APR-26	15-JUL-26	C	16
17	Project Rho	01-MAY-26	31-AUG-26	A	13
18	Project Sigma	15-JUN-24	15-SEP-24	A	5
19	Project Tau	01-JUL-26	30-NOV-26	B	17
20	Project Upsilon	01-AUG-25	15-OCT-25	C	10

Figure 22 Displaying Project table



Figure 23 Inserting values in Milestone table

MILESTONE_ID	MILESTONE_NAME	MILESTONE_PERIOD	PROJECT_PROJECT_ID
1	1 Initiation	15-JAN-25	1
2	2 Planning	15-FEB-25	1
3	3 Execution	15-MAR-25	1
4	4 Monitoring	15-APR-25	1
5	5 Closure	15-MAY-25	1
6	6 Requirements Gathering	20-JUN-25	2
7	7 Design Phase	20-JUL-25	2
8	8 Implementation	20-AUG-25	2
9	9 Testing Phase	20-SEP-25	2
10	10 Deployment	20-OCT-25	2
11	11 Requirements Review	25-JAN-26	3
12	12 Prototype	25-FEB-26	3
13	13 Final Review	25-MAR-26	3
14	14 User Acceptance Testing	25-APR-26	3
15	15 Go Live	25-MAY-26	3
16	16 Research Phase	30-JUN-26	4
17	17 Development Phase	25-FEB-25	4
18	18 Testing and Validation	30-MAR-26	4
19	19 Launch	30-APR-26	4
20	20 Post-Launch Review	30-MAY-26	4

Script Output: Query Result: All Rows Fetched: 20 in 0.094 seconds

Figure 24 Displaying Milestone table

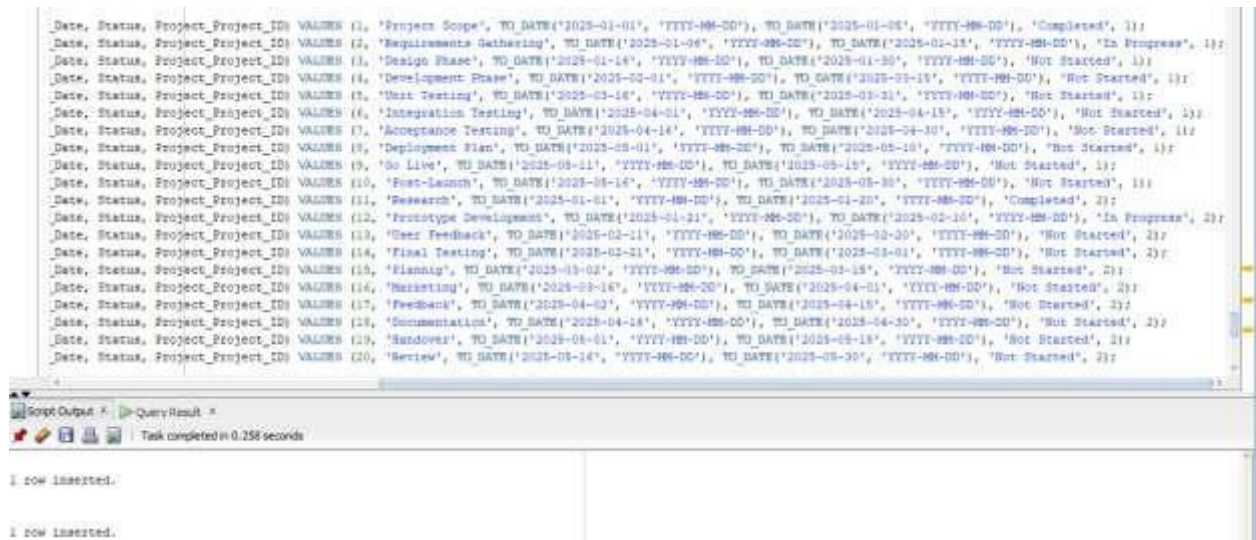


Figure 25 Inserting values in Task_Details table

select * from Task_Details;

Script Output: Query Result

All Rows Fetched: 20 in 0.07 seconds

TASK_ID	TASK_NAME	START_DATE	DUE_DATE	STATUS	PROJECT_PROJECT_ID
1	Project Scope	01-JAN-25	05-JAN-25	Completed	1
2	Requirements Gathering	06-JAN-25	15-JAN-25	In Progress	1
3	Design Phase	16-JAN-25	30-JAN-25	Not Started	1
4	Development Phase	01-FEB-25	15-MAR-25	Not Started	1
5	Unit Testing	16-MAR-25	31-MAR-25	Not Started	1
6	Integration Testing	01-APR-25	15-APR-25	Not Started	1
7	Acceptance Testing	16-APR-25	30-APR-25	Not Started	1
8	Deployment Plan	01-MAY-25	10-MAY-25	Not Started	1
9	Go Live	11-MAY-25	15-MAY-25	Not Started	1
10	Post-Launch	16-MAY-25	30-MAY-25	Not Started	1
11	Research	01-JAN-25	20-JAN-25	Completed	2
12	Prototype Development	21-JAN-25	10-FEB-25	In Progress	2
13	User Feedback	11-FEB-25	20-FEB-25	Not Started	2
14	Final Testing	21-FEB-25	01-MAR-25	Not Started	2
15	Planning	02-MAR-25	15-MAR-25	Not Started	2
16	Marketing	16-MAR-25	01-APR-25	Not Started	2
17	Feedback	01-APR-25	15-APR-25	Not Started	2
18	Documentation	16-APR-25	30-APR-25	Not Started	2
19	Handover	01-MAY-25	15-MAY-25	Not Started	2
20	Review	16-MAY-25	30-MAY-25	Not Started	2

Figure 26 Displaying Task_Details



Figure 27 Inserting values in SubTask table

select * from SubTask

Script Output | Query Result

All Rows Fetched: 20 in 0.309 seconds

SUBTASK_ID	SUBTASK_NAME	SUBTASK_STATUS	TASK_DETAILS_TASK_ID
1	1 Design UI	In Progress	1
2	2 Develop API	Not Started	2
3	3 Create Database Schema	Completed	2
4	4 Write Unit Tests	Not Started	3
5	5 Conduct Integration Tests	In Progress	3
6	6 Prepare Deployment Plan	Not Started	4
7	7 Gather User Feedback	Not Started	4
8	8 Finalize Documentation	In Progress	5
9	9 Conduct User Training	Not Started	6
10	10 Create Marketing Materials	Not Started	7
11	11 Set Up Monitoring	In Progress	8
12	12 Post-Launch Review Meeting	Not Started	9
13	13 Follow Up on Feedback	Not Started	10
14	14 Plan Next Phase	Not Started	11
15	15 Conduct Stakeholder Meeting	In Progress	12
16	16 Review Project Outcomes	Not Started	13
17	17 Identify Improvements	Not Started	14
18	18 Archive Project Documents	Not Started	15
19	19 Set Up Post-Mortem Analysis	Not Started	16
20	20 Prepare Final Report	Not Started	17

Figure 28 Displaying SubTask table

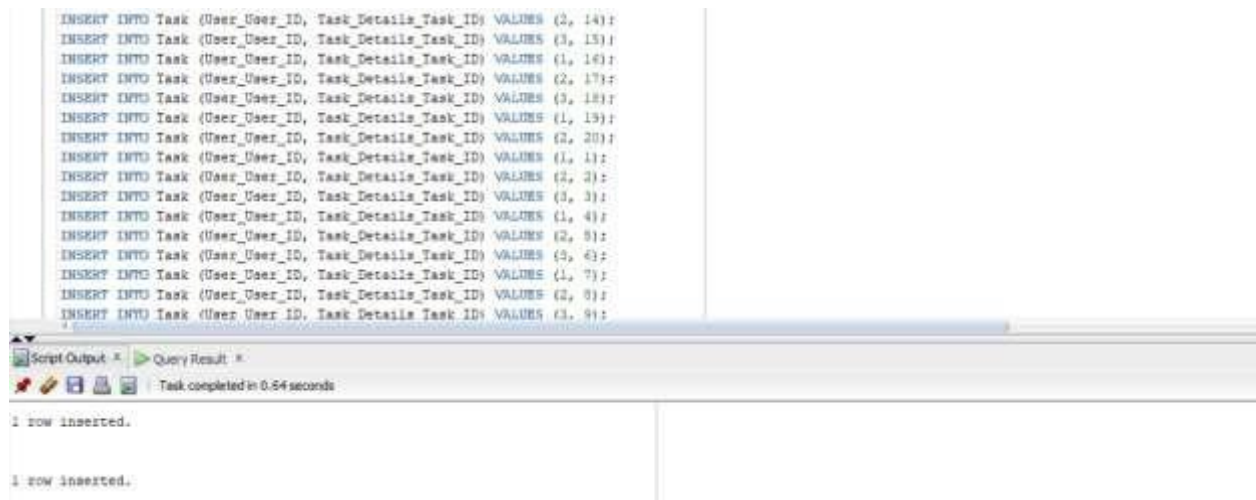


Figure 29 Inserting values in Task table

select * from Task;

Script Output x Query Result x

All Rows Fetched: 35 in 0.156 seconds

	USER_USER_ID	TASK_DETAILS_TASK_ID
1	1	1
2	2	2
3	3	3
4	1	4
5	2	5
6	3	6
7	1	7
8	2	8
9	3	9
10	1	10
11	2	11
12	3	12
13	1	13
14	2	14
15	3	15
16	1	16
17	2	17
18	3	18
19	1	19
20	2	20
21	3	21

Figure 30 Figure 34 Displaying Task table (1)

22	4	4
23	3	3
24	1	4
25	2	5
26	3	6
27	1	7
28	2	8
29	3	9
30	1	10
31	2	11
32	3	12
33	1	13
34	2	14
35	3	15

Figure 31 Displaying Task table (2)

8 Web-based Database Application

8.1 Basic Webforms

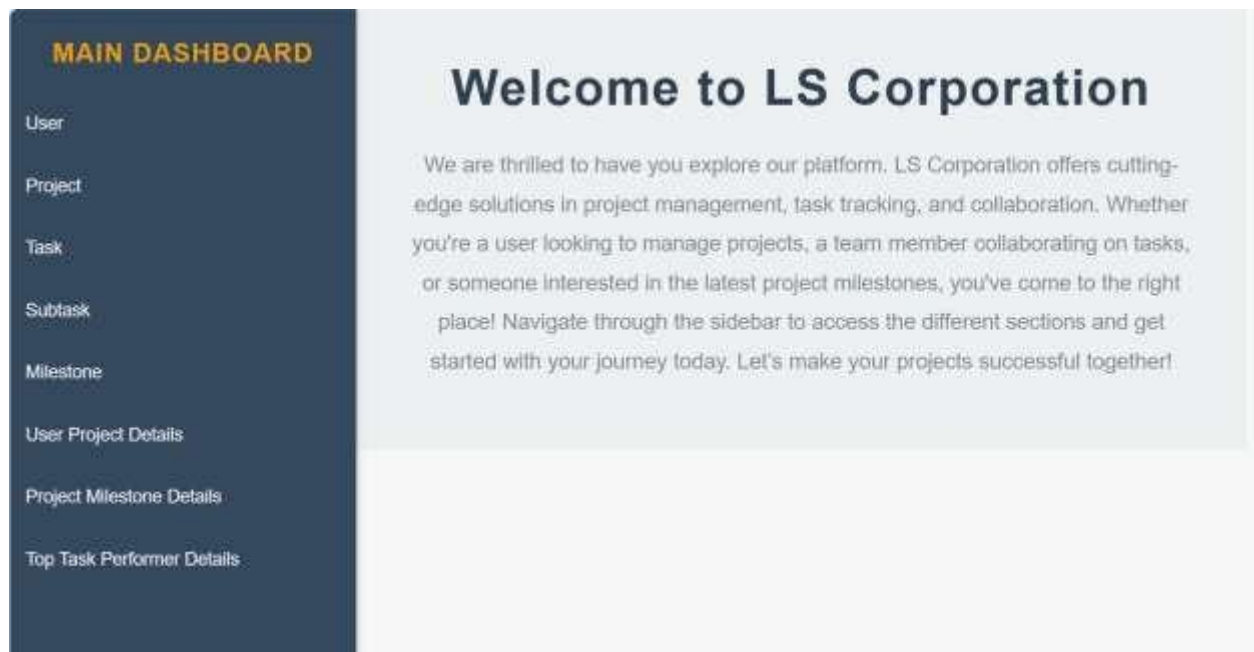


Figure 32 Dashboard page

USER_ID:	
USER_NAME:	
USER_EMAIL:	
USER_CONTACT:	
USER_ROLE:	
Insert Cancel	

	USER_ID	USER_NAME	USER_EMAIL	USER_CONTACT	USER_ROLE
Edit Delete	1	Alice	alice@gmail.com	9812345678	Developer
Edit Delete	2	Bibisha	bibisha@gmail.com	9822345678	Designer
Edit Delete	3	Charlie	charlie@gmail.com	9812345675	Project Manager
Edit Delete	4	Sugam	sugam@gmail.com	9812342378	Developer
Edit Delete	5	Sujan	sujan@gmail.com	9892345678	Designer
Edit Delete	6	Anjila	anjila@gmail.com	9712345678	Developer
Edit Delete	7	Bob	bob@gmail.com	9812345654	Designer
Edit Delete	8	Ram	ram@gmail.com	9711345678	Project Manager
Edit Delete	9	Diana	diana@gmail.com	9711345677	Developer
Edit Delete	10	Neha	neha@gmail.com	9711345688	Tester
Edit Delete	11	Numa	numa@gmail.com	9711555678	Developer
Edit Delete	12	Krish	krish@gmail.com	9701245678	Designer

Figure 33 User page

PROJECT_ID:	
PROJECT_NAME:	
PROJECT_START_DATE:	
PROJECT_DUE_DATE:	
PROJECT_STATUS:	
USER_USER_ID:	
Insert Cancel	

	PROJECT_ID	PROJECT_NAME	PROJECT_START_DATE	PROJECT_DUE_DATE	PROJECT_STATUS	USER_USER_ID
Edit Delete	24	Betaa	1/10/2025 12:00:00 AM	12/11/2025 12:00:00 AM	A	2
Edit Delete	22	Project Gama	1/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	C	1
Edit Delete	23	Z	1/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	C	1
Edit Delete	1	Project Alpha	1/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	A	3
Edit Delete	2	Project B	2/20/2025 12:00:00 AM	11/15/2025 12:00:00 AM	B	7
Edit Delete	3	Project C	3/5/2025 12:00:00 AM	10/20/2025 12:00:00 AM	C	12
Edit Delete	4	Project D	4/10/2024 12:00:00 AM	9/25/2024 12:00:00 AM	D	1
Edit Delete	5	Project E	5/15/2025 12:00:00 AM	8/10/2025 12:00:00 AM	E	20
Edit Delete	6	Project F	6/20/2025 12:00:00 AM	7/30/2025 12:00:00 AM	F	15
Edit Delete	7	Project G	7/25/2025 12:00:00 AM	12/5/2025 12:00:00 AM	G	9
Edit Delete	8	Project H	8/15/2023 12:00:00 AM	10/15/2023 12:00:00 AM	H	4
Edit Delete	9	Project I	9/10/2025 12:00:00 AM	11/20/2025 12:00:00 AM	I	18
Edit Delete	10	Project J	10/5/2024 12:00:00 AM	12/20/2024 12:00:00 AM	J	14
Edit Delete	11	Project K	11/10/2025 12:00:00 AM	1/10/2026 12:00:00 AM	K	6
Edit Delete	12	Project L	12/10/2022 12:00:00 AM	2/28/2023 12:00:00 AM	L	8
Edit Delete	13	Project M	1/20/2026 12:00:00 AM	3/25/2026 12:00:00 AM	M	19

Figure 34 Project page

TASK_ID: TASK_NAME: START_DATE: DUE_DATE: STATUS: PROJECT_PROJECT_ID: Insert Cancel						
	TASK_ID	TASK_NAME	START_DATE	DUE_DATE	STATUS	PROJECT_PROJECT_ID
Edit Delete	1	Project Scope	1/1/2025 12:00:00 AM	1/5/2025 12:00:00 AM	Completed	1
Edit Delete	2	Project Kickoff	1/6/2025 12:00:00 AM	1/10/2025 12:00:00 AM	In Progress	1
Edit Delete	3	Feasibility Study	1/11/2025 12:00:00 AM	1/15/2025 12:00:00 AM	Pending	1
Edit Delete	4	Project Budgeting	2/1/2025 12:00:00 AM	2/5/2025 12:00:00 AM	Completed	2
Edit Delete	5	Design Documentation	2/6/2025 12:00:00 AM	2/10/2025 12:00:00 AM	In Progress	2
Edit Delete	6	Integration Testing	4/1/2025 12:00:00 AM	4/15/2025 12:00:00 AM	Not Started	1
Edit Delete	7	Acceptance Testing	4/16/2025 12:00:00 AM	4/30/2025 12:00:00 AM	Not Started	1
Edit Delete	8	Deployment Plan	5/1/2025 12:00:00 AM	5/10/2025 12:00:00 AM	Not Started	1
Edit Delete	9	Go Live	5/11/2025 12:00:00 AM	5/15/2025 12:00:00 AM	Not Started	1
Edit Delete	10	Post-Launch	5/16/2025 12:00:00 AM	5/30/2025 12:00:00 AM	Not Started	1
Edit Delete	11	Research	1/1/2025 12:00:00 AM	1/20/2025 12:00:00 AM	Completed	2
Edit Delete	12	Prototype Development	1/21/2025 12:00:00 AM	2/10/2025 12:00:00 AM	In Progress	2
Edit Delete	13	User Feedback	2/11/2025 12:00:00 AM	2/20/2025 12:00:00 AM	Not Started	2

Figure 35 Task page

SUBTASK_ID: SUBTASK_NAME: SUBTASK_STATUS: TASK_DETAILS_TASK_ID: Insert Cancel				
	SUBTASK_ID	SUBTASK_NAME	SUBTASK_STATUS	TASK_DETAILS_TASK_ID
Edit Delete	1	Design UI	In Progress	1
Edit Delete	2	Develop API	Not Started	2
Edit Delete	3	Create Database Schema	Completed	2
Edit Delete	4	Write Unit Tests	Not Started	3
Edit Delete	5	Conduct Integration Tests	In Progress	3
Edit Delete	6	Prepare Deployment Plan	Not Started	4
Edit Delete	7	Gather User Feedback	Not Started	4
Edit Delete	8	Finalize Documentation	In Progress	5
Edit Delete	9	Conduct User Training	Not Started	6
Edit Delete	10	Create Marketing Materials	Not Started	7
Edit Delete	11	Set Up Monitoring	In Progress	8
Edit Delete	12	Post-Launch Review Meeting	Not Started	9
Edit Delete	13	Follow Up on Feedback	Not Started	10
Edit Delete	14	Plan Next Phase	Not Started	11
Edit Delete	15	Conduct Stakeholder Meeting	Not Started	12

Figure 36 SubTask page

MILESTONE_ID:
MILESTONE_NAME:
MILESTONE_PERIOD:
PROJECT_PROJECT_ID:
[Insert](#) [Cancel](#)

	MILESTONE_ID	MILESTONE_NAME	MILESTONE_PERIOD	PROJECT_PROJECT_ID
Edit Delete	16	Testing	1/19/2025 12:30:00 AM	1
Edit Delete	1	Initiation	1/15/2025 12:00:00 AM	1
Edit Delete	2	Planning	2/15/2025 12:00:00 AM	1
Edit Delete	3	Execution	3/15/2025 12:00:00 AM	1
Edit Delete	4	Monitoring	4/15/2025 12:00:00 AM	1
Edit Delete	5	Closure	5/15/2025 12:00:00 AM	1
Edit Delete	6	Requirements Gathering	1/20/2025 12:00:00 AM	2
Edit Delete	7	Design Phase	2/20/2025 12:00:00 AM	2
Edit Delete	9	Testing Phase	4/20/2025 12:00:00 AM	2
Edit Delete	10	Deployment	5/20/2025 12:00:00 AM	2
Edit Delete	11	Requirements Review	1/25/2025 12:00:00 AM	3
Edit Delete	12	Prototype	2/25/2025 12:00:00 AM	3
Edit Delete	13	Final Review	3/25/2025 12:00:00 AM	3
Edit Delete	14	User Acceptance Testing	4/25/2025 12:00:00 AM	3
Edit Delete	15	Go Live	5/25/2025 12:00:00 AM	3

Figure 37 Milestone page

8.2 Complex WebForms

User Project Details SQL

Query Logic:

```

SELECT U.User_Id, U.User_Name, U.User_Email, U.User_Contact, U.User_Role, P.Project_Id,
P.Project_Name, P.Project_Start_Date, P.Project_Due_Date, P.Project_Status
FROM
"User" U
JOIN
UserProject UP ON U.User_Id = UP.User_Id
JOIN
Project P ON UP.Project_Id = P.Project_Id
WHERE
U.User_Id = :USER_Name;

```

Alice								
PROJECT_NAME	PROJECT_START_DATE	PROJECT_DUE_DATE	PROJECT_STATUS	USER_NAME	USER_EMAIL	USER_CONTACT	USER_ROLE	
Octagon	9/5/2025 10:00:00 AM	11/11/2025 10:00:00 AM	E	Alice	alice@gmail.com	0012345678	Developer	
Project Gamma	1/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	C	Alice	alice@gmail.com	0012345678	Developer	
Zipper	1/15/2025 12:30:00 AM	11/21/2025 12:30:00 AM	A	Alice	alice@gmail.com	0012345678	Developer	
Project D	4/10/2024 12:00:00 AM	9/25/2024 12:00:00 AM	D	Alice	alice@gmail.com	0012345678	Developer	
Project U	1/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	H	Alice	alice@gmail.com	0012345678	Developer	

Figure 38 User Project Details

Project Milestone Details SQL

Query Logic:

SELECT m.Milestone_ID, m.Milestone_Name, m.Milestone_Period, m.Project_ID,
p.Project_Start_Date, , p.Project_Due_Date, p.Project_Status

FROM Milestone m

JOIN Project p

ON m.Project_ID = p.Project_ID

WHERE p.Project_Name = :Project_Name;

Project Alpha								
MILESTONE_ID	MILESTONE_NAME	MILESTONE_PERIOD	PROJECT_PROJECT_ID	PROJECT_NAME	PROJECT_START_DATE	PROJECT_DUE_DATE	PROJECT_STATUS	
1	Initiation	1/15/2025 12:00:00 AM	1	Project Alpha	5/5/2025 12:00:00 AM	12/31/2025 12:00:00 AM	A	
2	Planning	2/15/2025 12:00:00 AM	1	Project Alpha	5/5/2025 12:00:00 AM	12/31/2025 12:00:00 AM	A	
3	Execution	3/15/2025 12:00:00 AM	1	Project Alpha	5/5/2025 12:00:00 AM	12/31/2025 12:00:00 AM	A	
4	Monitoring	4/15/2025 12:00:00 AM	1	Project Alpha	5/5/2025 12:00:00 AM	12/31/2025 12:00:00 AM	A	
5	Closure	5/15/2025 12:00:00 AM	1	Project Alpha	5/5/2025 12:00:00 AM	12/31/2025 12:00:00 AM	A	

Figure 39 Project Milestone Details

Top Performer Details

SQL Query Logic:

```
SELECT U.USER_ID, U.USER_NAME, U.USER_ROLE, U.USER_CONTACT, P.PROJECT_ID,
P.PROJECT_NAME, COUNT(PUT.TASK_ID) AS CompletedTaskCount
```

```
FROM PUT_DETAILS PUT, "User" U, TASK_DETAILS T, PROJECT P
```

```
WHERE PUT.USER_ID = U.USER_ID AND PUT.TASK_ID = T.TASK_ID AND
PUT.PROJECT_ID = P.PROJECT_ID AND (T.STATUS <> 'In Progress')
```

```
AND (P.PROJECT_ID = :PROJECT_ID)
```

```
GROUP BY U.USER_ID, U.USER_NAME, P.PROJECT_ID, P.PROJECT_NAME,
U.USER_ROLE, U.USER_CONTACT
```

```
ORDER BY CompletedTaskCount DESC;
```

Project Alpha ▼

USER_ID	USER_NAME	USER_ROLE	USER_CONTACT	PROJECT_ID	PROJECT_NAME	COMPLETEDTASKCOUNT
3	Charlie Brown	Project Manager	123-456-7892	1	Project Alpha	2
1	Alice Johnson	Developer	123-456-7890	1	Project Alpha	2
2	Bob Smith	Tester	123-456-7891	1	Project Alpha	1

Figure 40 Top Performer Details

9 Testing

9.1 User Details

Table 8 User Details Test Table

Test	Create a user while updating and deleting the existing user from the table.
Action	Navigate to the user management form, fill out the required fields for a new user, update an existing user by clicking edit button in the user table, and delete an existing user from the table.
Expected Result	A new user is created successfully, and the existing user data is updated along with the deletion of an existing user.
Actual Result	Successfully created a new user, edited existing user data and deleted a user.
Conclusion	User details CRUD operation test is successfully performed.

9.1.1 Evidence of User Details

	USER_ID	USER_NAME	USER_EMAIL	USER_CONTACT	USER_ROLE
Edit Delete	1	Alice	alice@gmail.com	9812345678	Developer
Edit Delete	2	Bibisha	bibisha@gmail.com	9822345678	Designer
Edit Delete	3	Charlie	charlie@gmail.com	9812345675	Project Manager
Edit Delete	4	Sugam	sugam@gmail.com	9812342378	Developer
Edit Delete	5	Sujan	sujan@gmail.com	9892345678	Designer
Edit Delete	6	Anjila	anjila@gmail.com	9712345678	Developer
Edit Delete	7	Bob	bob@gmail.com	9812345654	Designer
Edit Delete	8	Ram	ram@gmail.com	9711345678	Project Manager
Edit Delete	9	Diana	diana@gmail.com	9711345677	Developer
Edit Delete	10	Neha	neha@gmail.com	9711345688	Tester
Edit Delete	11	Numa	numa@gmail.com	9711555678	Developer
Edit Delete	12	Krish	krish@gmail.com	9701345678	Designer
Edit Delete	13	Evan	evan@gmail.com	9711345666	Tester
Edit Delete	14	Aisha	aisha@gmail.com	9811345666	Developer
Edit Delete	15	Binu	binu@gmail.com	9711335666	Designer
Edit Delete	16	Akriti	akriti@gmail.com	9811845666	Developer
Edit Delete	17	Deepson	deepson@gmail.com	9781345666	Designer
Edit Delete	18	Smarika	smarika@gmail.com	9711345606	Tester
Edit Delete	19	Anna	anna@gmail.com	9711346066	Developer
Edit Delete	20	Laxman	laxman@gmail.com	9710045666	Designer

Figure 41 Initial User Details table

PROJECT_ID:

PROJECT_NAME:

PROJECT_START_DATE:

PROJECT_DUE_DATE:

PROJECT_STATUS:

USER_USER_ID:

[Insert](#) [Cancel](#)

	PROJECT_ID	PROJECT_NAME	PROJECT_START_DATE	P
Edit Delete	24	Betaa	1/10/2025 12:00:00 AM	12

Figure 42 User Management Table

USER_ID: 21
USER_NAME: Ningwa
USER_EMAIL: ningwa@gmail.com
USER_CONTACT: 9876543211
USER_ROLE: Designer
[Insert](#) [Cancel](#)

USER_ID	USER_NAME	USER_EMAIL
---------	-----------	------------

Figure 43 Inserting User Details data in form

	USER_ID	USER_NAME	USER_EMAIL	USER_CONTACT	USER_ROLE
Edit Delete	21	Ningwa	ningwa@gmail.com	9876543211	Designer
Edit Delete	1	Alice	alice@gmail.com	9812345678	Developer
Edit Delete	2	Bibisha	bibisha@gmail.com	9822345678	Designer
Edit Delete	3	Charlie	charlie@gmail.com	9812345675	Project Manager
Edit Delete	4	Sugam	sugam@gmail.com	9812342378	Developer
Edit Delete	5	Sujan	sujan@gmail.com	9892345678	Designer
Edit Delete	6	Anjila	anjila@gmail.com	9712345678	Developer
Edit Delete	7	Bob	bob@gmail.com	9812345654	Designer
Edit Delete	8	Ram	ram@gmail.com	9711345678	Project Manager
Edit Delete	9	Diana	diana@gmail.com	9711345677	Developer
Edit Delete	10	Neha	neha@gmail.com	9711345688	Tester
Edit Delete	11	Numa	numa@gmail.com	9711555678	Developer
Edit Delete	12	Krish	krish@gmail.com	9701345678	Designer
Edit Delete	13	Evan	evan@gmail.com	9711345666	Tester
Edit Delete	14	Aisha	aisha@gmail.com	9811345666	Developer
Edit Delete	15	Binu	binu@gmail.com	9711335666	Designer
Edit Delete	16	Akriti	akriti@gmail.com	9811845666	Developer
Edit Delete	17	Deepson	deepson@gmail.com	9781345666	Designer
Edit Delete	18	Smarika	smarika@gmail.com	9711345606	Tester
Edit Delete	19	Anna	anna@gmail.com	9711346066	Developer
Edit Delete	20	Laxman	laxman@gmail.com	9710045666	Designer

Figure 44 New User Details inserted

USER_ROLE:

[Insert](#) [Cancel](#)

	USER_ID	USER_NAME	USER_EMAIL	USER_CONTACT	USER_ROLE
Edit Delete	21	Ningwa	ningwa@gmail.com	9876543211	Designer
Edit Delete	1	Alice	alice@gmail.com	9812345678	Developer
Update Cancel	2	Bibisha	bibisha12@gmail.com	9822344678	Tester
Edit Delete	3	Charlie	charlie@gmail.com	9812345675	Project Manager
Edit Delete	4	Sugam	sugam@gmail.com	9812342378	Developer
Edit Delete	5	Sujan	sujan@gmail.com	9892345678	Designer
Edit Delete	6	Anjila	anjila@gmail.com	9712345678	Developer

Figure 45 Updating Existing User data

USER_ROLE:

[Insert](#) [Cancel](#)

	USER_ID	USER_NAME	USER_EMAIL	USER_CONTACT	USER_ROLE
Edit Delete	21	Ningwa	ningwa@gmail.com	9876543211	Designer
Edit Delete	1	Alice	alice@gmail.com	9812345678	Developer
Edit Delete	2	Bibisha	bibisha12@gmail.com	9822344678	Tester
Edit Delete	3	Charlie	charlie@gmail.com	9812345675	Project Manager
Edit Delete	4	Sugam	sugam@gmail.com	9812342378	Developer
Edit Delete	5	Sujan	sujan@gmail.com	9892345678	Designer
Edit Delete	6	Anjila	anjila@gmail.com	9712345678	Developer

Figure 46 Existing User Details Updated

	USER_ID	USER_NAME	USER_EMAIL	USER_CONTACT	USER_ROLE
Edit Delete	1	Alice	alice@gmail.com	9812345678	Developer
Edit Delete	2	Bibisha	bibisha@gmail.com	9822345678	Designer
Edit Delete	3	Charlie	charlie@gmail.com	9812345675	Project Manager
Edit Delete	4	Sugam	sugam@gmail.com	9812342378	Developer
Edit Delete	5	Sujan	sujan@gmail.com	9892345678	Designer
Edit Delete	6	Anjila	anjila@gmail.com	9712345678	Developer
Edit Delete	7	Bob	bob@gmail.com	9812345654	Designer
Edit Delete	8	Ram	ram@gmail.com	9711345678	Project Manager
Edit Delete	9	Diana	diana@gmail.com	9711345677	Developer
Edit Delete	10	Neha	neha@gmail.com	9711345688	Tester
Edit Delete	11	Numa	numa@gmail.com	9711555678	Developer
Edit Delete	12	Krish	krish@gmail.com	9701345678	Designer
Edit Delete	13	Pranav	pranav@gmail.com	9711345666	Tester

Figure 47 Deleting User_ID 6

	USER_ID	USER_NAME	USER_EMAIL	USER_CONTACT	USER_ROLE
Edit Delete	21	Ningwa	ningwa@gmail.com	9876543211	Designer
Edit Delete	1	Alice	alice@gmail.com	9812345678	Developer
Edit Delete	2	Bibisha	bibisha12@gmail.com	9822344678	Tester
Edit Delete	3	Charlie	charlie@gmail.com	9812345675	Project Manager
Edit Delete	4	Sugam	sugam@gmail.com	9812342378	Developer
Edit Delete	5	Sujan	sujan@gmail.com	9892345678	Designer
Edit Delete	7	Bob	bob@gmail.com	9812345654	Designer
Edit Delete	8	Ram	ram@gmail.com	9711345678	Project Manager
Edit Delete	9	Diana	diana@gmail.com	9711345677	Developer
Edit Delete	10	Neha	neha@gmail.com	9711345688	Tester
Edit Delete	11	Numa	numa@gmail.com	9711555678	Developer
Edit Delete	12	Krish	krish@gmail.com	9701345678	Designer
Edit Delete	13	Evan	evan@gmail.com	9711345666	Tester
Edit Delete	14	Aisha	aisha@gmail.com	9811345666	Developer
Edit Delete	15	Binu	binu@gmail.com	9711335666	Designer
Edit Delete	16	Akriti	akriti@gmail.com	9811845666	Developer
Edit Delete	17	Deepson	deepson@gmail.com	9781345666	Designer
Edit Delete	18	Smarika	smarika@gmail.com	9711345606	Tester
Edit Delete	19	Anna	anna@gmail.com	9711346066	Developer
Edit Delete	20	Laxman	laxman@gmail.com	9710045666	Designer

Figure 48 Final User details table after CRUD

9.2 Project Details

Table 9 Project Details Test Table

Test	Create a project while updating and deleting the existing project data from the table.
Action	Navigate to the project management form, fill out the required fields for a new project data, update an existing project by clicking edit button in the project table, and delete an existing project from the table.
Expected Result	A new project is created successfully, and
	the existing project data is updated along with the deletion of an existing project details.
Actual Result	Successfully created a new project, edited existing project data and deleted a project.
Conclusion	Project details CRUD operation test is successfully performed.

9.2.1 Evidence of Project Details

	PROJECT_ID	PROJECT_NAME	PROJECT_START_DATE	PROJECT_DUE_DATE	PROJECT_STATUS	USER_USER_ID
Edit Delete	24	Betaa	1/10/2025 12:00:00 AM	12/11/2025 12:00:00 AM	A	2
Edit Delete	22	Project Gama	1/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	C	1
Edit Delete	23	Z	1/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	C	1
Edit Delete	1	Project Alpha	1/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	A	3
Edit Delete	2	Project B	2/20/2025 12:00:00 AM	11/15/2025 12:00:00 AM	B	7
Edit Delete	3	Project C	3/5/2025 12:00:00 AM	10/20/2025 12:00:00 AM	C	12
Edit Delete	4	Project D	4/10/2024 12:00:00 AM	9/25/2024 12:00:00 AM	D	1
Edit Delete	5	Project E	5/15/2025 12:00:00 AM	8/10/2025 12:00:00 AM	E	20
Edit Delete	6	Project F	6/20/2025 12:00:00 AM	7/30/2025 12:00:00 AM	F	15
Edit Delete	7	Project G	7/25/2025 12:00:00 AM	12/5/2025 12:00:00 AM	G	9
Edit Delete	8	Project H	8/15/2023 12:00:00 AM	10/15/2023 12:00:00 AM	H	4
Edit Delete	9	Project I	9/10/2025 12:00:00 AM	11/20/2025 12:00:00 AM	I	18
Edit Delete	10	Project J	10/5/2024 12:00:00 AM	12/20/2024 12:00:00 AM	J	14
Edit Delete	11	Project K	11/10/2025 12:00:00 AM	1/10/2026 12:00:00 AM	K	6
Edit Delete	12	Project L	12/10/2022 12:00:00 AM	2/28/2023 12:00:00 AM	L	8
Edit Delete	13	Project M	1/20/2026 12:00:00 AM	3/25/2026 12:00:00 AM	M	19
Edit Delete	14	Project N	2/15/2026 12:00:00 AM	5/10/2026 12:00:00 AM	N	2
Edit Delete	15	Project O	3/5/2026 12:00:00 AM	6/15/2026 12:00:00 AM	O	11
Edit Delete	16	Project P	4/5/2026 12:00:00 AM	7/10/2026 12:00:00 AM	P	16
Edit Delete	17	Project Q	5/10/2026 12:00:00 AM	8/15/2026 12:00:00 AM	Q	13
Edit Delete	18	Project R	6/20/2024 12:00:00 AM	9/20/2024 12:00:00 AM	R	5
Edit Delete	19	Project S	7/15/2026 12:00:00 AM	11/15/2026 12:00:00 AM	S	17
Edit Delete	20	Project T	8/10/2025 12:00:00 AM	10/20/2025 12:00:00 AM	T	10
Edit Delete	21	Project U	1/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	H	1

Figure 49 Initial Project details table

PROJECT_ID: 25

PROJECT_NAME: Octagon

PROJECT_START_DATE: 09/05/2025 10:00:00 AM

PROJECT_DUE_DATE: 11/11/2025 10:00:00 AM

PROJECT_STATUS: E

USER_USER_ID: 1

[Insert](#)
[Cancel](#)

Figure 50 Inserting Project Details data in form

	PROJECT_ID	PROJECT_NAME	PROJECT_START_DATE	PROJECT_DUE_DATE	PROJECT_STATUS	USER_USER_ID
Edit Delete	24	Betaa	1/10/2025 12:00:00 AM	12/11/2025 12:00:00 AM	A	2
Edit Delete	25	Octagon	9/5/2025 10:00:00 AM	11/11/2025 10:00:00 AM	E	1
Edit Delete	22	Project Gama	1/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	C	1
Edit Delete	23	Z	1/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	C	1
Edit Delete	1	Project Alpha	1/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	A	3
Edit Delete	2	Project B	2/20/2025 12:00:00 AM	11/15/2025 12:00:00 AM	B	7
Edit Delete	3	Project C	3/5/2025 12:00:00 AM	10/20/2025 12:00:00 AM	C	12
Edit Delete	4	Project D	4/10/2024 12:00:00 AM	9/25/2024 12:00:00 AM	D	1
Edit Delete	5	Project E	5/15/2025 12:00:00 AM	8/10/2025 12:00:00 AM	E	20
Edit Delete	6	Project F	6/20/2025 12:00:00 AM	7/30/2025 12:00:00 AM	F	15
Edit Delete	7	Project G	7/25/2025 12:00:00 AM	12/5/2025 12:00:00 AM	G	9
Edit Delete	8	Project H	8/15/2023 12:00:00 AM	10/15/2023 12:00:00 AM	H	4
Edit Delete	9	Project I	9/10/2025 12:00:00 AM	11/20/2025 12:00:00 AM	I	18
Edit Delete	10	Project J	10/5/2024 12:00:00 AM	12/20/2024 12:00:00 AM	J	14
Edit Delete	11	Project K	11/10/2025 12:00:00 AM	1/10/2026 12:00:00 AM	K	6
Edit Delete	12	Project L	12/10/2022 12:00:00 AM	2/28/2023 12:00:00 AM	L	8
Edit Delete	13	Project M	1/20/2026 12:00:00 AM	3/25/2026 12:00:00 AM	M	19
Edit Delete	14	Project N	2/15/2026 12:00:00 AM	4/10/2026 12:00:00 AM	N	3

Figure 51 New Project Details Inserted

USER_USER_ID:

[Insert](#) [Cancel](#)

	PROJECT_ID	PROJECT_NAME	PROJECT_START_DATE	PROJECT_DUE_DATE	PROJECT_STATUS	USER_USER_ID
Edit Delete	24	Betaa	1/10/2025 12:00:00 AM	12/11/2025 12:00:00 AM	A	2
Edit Delete	25	Octagon	9/5/2025 10:00:00 AM	11/11/2025 10:00:00 AM	E	1
Edit Delete	22	Project Gama	1/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	C	1
Update Cancel	23	Zipper	1/15/2025 12:30:00 AM	11/21/2025 12:30:00 AM	A	1
Edit Delete	1	Project Alpha	1/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	A	3
Edit Delete	2	Project B	2/20/2025 12:00:00 AM	11/15/2025 12:00:00 AM	B	7
Edit Delete	3	Project C	3/5/2025 12:00:00 AM	10/20/2025 12:00:00 AM	C	12
Edit Delete	4	Project D	4/10/2024 12:00:00 AM	9/25/2024 12:00:00 AM	D	1
Edit Delete	5	Project E	5/15/2025 12:00:00 AM	8/10/2025 12:00:00 AM	E	20
Edit Delete	6	Project F	6/20/2025 12:00:00 AM	7/30/2025 12:00:00 AM	F	15

Figure 52 Updating Existing Project Data

PROJECT STATUS:

USER_USER_ID:

[Insert](#) [Cancel](#)

	PROJECT_ID	PROJECT_NAME	PROJECT_START_DATE	PROJECT_DUE_DATE	PROJECT_STATUS	USER_USER_ID
Edit Delete	24	Betaa	1/10/2025 12:00:00 AM	12/11/2025 12:00:00 AM	A	2
Edit Delete	25	Octagon	9/5/2025 10:00:00 AM	11/11/2025 10:00:00 AM	E	1
Edit Delete	22	Project Gama	1/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	C	1
Edit Delete	23	Zipper	1/15/2025 12:30:00 AM	11/21/2025 12:30:00 AM	A	1
Edit Delete	1	Project Alpha	1/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	A	3
Edit Delete	2	Project B	2/20/2025 12:00:00 AM	11/15/2025 12:00:00 AM	B	7
Edit Delete	3	Project C	3/5/2025 12:00:00 AM	10/20/2025 12:00:00 AM	C	12
Edit Delete	4	Project D	4/10/2024 12:00:00 AM	9/25/2024 12:00:00 AM	D	1

Figure 53 Existing Project Details Updated

	PROJECT_ID	PROJECT_NAME	PROJECT_START_DATE	PROJECT_DUE_DATE	PROJECT_STATUS	USER_USER_ID
Edit Delete	24	Betaa	1/10/2025 12:00:00 AM	12/11/2025 12:00:00 AM	A	2
Edit Delete	25	Octagon	9/5/2025 10:00:00 AM	11/11/2025 10:00:00 AM	E	1
Edit Delete	22	Project Gama	1/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	C	1
Edit Delete	23	Zipper	1/15/2025 12:30:00 AM	11/21/2025 12:30:00 AM	A	1
Edit Delete	1	Project Alpha	1/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	A	3
Edit Delete	2	Project B	2/20/2025 12:00:00 AM	11/15/2025 12:00:00 AM	B	7
Edit Delete	3	Project C	3/5/2025 12:00:00 AM	10/20/2025 12:00:00 AM	C	12
Edit Delete	4	Project D	4/10/2024 12:00:00 AM	9/25/2024 12:00:00 AM	D	1
Edit Delete	5	Project E	5/15/2025 12:00:00 AM	8/10/2025 12:00:00 AM	E	20
Edit Delete	6	Project F	6/20/2025 12:00:00 AM	7/30/2025 12:00:00 AM	F	15
Edit Delete	7	Project G	7/25/2025 12:00:00 AM	12/5/2025 12:00:00 AM	G	9
Edit Delete	8	Project H	8/15/2023 12:00:00 AM	10/15/2023 12:00:00 AM	H	4

Figure 54 Deleting Project_ID 6

	PROJECT_ID	PROJECT_NAME	PROJECT_START_DATE	PROJECT_DUE_DATE	PROJECT_STATUS	USER_USER_ID
Edit Delete	24	Betaa	1/10/2025 12:00:00 AM	12/11/2025 12:00:00 AM	A	2
Edit Delete	25	Octagon	9/5/2025 10:00:00 AM	11/11/2025 10:00:00 AM	E	1
Edit Delete	22	Project Gama	1/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	C	1
Edit Delete	23	Zipper	1/15/2025 12:30:00 AM	11/21/2025 12:30:00 AM	A	1
Edit Delete	1	Project Alpha	1/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	A	3
Edit Delete	2	Project B	2/20/2025 12:00:00 AM	11/15/2025 12:00:00 AM	B	7
Edit Delete	3	Project C	3/5/2025 12:00:00 AM	10/20/2025 12:00:00 AM	C	12
Edit Delete	4	Project D	4/10/2024 12:00:00 AM	9/25/2024 12:00:00 AM	D	1
Edit Delete	5	Project E	5/15/2025 12:00:00 AM	8/10/2025 12:00:00 AM	E	20
Edit Delete	7	Project G	7/25/2025 12:00:00 AM	12/5/2025 12:00:00 AM	G	9
Edit Delete	8	Project H	8/15/2023 12:00:00 AM	10/15/2023 12:00:00 AM	H	4
Edit Delete	9	Project I	9/10/2025 12:00:00 AM	11/20/2025 12:00:00 AM	I	18
Edit Delete	10	Project J	10/5/2024 12:00:00 AM	12/20/2024 12:00:00 AM	J	14
Edit Delete	11	Project K	11/10/2025 12:00:00 AM	1/10/2026 12:00:00 AM	K	6
Edit Delete	12	Project L	12/10/2022 12:00:00 AM	2/28/2023 12:00:00 AM	L	8
Edit Delete	13	Project M	1/20/2026 12:00:00 AM	3/25/2026 12:00:00 AM	M	19
Edit Delete	14	Project N	2/15/2026 12:00:00 AM	5/10/2026 12:00:00 AM	N	2
Edit Delete	15	Project O	3/5/2026 12:00:00 AM	6/15/2026 12:00:00 AM	O	11
Edit Delete	16	Project P	4/5/2026 12:00:00 AM	7/10/2026 12:00:00 AM	P	16
Edit Delete	17	Project Q	5/10/2026 12:00:00 AM	8/15/2026 12:00:00 AM	Q	13
Edit Delete	18	Project R	6/20/2024 12:00:00 AM	9/20/2024 12:00:00 AM	R	5
Edit Delete	19	Project S	7/15/2026 12:00:00 AM	11/15/2026 12:00:00 AM	S	17
Edit Delete	20	Project T	8/10/2025 12:00:00 AM	10/20/2025 12:00:00 AM	T	10
Edit Delete	21	Project U	1/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	H	1

Figure 55 Final Project Details table after CRUD

9.3 Task Details

Table 10 Task Details Test Table

Test	Create a task while updating and deleting the existing task data from the table.
Action	Navigate to the task management form, fill out the required fields for a new task data, update an existing task by clicking edit button in the task table, and delete an existing task from the table.
Expected Result	A new task is created successfully, and the existing task data is updated along with the deletion of an existing task details.
Actual Result	Successfully created a new task, edited existing task data and deleted a task.
Conclusion	Task details CRUD operation test is successfully performed.

9.3.1 Evidence of Task Details

	TASK_ID	TASK_NAME	START_DATE	DUE_DATE	STATUS	PROJECT_PROJECT_ID
Edit Delete	1	Project Scope	1/1/2025 12:00:00 AM	1/5/2025 12:00:00 AM	Completed	1
Edit Delete	2	Project Kickoff	1/6/2025 12:00:00 AM	1/10/2025 12:00:00 AM	In Progress	1
Edit Delete	3	Feasibility Study	1/11/2025 12:00:00 AM	1/15/2025 12:00:00 AM	Pending	1
Edit Delete	4	Project Budgeting	2/1/2025 12:00:00 AM	2/5/2025 12:00:00 AM	Completed	2
Edit Delete	5	Design Documentation	2/6/2025 12:00:00 AM	2/10/2025 12:00:00 AM	In Progress	2
Edit Delete	6	Integration Testing	4/1/2025 12:00:00 AM	4/15/2025 12:00:00 AM	Not Started	1
Edit Delete	7	Acceptance Testing	4/16/2025 12:00:00 AM	4/30/2025 12:00:00 AM	Not Started	1
Edit Delete	8	Deployment Plan	5/1/2025 12:00:00 AM	5/10/2025 12:00:00 AM	Not Started	1
Edit Delete	9	Go Live	5/11/2025 12:00:00 AM	5/15/2025 12:00:00 AM	Not Started	1
Edit Delete	10	Post-Launch	5/16/2025 12:00:00 AM	5/30/2025 12:00:00 AM	Not Started	1
Edit Delete	11	Research	1/1/2025 12:00:00 AM	1/20/2025 12:00:00 AM	Completed	2
Edit Delete	12	Prototype Development	1/21/2025 12:00:00 AM	2/10/2025 12:00:00 AM	In Progress	2
Edit Delete	13	User Feedback	2/11/2025 12:00:00 AM	2/20/2025 12:00:00 AM	Not Started	2
Edit Delete	14	Final Testing	2/21/2025 12:00:00 AM	3/1/2025 12:00:00 AM	Not Started	2
Edit Delete	15	Planning	3/2/2025 12:00:00 AM	3/15/2025 12:00:00 AM	Not Started	2
Edit Delete	16	Marketing	3/16/2025 12:00:00 AM	4/1/2025 12:00:00 AM	Not Started	2
Edit Delete	17	Feedback	4/2/2025 12:00:00 AM	4/15/2025 12:00:00 AM	Not Started	2
Edit Delete	18	Documentation	4/16/2025 12:00:00 AM	4/30/2025 12:00:00 AM	Not Started	2
Edit Delete	19	Handover	5/1/2025 12:00:00 AM	5/15/2025 12:00:00 AM	Not Started	2
Edit Delete	20	Review	5/16/2025 12:00:00 AM	5/30/2025 12:00:00 AM	Not Started	2

Figure 56 Initial Task details table

TASK_ID: 21

TASK_NAME: Development

START_DATE: 5/10/2025 1:00:00 PM

DUE_DATE: 7/03/2025 4:30:00 PM

STATUS: In progress

PROJECT_PROJECT_ID: 1

[Insert](#) [Cancel](#)

Figure 57 Inserting Task Details data in form

PROJECT_PROJECT_ID:						
Insert Cancel						
	TASK_ID	TASK_NAME	START_DATE	DUE_DATE	STATUS	PROJECT_PROJECT_ID
Edit Delete	21	Development	5/10/2025 1:00:00 PM	7/3/2025 4:30:00 PM	In progress	1
Edit Delete	1	Project Scope	1/1/2025 12:00:00 AM	1/5/2025 12:00:00 AM	Completed	1
Edit Delete	2	Project Kickoff	1/6/2025 12:00:00 AM	1/10/2025 12:00:00 AM	In Progress	1
Edit Delete	3	Feasibility Study	1/11/2025 12:00:00 AM	1/15/2025 12:00:00 AM	Pending	1
Edit Delete	4	Project Budgeting	2/1/2025 12:00:00 AM	2/5/2025 12:00:00 AM	Completed	2
Edit Delete	5	Design Documentation	2/6/2025 12:00:00 AM	2/10/2025 12:00:00 AM	In Progress	2
Edit Delete	6	Integration Testing	4/1/2025 12:00:00 AM	4/15/2025 12:00:00 AM	Not Started	1
Edit Delete	7	Acceptance Testing	4/16/2025 12:00:00 AM	4/30/2025 12:00:00 AM	Not Started	1
Edit Delete	8	Deployment Plan	5/1/2025 12:00:00 AM	5/10/2025 12:00:00 AM	Not Started	1
Edit Delete	9	Go Live	5/11/2025 12:00:00 AM	5/15/2025 12:00:00 AM	Not Started	1
Edit Delete	10	Post-Launch	5/16/2025 12:00:00 AM	5/30/2025 12:00:00 AM	Not Started	1
Edit Delete	11	Research	1/1/2025 12:00:00 AM	1/20/2025 12:00:00 AM	Completed	2
Edit Delete	12	Prototype Development	1/21/2025 12:00:00 AM	2/10/2025 12:00:00 AM	In Progress	2

Figure 58 New Task Details Inserted

PROJECT_PROJECT_ID:						
Insert Cancel						
	TASK_ID	TASK_NAME	START_DATE	DUE_DATE	STATUS	PROJECT_PROJECT_ID
Edit Delete	21	Development	5/10/2025 1:00:00 PM	7/3/2025 4:30:00 PM	In progress	1
Edit Delete	1	Project Scope	1/1/2025 12:00:00 AM	1/5/2025 12:00:00 AM	Completed	1
Edit Delete	2	Project Kickoff	1/6/2025 12:00:00 AM	1/10/2025 12:00:00 AM	In Progress	1
Edit Delete	3	Feasibility Study	1/11/2025 12:00:00 AM	1/15/2025 12:00:00 AM	Pending	1
Edit Delete	4	Project Budgeting	2/1/2025 12:00:00 AM	2/5/2025 12:00:00 AM	Completed	2
Edit Delete	5	Design Documentation	2/6/2025 12:00:00 AM	2/10/2025 12:00:00 AM	In Progress	2
Edit Delete	6	Integration Testing	4/1/2025 12:00:00 AM	4/15/2025 12:00:00 AM	Not Started	1
Edit Delete	7	Acceptance Testing	4/16/2025 12:00:00 AM	4/30/2025 12:00:00 AM	Not Started	1
Edit Delete	8	Deployment Plan	5/1/2025 12:00:00 AM	5/10/2025 12:00:00 AM	Not Started	1
Update Cancel	9	Ground Research	5/11/2025 2:30:00 PM	5/15/2025 11:00:00 PM	Completed	1
Edit Delete	10	Post-Launch	5/16/2025 12:00:00 AM	5/30/2025 12:00:00 AM	Not Started	1
Edit Delete	11	Research	1/1/2025 12:00:00 AM	1/20/2025 12:00:00 AM	Completed	2
Edit Delete	12	Prototype Development	1/21/2025 12:00:00 AM	2/10/2025 12:00:00 AM	In Progress	2

Figure 59 Updating Existing Task Data

Edit Delete	6	Integration Testing	4/1/2025 12:00:00 AM	4/15/2025 12:00:00 AM	Not Started	1
Edit Delete	7	Acceptance Testing	4/16/2025 12:00:00 AM	4/30/2025 12:00:00 AM	Not Started	1
Edit Delete	8	Deployment Plan	5/1/2025 12:00:00 AM	5/10/2025 12:00:00 AM	Not Started	1
Edit Delete	9	Ground Research	5/11/2025 2:30:00 PM	5/15/2025 11:00:00 PM	Completed	1
Edit Delete	10	Post-Launch	5/16/2025 12:00:00 AM	5/30/2025 12:00:00 AM	Not Started	1
Edit Delete	11	Research	1/1/2025 12:00:00 AM	1/20/2025 12:00:00 AM	Completed	2
Edit Delete	12	Prototype Development	1/21/2025 12:00:00 AM	2/10/2025 12:00:00 AM	In Progress	2
Edit Delete	13	User Feedback	2/11/2025 12:00:00 AM	2/20/2025 12:00:00 AM	Not Started	2
Edit Delete	14	Final Testing	2/21/2025 12:00:00 AM	3/1/2025 12:00:00 AM	Not Started	2

Figure 60 Existing Task Details Updated

Edit Delete	13	User Feedback	2/11/2025 12:00:00 AM	2/20/2025 12:00:00 AM	Not Started	2
Edit Delete	14	Final Testing	2/21/2025 12:00:00 AM	3/1/2025 12:00:00 AM	Not Started	2
Edit Delete	15	Planning	3/2/2025 12:00:00 AM	3/15/2025 12:00:00 AM	Not Started	2
Edit Delete	16	Marketing	3/16/2025 12:00:00 AM	4/1/2025 12:00:00 AM	Not Started	2
Edit Delete	17	Feedback	4/2/2025 12:00:00 AM	4/15/2025 12:00:00 AM	Not Started	2
Edit Delete	18	Documentation	4/16/2025 12:00:00 AM	4/30/2025 12:00:00 AM	Not Started	2

Figure 61 Deleting the Task_ID 15

	TASK_ID	TASK_NAME	START_DATE	DUE_DATE	STATUS	PROJECT_PROJECT_ID
Edit Delete	21	Development	5/10/2025 1:00:00 PM	7/3/2025 4:30:00 PM	In progress	1
Edit Delete	1	Project Scope	1/1/2025 12:00:00 AM	1/5/2025 12:00:00 AM	Completed	1
Edit Delete	2	Project Kickoff	1/6/2025 12:00:00 AM	1/10/2025 12:00:00 AM	In Progress	1
Edit Delete	3	Feasibility Study	1/11/2025 12:00:00 AM	1/15/2025 12:00:00 AM	Pending	1
Edit Delete	4	Project Budgeting	2/1/2025 12:00:00 AM	2/5/2025 12:00:00 AM	Completed	2
Edit Delete	5	Design Documentation	2/6/2025 12:00:00 AM	2/10/2025 12:00:00 AM	In Progress	2
Edit Delete	6	Integration Testing	4/1/2025 12:00:00 AM	4/15/2025 12:00:00 AM	Not Started	1
Edit Delete	7	Acceptance Testing	4/16/2025 12:00:00 AM	4/30/2025 12:00:00 AM	Not Started	1
Edit Delete	8	Deployment Plan	5/1/2025 12:00:00 AM	5/10/2025 12:00:00 AM	Not Started	1
Edit Delete	9	Ground Research	5/11/2025 2:30:00 PM	5/15/2025 11:00:00 PM	Completed	1
Edit Delete	10	Post-Launch	5/16/2025 12:00:00 AM	5/30/2025 12:00:00 AM	Not Started	1
Edit Delete	11	Research	1/1/2025 12:00:00 AM	1/20/2025 12:00:00 AM	Completed	2
Edit Delete	12	Prototype Development	1/21/2025 12:00:00 AM	2/10/2025 12:00:00 AM	In Progress	2
Edit Delete	13	User Feedback	2/11/2025 12:00:00 AM	2/20/2025 12:00:00 AM	Not Started	2
Edit Delete	14	Final Testing	2/21/2025 12:00:00 AM	3/1/2025 12:00:00 AM	Not Started	2
Edit Delete	16	Marketing	3/16/2025 12:00:00 AM	4/1/2025 12:00:00 AM	Not Started	2
Edit Delete	17	Feedback	4/2/2025 12:00:00 AM	4/15/2025 12:00:00 AM	Not Started	2
Edit Delete	18	Documentation	4/16/2025 12:00:00 AM	4/30/2025 12:00:00 AM	Not Started	2
Edit Delete	19	Handover	5/1/2025 12:00:00 AM	5/15/2025 12:00:00 AM	Not Started	2
Edit Delete	20	Review	5/16/2025 12:00:00 AM	5/30/2025 12:00:00 AM	Not Started	2

Figure 62 Final Task Details table after CRUD

9.4 SubTask Details

Table 11 SubTask Details Test Table

Test	Create a subtask while updating and deleting the existing subtask data from the table.
Action	Navigate to the subtask management form, fill out the required fields for a new subtask data, update an existing subtask by clicking edit button in the subtask table, and delete an existing subtask from the table.
Expected Result	A new subtask is created successfully, and the existing subtask data is updated along with the deletion of existing subtask details of a project.
Actual Result	Successfully created a new subtask, edited existing subtask data and deleted the subtask of a project.
Conclusion	Subtask details CRUD operation test is successfully performed.

9.4.1 Evidence of SubTask Details

	SUBTASK_ID	SUBTASK_NAME	SUBTASK_STATUS	TASK_DETAILS_TASK_ID
Edit Delete	1	Design UI	In Progress	1
Edit Delete	2	Develop API	Not Started	2
Edit Delete	3	Create Database Schema	Completed	2
Edit Delete	4	Write Unit Tests	Not Started	3
Edit Delete	5	Conduct Integration Tests	In Progress	3
Edit Delete	6	Prepare Deployment Plan	Not Started	4
Edit Delete	7	Gather User Feedback	Not Started	4
Edit Delete	8	Finalize Documentation	In Progress	5
Edit Delete	9	Conduct User Training	Not Started	6
Edit Delete	10	Create Marketing Materials	Not Started	7
Edit Delete	11	Set Up Monitoring	In Progress	8
Edit Delete	12	Post-Launch Review Meeting	Not Started	9
Edit Delete	13	Follow Up on Feedback	Not Started	10
Edit Delete	14	Plan Next Phase	Not Started	11
Edit Delete	15	Conduct Stakeholder Meeting	In Progress	12
Edit Delete	16	Review Project Outcomes	Not Started	13
Edit Delete	17	Identify Improvements	Not Started	14
Edit Delete	18	Archive Project Documents	Not Started	15
Edit Delete	19	Set Up Post-Mortem Analysis	Not Started	16
Edit Delete	20	Prepare Final Report	Not Started	17

Figure 63 Initial SubTask details table

SUBTASK_ID:

SUBTASK_NAME:

SUBTASK_STATUS:

TASK_DETAILS_TASK_ID:

[Insert](#) [Cancel](#)

	SUBTASK_ID	SUBTASK_NAME	SUBTASK ST
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Figure 64 Inserting SubTask Details data in form

SUBTASK STATUS:

TASK_DETAILS_TASK_ID:

[Insert](#) [Cancel](#)

	SUBTASK_ID	SUBTASK_NAME	SUBTASK_STATUS	TASK_DETAILS_TASK_ID
Edit Delete	21	AI research	Completed	1
Edit Delete	1	Design UI	In Progress	1
Edit Delete	2	Develop API	Not Started	2
Edit Delete	3	Create Database Schema	Completed	2
Edit Delete	4	Write Unit Tests	Not Started	3
Edit Delete	5	Conduct Integration Tests	In Progress	3
Edit Delete	6	Prepare Deployment Plan	Not Started	4
Edit Delete	7	Gather User Feedback	Not Started	4

Figure 65 New SubTask Details Inserted

[Insert](#) [Cancel](#)

	SUBTASK_ID	SUBTASK_NAME	SUBTASK_STATUS	TASK_DETAILS_TASK_ID
Edit Delete	21	AI research	Completed	1
Edit Delete	1	Design UI	In Progress	1
Edit Delete	2	Develop API	Not Started	2
Edit Delete	3	Create Database Schema	Completed	2
Update Cancel	4	Integrate Unit Test	In progress	3
Edit Delete	5	Conduct Integration Tests	In Progress	3
Edit Delete	6	Prepare Deployment Plan	Not Started	4
Edit Delete	7	Gather User Feedback	Not Started	4
Edit Delete	8	Finalize Documentation	In Progress	5

Figure 66 Updating Existing SubTask Data

SUBTASK STATUS:

TASK_DETAILS_TASK_ID:

[Insert](#) [Cancel](#)

	SUBTASK_ID	SUBTASK_NAME	SUBTASK_STATUS	TASK_DETAILS_TASK_ID
Edit Delete	21	AI research	Completed	1
Edit Delete	1	Design UI	In Progress	1
Edit Delete	2	Develop API	Not Started	2
Edit Delete	3	Create Database Schema	Completed	2
Edit Delete	4	Integrate Unit Test	In progress	3
Edit Delete	5	Conduct Integration Tests	In Progress	3
Edit Delete	6	Prepare Deployment Plan	Not Started	4
Edit Delete	7	Gather User Feedback	Not Started	4
Edit Delete	8	Finalize Documentation	In Progress	5
Edit Delete	9	Conduct User Testing	Not Started	6

Figure 67 Existing SubTask Details Updated

TASK DETAILS TASK ID:				
Insert Cancel				
	SUBTASK_ID	SUBTASK_NAME	SUBTASK_STATUS	TASK_DETAILS_TASK_ID
Edit Delete	21	AI research	Completed	1
Edit Delete	1	Design UI	In Progress	1
Edit Delete	2	Develop API	Not Started	2
Edit Delete	3	Create Database Schema	Completed	2
Edit Delete	4	Integrate Unit Test	In progress	3
Edit Delete	5	Conduct Integration Tests	In Progress	3
Edit Delete	6	Prepare Deployment Plan	Not Started	4
Edit Delete	7	Gather User Feedback	Not Started	4
Edit Delete	8	Finalize Documentation	In Progress	5
Edit Delete	9	Conduct User Training	Not Started	6
Edit Delete	10	Create Marketing Materials	Not Started	7
Edit Delete	11	Set Up Monitoring	In Progress	8

Figure 68 Deleting the SubTask_ID 10

	SUBTASK_ID	SUBTASK_NAME	SUBTASK_STATUS	TASK_DETAILS_TASK_ID
Edit Delete	21	AI research	Completed	1
Edit Delete	1	Design UI	In Progress	1
Edit Delete	2	Develop API	Not Started	2
Edit Delete	3	Create Database Schema	Completed	2
Edit Delete	4	Integrate Unit Test	In progress	3
Edit Delete	5	Conduct Integration Tests	In Progress	3
Edit Delete	6	Prepare Deployment Plan	Not Started	4
Edit Delete	7	Gather User Feedback	Not Started	4
Edit Delete	8	Finalize Documentation	In Progress	5
Edit Delete	9	Conduct User Training	Not Started	6
Edit Delete	11	Set Up Monitoring	In Progress	8
Edit Delete	12	Post-Launch Review Meeting	Not Started	9
Edit Delete	13	Follow Up on Feedback	Not Started	10
Edit Delete	14	Plan Next Phase	Not Started	11
Edit Delete	15	Conduct Stakeholder Meeting	In Progress	12
Edit Delete	16	Review Project Outcomes	Not Started	13
Edit Delete	17	Identify Improvements	Not Started	14
Edit Delete	18	Archive Project Documents	Not Started	15
Edit Delete	19	Set Up Post-Mortem Analysis	Not Started	16
Edit Delete	20	Prepare Final Report	Not Started	17

Figure 69 Final SubTask Details table after CRUD

9.5 Milestone Details

Table 12 Milestone Details Test Table

Test	Create milestones while updating and deleting the existing milestone data from the table.
Action	Navigate to the milestone management form, fill out the required fields for a new milestone data, update an existing milestone by clicking edit button in the milestone table, and delete an existing milestone from the table.
Expected Result	A new milestone is created successfully, and the existing milestone data is updated along with the deletion of existing milestone details of a project.
Actual Result	Successfully created a new milestone, edited existing milestone data and deleted the milestone of a project.
Conclusion	Milestone details CRUD operation test is successfully performed.

9.5.1 Evidence of Milestone Details

	<u>MILESTONE_ID</u>	<u>MILESTONE_NAME</u>	<u>MILESTONE_PERIOD</u>	<u>PROJECT_PROJECT_ID</u>
Edit Delete	16	Testing	1/19/2025 12:30:00 AM	1
Edit Delete	1	Initiation	1/15/2025 12:00:00 AM	1
Edit Delete	2	Planning	2/15/2025 12:00:00 AM	1
Edit Delete	3	Execution	3/15/2025 12:00:00 AM	1
Edit Delete	4	Monitoring	4/15/2025 12:00:00 AM	1
Edit Delete	5	Closure	5/15/2025 12:00:00 AM	1
Edit Delete	6	Requirements Gathering	1/20/2025 12:00:00 AM	2
Edit Delete	7	Design Phase	2/20/2025 12:00:00 AM	2
Edit Delete	9	Testing Phase	4/20/2025 12:00:00 AM	2
Edit Delete	10	Deployment	5/20/2025 12:00:00 AM	2
Edit Delete	11	Requirements Review	1/25/2025 12:00:00 AM	3
Edit Delete	12	Prototype	2/25/2025 12:00:00 AM	3
Edit Delete	13	Final Review	3/25/2025 12:00:00 AM	3
Edit Delete	14	User Acceptance Testing	4/25/2025 12:00:00 AM	3
Edit Delete	15	Go Live	5/25/2025 12:00:00 AM	3

Figure 70 Initial Milestone details table

MILESTONE_ID:

MILESTONE_NAME:

MILESTONE_PERIOD:

PROJECT_PROJECT_ID:

[Insert](#) [Cancel](#)

<u>MILESTONE_ID</u>	<u>MILESTONE_NAME</u>	<u>MILE</u>
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Figure 71 Inserting Milestone Details data in form

	MILESTONE_ID	MILESTONE_NAME	MILESTONE_PERIOD	PROJECT_PROJECT_ID
Edit Delete	17	Meeting	3/15/2025 12:20:00 PM	1
Edit Delete	16	Testing	1/19/2025 12:30:00 AM	1
Edit Delete	1	Initiation	1/15/2025 12:00:00 AM	1
Edit Delete	2	Planning	2/15/2025 12:00:00 AM	1
Edit Delete	3	Execution	3/15/2025 12:00:00 AM	1
Edit Delete	4	Monitoring	4/15/2025 12:00:00 AM	1
Edit Delete	5	Closure	5/15/2025 12:00:00 AM	1
Edit Delete	6	Requirements Gathering	1/20/2025 12:00:00 AM	2
Edit Delete	7	Design Phase	2/20/2025 12:00:00 AM	2
Edit Delete	9	Testing Phase	4/20/2025 12:00:00 AM	2
Edit Delete	10	Deployment	5/20/2025 12:00:00 AM	2
Edit Delete	11	Requirements Review	1/25/2025 12:00:00 AM	3
Edit Delete	12	Prototype	2/25/2025 12:00:00 AM	3
Edit Delete	13	Final Review	3/25/2025 12:00:00 AM	3
Edit Delete	14	User Acceptance Testing	4/25/2025 12:00:00 AM	3
Edit Delete	15	Go Live	5/25/2025 12:00:00 AM	3

Figure 72 New Milestone Details Inserted

Insert Cancel				
	MILESTONE_ID	MILESTONE_NAME	MILESTONE_PERIOD	PROJECT_PROJECT_ID
Edit Delete	17	Meeting	3/15/2025 12:20:00 PM	1
Edit Delete	16	Testing	1/19/2025 12:30:00 AM	1
Edit Delete	1	Initiation	1/15/2025 12:00:00 AM	1
Edit Delete	2	Planning	2/15/2025 12:00:00 AM	1
Edit Delete	3	Execution	3/15/2025 12:00:00 AM	1
Edit Delete	4	Monitoring	4/15/2025 12:00:00 AM	1
Edit Delete	5	Closure	5/15/2025 12:00:00 AM	1
Edit Delete	6	Requirements Gathering	1/20/2025 12:00:00 AM	2
Edit Delete	7	Design Phase	2/20/2025 12:00:00 AM	2
Edit Delete	9	Testing Phase	4/20/2025 12:00:00 AM	2
Update Cancel	10	Final Deployment	7/27/2025 04:30:00 PM	2
Edit Delete	11	Requirements Review	1/25/2025 12:00:00 AM	3
Edit Delete	12	Prototype	2/25/2025 12:00:00 AM	3
Edit Delete	13	Final Review	3/25/2025 12:00:00 AM	3
Edit Delete	14	User Acceptance Testing	4/25/2025 12:00:00 AM	3
Edit Delete	15	Go Live	5/25/2025 12:00:00 AM	3

Figure 73 Updating Existing Milestone Data

	MILESTONE_ID	MILESTONE_NAME	MILESTONE_PERIOD	PROJECT_PROJECT_ID
Edit Delete	17	Meeting	3/15/2025 12:20:00 PM	1
Edit Delete	16	Testing	1/19/2025 12:30:00 AM	1
Edit Delete	1	Initiation	1/15/2025 12:00:00 AM	1
Edit Delete	2	Planning	2/15/2025 12:00:00 AM	1
Edit Delete	3	Execution	3/15/2025 12:00:00 AM	1
Edit Delete	4	Monitoring	4/15/2025 12:00:00 AM	1
Edit Delete	5	Closure	5/15/2025 12:00:00 AM	1
Edit Delete	6	Requirements Gathering	1/20/2025 12:00:00 AM	2
Edit Delete	7	Design Phase	2/20/2025 12:00:00 AM	2
Edit Delete	9	Testing Phase	4/20/2025 12:00:00 AM	2
Edit Delete	10	Final Deployment	7/27/2025 4:30:00 PM	2
Edit Delete	11	Requirements Review	1/25/2025 12:00:00 AM	3
Edit Delete	12	Prototype	2/25/2025 12:00:00 AM	3
Edit Delete	13	Final Review	3/25/2025 12:00:00 AM	3
Edit Delete	14	User Acceptance Testing	4/25/2025 12:00:00 AM	3
Edit Delete	15	Go Live	5/25/2025 12:00:00 AM	3

Figure 74 Existing Milestone Details Updated

	MILESTONE_ID	MILESTONE_NAME	MILESTONE_PERIOD	PROJECT_PROJECT_ID
Edit Delete	17	Meeting	3/15/2025 12:20:00 PM	1
Edit Delete	16	Testing	1/19/2025 12:30:00 AM	1
Edit Delete	1	Initiation	1/15/2025 12:00:00 AM	1
Edit Delete	2	Planning	2/15/2025 12:00:00 AM	1
Edit Delete	3	Execution	3/15/2025 12:00:00 AM	1
Edit Delete	4	Monitoring	4/15/2025 12:00:00 AM	1
Edit Delete	5	Closure	5/15/2025 12:00:00 AM	1
Edit Delete	6	Requirements Gathering	1/20/2025 12:00:00 AM	2
Edit Delete	7	Design Phase	2/20/2025 12:00:00 AM	2

Figure 75 Deleting the Milestone_ID 5

	MILESTONE_ID	MILESTONE_NAME	MILESTONE_PERIOD	PROJECT_PROJECT_ID
Edit Delete	17	Meeting	3/15/2025 12:20:00 PM	1
Edit Delete	16	Testing	1/19/2025 12:30:00 AM	1
Edit Delete	1	Initiation	1/15/2025 12:00:00 AM	1
Edit Delete	2	Planning	2/15/2025 12:00:00 AM	1
Edit Delete	3	Execution	3/15/2025 12:00:00 AM	1
Edit Delete	4	Monitoring	4/15/2025 12:00:00 AM	1
Edit Delete	6	Requirements Gathering	1/20/2025 12:00:00 AM	2
Edit Delete	7	Design Phase	2/20/2025 12:00:00 AM	2
Edit Delete	9	Testing Phase	4/20/2025 12:00:00 AM	2
Edit Delete	10	Final Deployment	7/27/2025 4:30:00 PM	2
Edit Delete	11	Requirements Review	1/25/2025 12:00:00 AM	3
Edit Delete	12	Prototype	2/25/2025 12:00:00 AM	3
Edit Delete	13	Final Review	3/25/2025 12:00:00 AM	3
Edit Delete	14	User Acceptance Testing	4/25/2025 12:00:00 AM	3
Edit Delete	15	Go Live	5/25/2025 12:00:00 AM	3

Figure 76 Final Milestone Details table after CRUD

9.6 User Project Details

Table 13 User Project Details Test Table

Test	Display all the user details with the projects joined based on the selected username.
Action	Filter the project details along with a list of project details by choosing a specific user from the dropdown list.
Expected Result	It should display all user and project details associated with the selected user.
Actual Result	Successfully displayed user details along with their respective project details.

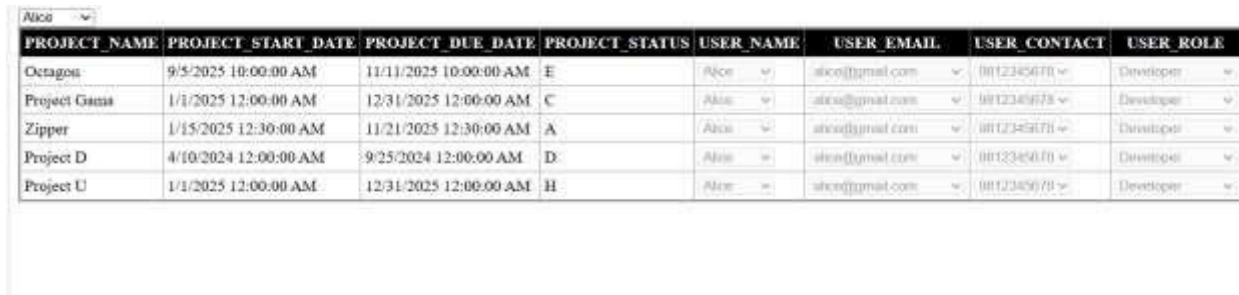
Conclusion	User Project details test is successfully performed showcasing filtered results as expected.
------------	--

9.6.1 Evidence of User Project Details



NAME	PROJECT_START_DATE	PROJECT_DUE_DATE	PROJECT_STATUS	USER_NAME	USER_EMAIL	USER_CONTACT	USER_ROLE
Alice	1/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	A	Charlie	charlie@gmail.com	0012345678	Project Manager

Figure 77 Filtering Username with dropdown option



PROJECT_NAME	PROJECT_START_DATE	PROJECT_DUE_DATE	PROJECT_STATUS	USER_NAME	USER_EMAIL	USER_CONTACT	USER_ROLE
Octagon	9/5/2025 10:00:00 AM	11/11/2025 10:00:00 AM	E	Alice	alice@gmail.com	0012345678	Developer
Project Gauss	1/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	C	Alice	alice@gmail.com	0012345678	Developer
Zipper	1/15/2025 12:30:00 AM	11/21/2025 12:30:00 AM	A	Alice	alice@gmail.com	0012345678	Developer
Project D	4/10/2024 12:00:00 AM	9/25/2024 12:00:00 AM	D	Alice	alice@gmail.com	0012345678	Developer
Project U	1/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	H	Alice	alice@gmail.com	0012345678	Developer

Figure 78 User Project details of Username Alice

Alice

Ningwa

Alice

Bibisha

Charlie

Sugam

Sujan

Anjita

Bob

Ram

Dane

Neha

Numa

Krish

Evan

Asha

Biju

Aknti

Deepson

Srujanika

Aneha

PROJECT_NAME	PROJECT_START_DATE	PROJECT_DUE_DATE	PROJECT_STATUS	USER_NAME	USER_EMAIL	USER_CONTACT	USER_ROLE
	9/5/2025 10:00:00 AM	11/11/2025 10:00:00 AM	E	Alice	alice@gmail.com	0012345678	Developer
ma	1/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	C	Alice	alice@gmail.com	0012345678	Developer
	1/15/2025 12:30:00 AM	11/21/2025 12:30:00 AM	A	Alice	alice@gmail.com	0012345678	Developer
	4/10/2024 12:00:00 AM	9/25/2024 12:00:00 AM	D	Alice	alice@gmail.com	0012345678	Developer
	1/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	H	Alice	alice@gmail.com	0012345678	Developer

Figure 79 Again Filtering Username with dropdown option

Bibisha

PROJECT_NAME	PROJECT_START_DATE	PROJECT_DUE_DATE	PROJECT_STATUS	USER_NAME	USER_EMAIL	USER_CONTACT	USER_ROLE
Betsa	1/10/2025 12:00:00 AM	12/11/2025 12:00:00 AM	A	Bibisha	bibisha12@gmail.com	0022344070	Tester
Project N	2/15/2026 12:00:00 AM	5/10/2026 12:00:00 AM	N	Bibisha	bibisha12@gmail.com	0022344070	Tester

Figure 80 User Project details of Username Bibisha

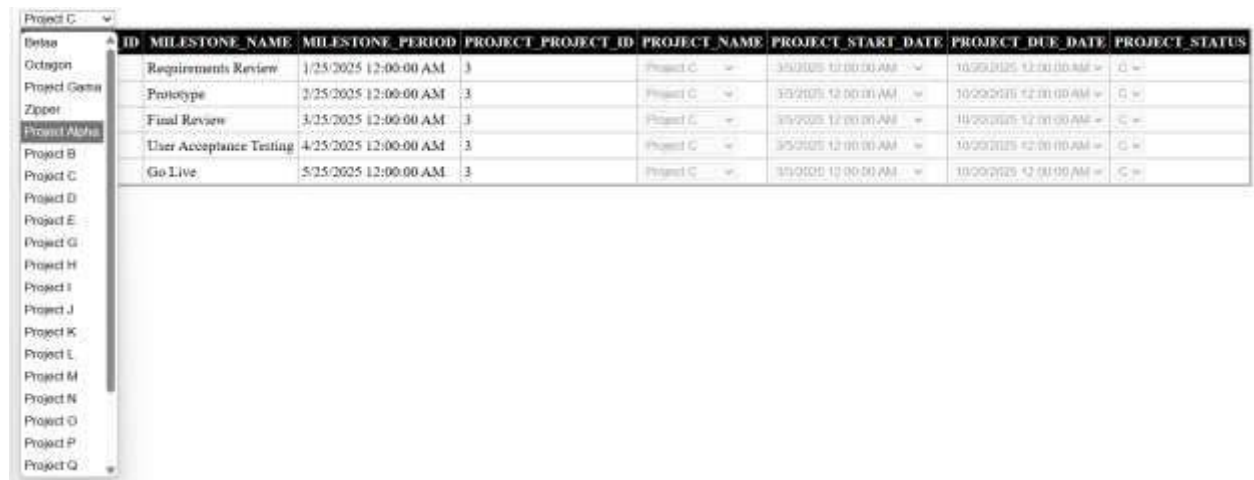
9.7 Project Milestone Details

Table 14 Project Milestone Details Test Table

Test	Display the project details with all their milestones based on the selected project.
Action	Filter project details and all milestones related to a specific project from the dropdown list.

Expected Result	It should display all project and their milestone details associated with the selected project name.
Actual Result	Successfully displayed project details along with their respective milestone details.
Conclusion	Project Milestone details test is successfully performed showcasing filtered results as expected.

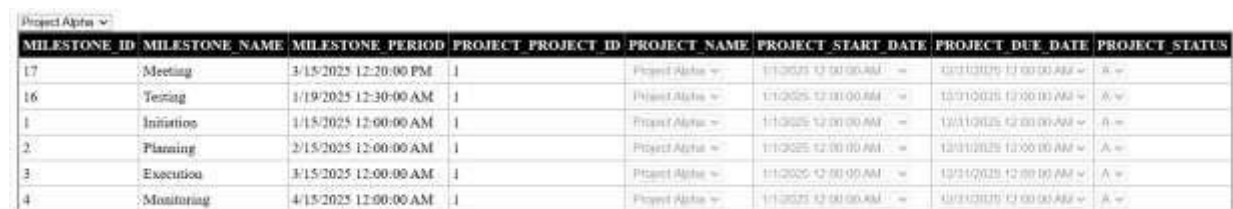
9.7.1 Evidence of Project Milestone Details



The screenshot shows a web application interface. On the left, there is a vertical list of project names from Project A to Project Q. A dropdown menu is open, showing 'Project C' as the selected option. To the right of the dropdown is a table displaying project milestone details for Project C.

ID	MILESTONE_NAME	MILESTONE_PERIOD	PROJECT_PROJECT_ID	PROJECT_NAME	PROJECT_START_DATE	PROJECT_DUE_DATE	PROJECT_STATUS
	Requirements Review	1/25/2025 12:00:00 AM	3	Project C	3/5/2025 12:00:00 AM	10/25/2025 12:00:00 AM	C
	Prototype	2/25/2025 12:00:00 AM	3	Project C	3/5/2025 12:00:00 AM	10/25/2025 12:00:00 AM	C
	Final Review	3/25/2025 12:00:00 AM	3	Project C	3/5/2025 12:00:00 AM	10/25/2025 12:00:00 AM	C
	User Acceptance Testing	4/25/2025 12:00:00 AM	3	Project C	3/5/2025 12:00:00 AM	10/25/2025 12:00:00 AM	C
	Go Live	5/25/2025 12:00:00 AM	3	Project C	3/5/2025 12:00:00 AM	10/25/2025 12:00:00 AM	C

Figure 81 Filtering Project with dropdown option



The screenshot shows a web application interface. On the left, there is a vertical list of project names from Project A to Project Q. A dropdown menu is open, showing 'Project Alpha' as the selected option. To the right of the dropdown is a table displaying project milestone details for Project Alpha.

MILESTONE_ID	MILESTONE_NAME	MILESTONE_PERIOD	PROJECT_PROJECT_ID	PROJECT_NAME	PROJECT_START_DATE	PROJECT_DUE_DATE	PROJECT_STATUS
17	Meeting	3/15/2025 12:20:00 PM	1	Project Alpha	3/15/2025 12:00:00 AM	12/31/2025 12:00:00 AM	A
16	Testing	1/19/2025 12:30:00 AM	1	Project Alpha	1/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	A
1	Initiation	1/15/2025 12:00:00 AM	1	Project Alpha	1/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	A
2	Planning	2/15/2025 12:00:00 AM	1	Project Alpha	1/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	A
3	Execution	3/15/2025 12:00:00 AM	1	Project Alpha	1/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	A
4	Monitoring	4/15/2025 12:00:00 AM	1	Project Alpha	1/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	A

Figure 82 Project Milestone details of Project name Alpha

Project Alpha								
Belaa								
Oclagon								
Project Gamma								
Zppor								
Project Alpha								
Project B								
Project C								
Project D								
Project E								
Project G								
Project H								
Project I								
Project J								
Project K								
Project L								
Project M								
Project N								
Project O								
Project P								
Project Q								

Figure 83 Again Filtering Project with dropdown option

Project B								
MILESTONE_ID	MILESTONE_NAME	MILESTONE_PERIOD	PROJECT_PROJECT_ID	PROJECT_NAME	PROJECT_START_DATE	PROJECT_DUE_DATE	PROJECT_STATUS	
6	Requirements Gathering	1/29/2025 12:00:00 AM	2	Project B	1/29/2025 12:00:00 AM	11/14/2025 12:00:00 AM	B	
7	Design Phase	2/26/2025 12:00:00 AM	2	Project B	2/26/2025 12:00:00 AM	11/16/2025 12:00:00 AM	B	
9	Testing Phase	4/29/2025 12:00:00 AM	2	Project B	4/29/2025 12:00:00 AM	11/18/2025 12:00:00 AM	B	
10	Final Deployment	7/27/2025 4:30:00 PM	2	Project B	7/27/2025 12:00:00 AM	11/19/2025 12:00:00 AM	B	

Figure 84 Project Milestone details of Project name B

9.8 Top Performer Details

Table 15 Test Table of Top Performer

Test	Display the details of top 3 users who have completed the most tasks for a project.
Action	Filter top 3 performer details from each project through the dropdown list based on task completion.

Expected Result	It should display details of the top 3 users who have completed the most tasks for the selected project.
Actual Result	Successfully displayed top performer details.
Conclusion	Top Performer test is successfully performed showcasing filtered results as expected.

9.8.1 Evidence of Top Performer

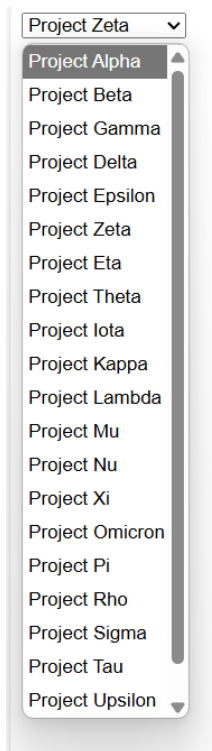
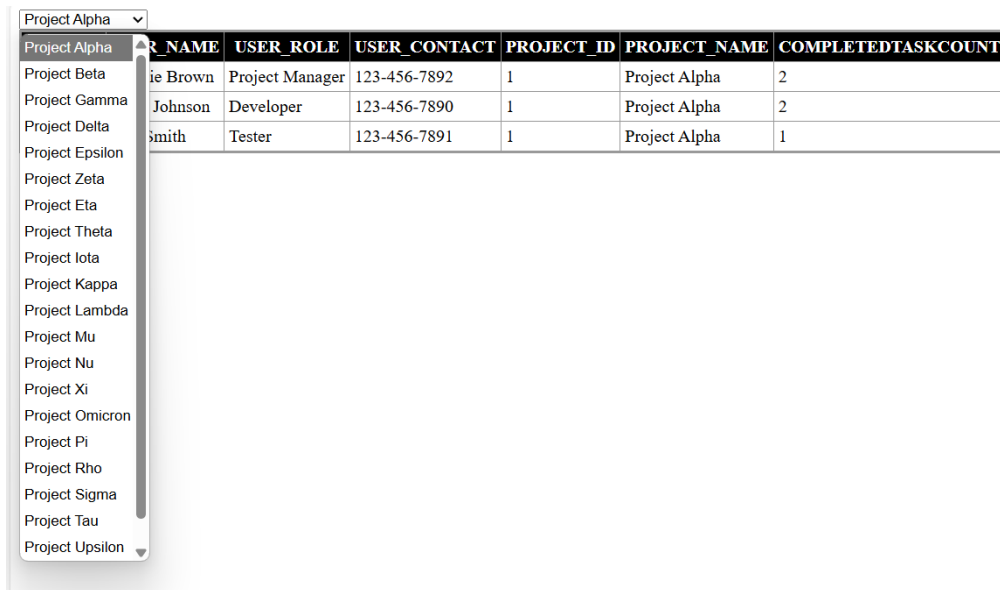


Figure 85 Filtering Top Performer with dropdown option



The screenshot shows a web application interface. On the left, there is a dropdown menu with 'Project Alpha' selected. Below the dropdown is a table with the following data:

USER_NAME	USER_ROLE	USER_CONTACT	PROJECT_ID	PROJECT_NAME	COMPLETEDTASKCOUNT
ie Brown	Project Manager	123-456-7892	1	Project Alpha	2
Johnson	Developer	123-456-7890	1	Project Alpha	2
smith	Tester	123-456-7891	1	Project Alpha	1

Figure 86 Top Performer details of Project name B

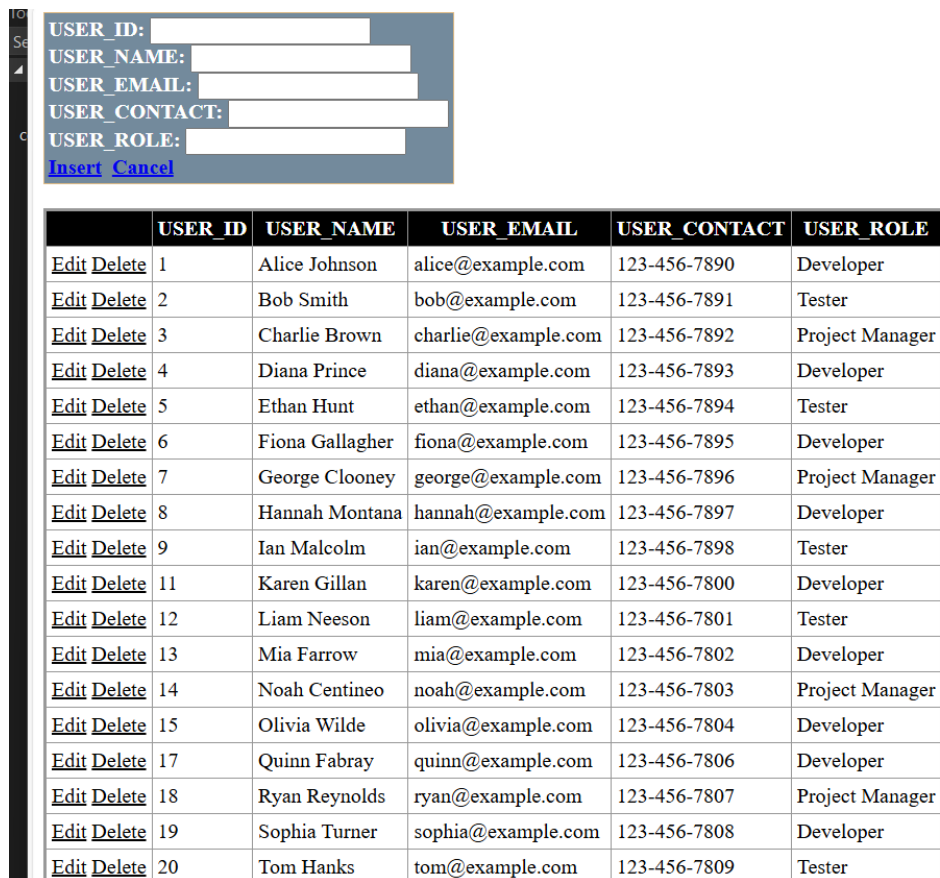
9.9 Input Validation Failures and Data Types

Table 16 Test table of Input Validation failures and data types

Test	Forms are submitted without entry data and incorrect data types are inserted to check for validation message.
Action	All required fields are left empty, and data are inserted with incorrect data types in form.
Expected Result	Appropriate validation messages indicating required fields and invalid input are displayed.
Actual Result	Error message indication required fields and invalid data type input is displayed.

Conclusion	Validation for empty fields and error handling for invalid data types are functioning as expected.
------------	--

9.9.1 Evidence of Input Validation Failures and Data Types



The screenshot displays a user registration form with the following fields: USER_ID, USER_NAME, USER_EMAIL, USER_CONTACT, and USER_ROLE. Each field is currently empty. Below the form are 'Insert' and 'Cancel' buttons. To the right of the form is a table listing existing users.

	USER_ID	USER_NAME	USER_EMAIL	USER_CONTACT	USER_ROLE
Edit Delete	1	Alice Johnson	alice@example.com	123-456-7890	Developer
Edit Delete	2	Bob Smith	bob@example.com	123-456-7891	Tester
Edit Delete	3	Charlie Brown	charlie@example.com	123-456-7892	Project Manager
Edit Delete	4	Diana Prince	diana@example.com	123-456-7893	Developer
Edit Delete	5	Ethan Hunt	ethan@example.com	123-456-7894	Tester
Edit Delete	6	Fiona Gallagher	fiona@example.com	123-456-7895	Developer
Edit Delete	7	George Clooney	george@example.com	123-456-7896	Project Manager
Edit Delete	8	Hannah Montana	hannah@example.com	123-456-7897	Developer
Edit Delete	9	Ian Malcolm	ian@example.com	123-456-7898	Tester
Edit Delete	11	Karen Gillan	karen@example.com	123-456-7800	Developer
Edit Delete	12	Liam Neeson	liam@example.com	123-456-7801	Tester
Edit Delete	13	Mia Farrow	mia@example.com	123-456-7802	Developer
Edit Delete	14	Noah Centineo	noah@example.com	123-456-7803	Project Manager
Edit Delete	15	Olivia Wilde	olivia@example.com	123-456-7804	Developer
Edit Delete	17	Quinn Fabray	quinn@example.com	123-456-7806	Developer
Edit Delete	18	Ryan Reynolds	ryan@example.com	123-456-7807	Project Manager
Edit Delete	19	Sophia Turner	sophia@example.com	123-456-7808	Developer
Edit Delete	20	Tom Hanks	tom@example.com	123-456-7809	Tester

Figure 87 Inserting the form with empty fields

Server Error in '/' Application.

ORA-01400: cannot insert NULL into ("LSCOR"."User"."USER_ID")

Description: An unhandled exception occurred during the execution of the current web request. Please review the stack trace for more information about the error and where it originated in the code.

Exception Details: System.Data.OracleClient.OracleException: ORA-01400: cannot insert NULL into ("LSCOR"."User"."USER_ID")

Source Error:

An unhandled exception was generated during the execution of the current web request. Information regarding the origin and location of the exception can be identified using the exception stack trace below.

Stack Trace:

```
[OracleException (0x80131938): ORA-01400: cannot insert NULL into ("LSCOR"."User"."USER_ID")
]
System.Data.OracleClient.OracleConnection.CheckError(OciErrorHandle errorHandle, Int32 rc) +404
System.Data.OracleClient.OracleCommand.Execute(OciStatementHandle statementHandle, CommandBehavior behavior, Boolean needRowid, OciRowidDescriptor& rowidDescriptor, ArrayList& resultP
System.Data.OracleClient.OracleCommand.ExecuteNonQueryInternal(Boolean needRowid, OciRowidDescriptor& rowidDescriptor) +670
System.Data.OracleClient.OracleCommand.ExecuteNonQuery() +146
System.Web.UI.WebControls.SqlDataSourceView.ExecuteDbCommand(DbCommand command, DataSourceOperation operation) +582
System.Web.UI.DataSourceView.Insert(IDictionary values, DataSourceViewOperationCallback callback) +167
System.Web.UI.WebControls.FormView.HandleInsert(String commandArg, Boolean causesValidation) +497
System.Web.UI.WebControls.FormView.HandleEvent(EventArgs e, Boolean causesValidation, String validationGroup) +706
System.Web.UI.Control.RaiseBubbleEvent(Object source, EventArgs args) +50
System.Web.UI.WebControls.FormViewRow.OnBubbleEvent(Object source, EventArgs e) +132
System.Web.UI.Control.RaiseBubbleEvent(Object source, EventArgs args) +50
System.Web.UI.Page.ProcessRequestMain(Boolean includeStagesBeforeAsyncPoint, Boolean includeStagesAfterAsyncPoint) +1959
```

Version Information: Microsoft .NET Framework Version 4.0.30319; ASP.NET Version 4.8.9277.0

Figure 88 Error message indicating null inserts

USER_ID:

USER_NAME:

USER_EMAIL:

USER_CONTACT:

USER_ROLE:

[Insert](#) [Cancel](#)

	USER_ID	USER_NAME	USER_EMAIL	USER_CONTACT	USER_ROLE
Edit Delete	1	Alice Johnson	alice@example.com	123-456-7890	Developer
Edit Delete	2	Bob Smith	bob@example.com	123-456-7891	Tester
Edit Delete	3	Charlie Brown	charlie@example.com	123-456-7892	Project Manager
Edit Delete	4	Diana Prince	diana@example.com	123-456-7893	Developer
Edit Delete	5	Ethan Hunt	ethan@example.com	123-456-7894	Tester
Edit Delete	6	Fiona Gallagher	fiona@example.com	123-456-7895	Developer
Edit Delete	7	George Clooney	george@example.com	123-456-7896	Project Manager
Edit Delete	8	Hannah Montana	hannah@example.com	123-456-7897	Developer

Figure 89 Inserting with mismatch data type in form

Server Error in '/' Application.*Input string was not in a correct format.***Description:** An unhandled exception occurred during the execution of the current web request. Please review the stack trace for more information about the error and where it originated in the code.**Exception Details:** System.FormatException: Input string was not in a correct format.**Source Error:**

An unhandled exception was generated during the execution of the current web request. Information regarding the origin and location of the exception can be identified using the exception stack trace below.

Stack Trace:

```
[FormatException: Input string was not in a correct format.]
  System.Number.StringToNumber(String str, NumberStyles options, NumberBuffer& number, NumberFormatInfo info, Boolean parseDecimal) +14608746
  System.Number.ParseDecimal(String value, NumberStyles options, NumberFormatInfo numfmt) +192
  System.Convert.ToDecimal(String value, IFormatProvider provider) +108
  System.Convert.ChangeType(Object value, TypeCode typeCode, IFormatProvider provider) +694
  System.Web.UI.WebControls.Parameter.GetValue(Object value, String defaultValue, TypeCode type, Boolean convertEmptyStringToNull, Boolean ignoreNullableTypeChanges) +151
  System.Web.UI.WebControls.SqlDataSourceView.AddParameters(DbCommand command, ParameterCollection reference, IDictionary parameters, IDictionary exclusionList, String oldValuesParameterNameKey) +94
  System.Web.UI.WebControls.SqlDataSourceView.ExecuteInsert(IDictionary values) +368
  System.Web.UI.DataSourceView.Insert(IDictionary values, DataSourceViewOperationCallback callback) +167
  System.Web.UI.WebControls.FormView.HandleInsert(String commandArg, Boolean causesValidation) +497
  System.Web.UI.WebControls.FormView.HandleEvent(EventArgs e, Boolean causesValidation, String validationGroup) +706
  System.Web.UI.Control.RaiseBubbleEvent(Object source, EventArgs args) +50
  System.Web.UI.WebControls.FormViewRow.OnBubbleEvent(Object source, EventArgs e) +132
  System.Web.UI.Control.RaiseBubbleEvent(Object source, EventArgs args) +50
  System.Web.UI.Page.ProcessRequestMain(Boolean includeStagesBeforeAsyncPoint, Boolean includeStagesAfterAsyncPoint) +1959
```

Version Information: Microsoft .NET Framework Version 4.0.30319; ASP.NET Version 4.8.9277.0*Figure 90 Error indicating data type incorrect format*

USER_ID:
USER_NAME:
USER_EMAIL:
USER_CONTACT:
USER_ROLE:
[Insert](#) [Cancel](#)

	USER_ID	USER_NAME	USER_EMAIL	USER_CONTACT	USER_ROLE
Edit Delete	1	Alice Johnson	alice@example.com	123-456-7890	Developer
Edit Delete	2	Bob Smith	bob@example.com	123-456-7891	Tester
Edit Delete	3	Charlie Brown	charlie@example.com	123-456-7892	Project Manager
Edit Delete	4	Diana Prince	diana@example.com	123-456-7893	Developer

Figure 91 Inserting all the form field with correct data types

USER_EMAIL: alice@example.com USER_CONTACT: 123-456-7890 USER_ROLE: Developer Edit Delete New					
	USER_ID	USER_NAME	USER_EMAIL	USER_CONTACT	USER_ROLE
Edit Delete	1	Alice Johnson	alice@example.com	123-456-7890	Developer
Edit Delete	2	Bob Smith	bob@example.com	123-456-7891	Tester
Edit Delete	3	Charlie Brown	charlie@example.com	123-456-7892	Project Manager
Edit Delete	4	Diana Prince	diana@example.com	123-456-7893	Developer
Edit Delete	5	Ethan Hunt	ethan@example.com	123-456-7894	Tester
Edit Delete	6	Fiona Gallagher	fiona@example.com	123-456-7895	Developer
Edit Delete	7	George Clooney	george@example.com	123-456-7896	Project Manager
Edit Delete	8	Hannah Montana	hannah@example.com	123-456-7897	Developer
Edit Delete	9	Ian Malcolm	ian@example.com	123-456-7898	Tester
Edit Delete	11	Karen Gillan	karen@example.com	123-456-7800	Developer
Edit Delete	12	Liam Neeson	liam@example.com	123-456-7801	Tester
Edit Delete	13	Mia Farrow	mia@example.com	123-456-7802	Developer
Edit Delete	14	Noah Centineo	noah@example.com	123-456-7803	Project Manager
Edit Delete	15	Olivia Wilde	olivia@example.com	123-456-7804	Developer
Edit Delete	17	Quinn Fabray	quinn@example.com	123-456-7806	Developer
Edit Delete	18	Ryan Reynolds	ryan@example.com	123-456-7807	Project Manager
Edit Delete	19	Sophia Turner	sophia@example.com	123-456-7808	Developer
Edit Delete	20	Tom Hanks	tom@example.com	123-456-7809	Tester
Edit Delete	21	Neww	ningwa12@gmail.com	9865432123	Manager
Edit Delete	22	Json	json@gmail.com	9898076543	Tester

Figure 92 Successfully inserted the data

9.10 Duplicate Entry Validation

Table 17 Test table for duplicate entry validation

Test	Duplicate records are being attempted where unique constraints apply.
Action	Duplicate data are inserted in the form and submitted.
Expected Result	An error message indicating uniqueness requirements is displayed.

Actual Result	An error message indicating requirement of unique constraints is displayed.
Conclusion	Duplicate entry validation is properly handled by the application.

9.10.1 Evidence of Duplicate Entry Validation

TASK_ID: 1 TASK_NAME: Task 1 START_DATE: 1/1/2023 12:00:00 AM DUE_DATE: 2/1/2023 12:00:00 AM STATUS: In Progress Edit Delete New					
	TASK_ID	TASK_NAME	START_DATE	DUE_DATE	STATUS
Edit Delete	1	Task 1	1/1/2023 12:00:00 AM	2/1/2023 12:00:00 AM	In Progress
Edit Delete	2	Task 2	1/15/2023 12:00:00 AM	2/15/2023 12:00:00 AM	Completed
Edit Delete	3	Task 3	2/1/2023 12:00:00 AM	3/1/2023 12:00:00 AM	Not Started
Edit Delete	4	Task 4	2/15/2023 12:00:00 AM	3/15/2023 12:00:00 AM	In Progress
Edit Delete	5	Task 5	3/1/2023 12:00:00 AM	4/1/2023 12:00:00 AM	Completed
Edit Delete	6	Task 6	3/15/2023 12:00:00 AM	4/15/2023 12:00:00 AM	Not Started
Edit Delete	7	Task 7	4/1/2023 12:00:00 AM	5/1/2023 12:00:00 AM	In Progress
Edit Delete	8	Task 8	4/15/2023 12:00:00 AM	5/15/2023 12:00:00 AM	Completed
Edit Delete	9	Task 9	5/1/2023 12:00:00 AM	6/1/2023 12:00:00 AM	Not Started
Edit Delete	10	Task 10	5/15/2023 12:00:00 AM	6/15/2023 12:00:00 AM	In Progress
Edit Delete	11	Task 11	1/1/2023 12:00:00 AM	2/1/2023 12:00:00 AM	Completed
Edit Delete	12	Task 12	1/15/2023 12:00:00 AM	2/15/2023 12:00:00 AM	In Progress
Edit Delete	13	Task 13	2/1/2023 12:00:00 AM	3/1/2023 12:00:00 AM	Not Started
Edit Delete	14	Task 14	2/15/2023 12:00:00 AM	3/15/2023 12:00:00 AM	Completed
Edit Delete	15	Task 15	3/1/2023 12:00:00 AM	4/1/2023 12:00:00 AM	In Progress
Edit Delete	16	Task 16	3/15/2023 12:00:00 AM	4/15/2023 12:00:00 AM	Completed
Edit Delete	17	Task 17	4/1/2023 12:00:00 AM	5/1/2023 12:00:00 AM	Not Started
Edit Delete	18	Task 18	4/15/2023 12:00:00 AM	5/15/2023 12:00:00 AM	In Progress
Edit Delete	19	Task 19	5/1/2023 12:00:00 AM	6/1/2023 12:00:00 AM	Completed
Edit Delete	20	Task 20	5/15/2023 12:00:00 AM	6/15/2023 12:00:00 AM	Not Started

Figure 93 Data table before duplicate validation

TASK_ID:
 TASK_NAME:
 START_DATE:
 DUE_DATE:
 STATUS:
[Insert](#) [Cancel](#)

	TASK_ID	TASK_NAME	START_DATE	DUE_DATE	STATUS
Edit Delete	1	Task 1	1/1/2023 12:00:00 AM	2/1/2023 12:00:00 AM	In Progress
Edit Delete	2	Task 2	1/15/2023 12:00:00 AM	2/15/2023 12:00:00 AM	Completed
Edit Delete	3	Task 3	2/1/2023 12:00:00 AM	3/1/2023 12:00:00 AM	Not Started
Edit Delete	4	Task 4	2/15/2023 12:00:00 AM	3/15/2023 12:00:00 AM	In Progress
Edit Delete	5	Task 5	3/1/2023 12:00:00 AM	4/1/2023 12:00:00 AM	Completed
Edit Delete	6	Task 6	3/15/2023 12:00:00 AM	4/15/2023 12:00:00 AM	Not Started
Edit Delete	7	Task 7	4/1/2023 12:00:00 AM	5/1/2023 12:00:00 AM	In Progress
Edit Delete	8	Task 8	4/15/2023 12:00:00 AM	5/15/2023 12:00:00 AM	Completed
Edit Delete	9	Task 9	5/1/2023 12:00:00 AM	6/1/2023 12:00:00 AM	Not Started

Figure 94 Inserting duplicate unique constraints in form

Server Error in '/' Application.

ORA-00001: unique constraint (LSCOR.TASK_DETAILS_PK) violated

Description: An unhandled exception occurred during the execution of the current web request. Please review the stack trace for more information about the error and where it originated in the code.

Exception Details: System.Data.OracleClient.OracleException: ORA-00001: unique constraint (LSCOR.TASK_DETAILS_PK) violated

Source Error:

An unhandled exception was generated during the execution of the current web request. Information regarding the origin and location of the exception can be identified using the exception stack trace below.

Stack Trace:

```
[OracleException (0x80131938): ORA-00001: unique constraint (LSCOR.TASK_DETAILS_PK) violated]
System.Data.OracleClient.OracleConnection.CheckError(OciErrorHandle errorHandle, Int32 rc) +404
System.Data.OracleClient.OracleCommand.Execute(OciStatementHandle statementHandle, CommandBehavior behavior, Boolean needRowid, OciRowidDescriptor& rowidDescriptor, ArrayList& resultPai
System.Data.OracleClient.OracleCommand.ExecuteNonQueryInternal(Boolean needRowid, OciRowidDescriptor& rowidDescriptor) +670
System.Data.OracleClient.OracleCommand.ExecuteNonQuery() +146
System.Web.UI.WebControls.SqlDataSourceView.ExecuteDbCommand(DbCommand command, DataSourceOperation operation) +582
System.Web.UI.DataSourceView.Insert(IDictionary values, DataSourceViewOperationCallback callback) +167
System.Web.UI.WebControls.FormView.HandleInsert(String commandArg, Boolean causesValidation) +497
System.Web.UI.WebControls.FormView.HandleEvent(EventArgs e, Boolean causesValidation, String validationGroup) +706
System.Web.UI.Control.RaiseBubbleEvent(Object source, EventArgs args) +50
System.Web.UI.WebControls.FormViewRow.OnBubbleEvent(Object source, EventArgs e) +132
System.Web.UI.Control.RaiseBubbleEvent(Object source, EventArgs args) +50
System.Web.UI.Page.ProcessRequestMain(Boolean includeStagesBeforeAsyncPoint, Boolean includeStagesAfterAsyncPoint) +1959
```

Version Information: Microsoft .NET Framework Version: 4.0.30319; ASP.NET Version: 4.8.9277.0

Figure 95 Error message indicating violation of unique constraints

10 User Manual

Step 1: The main dashboard is displayed containing various form pages.

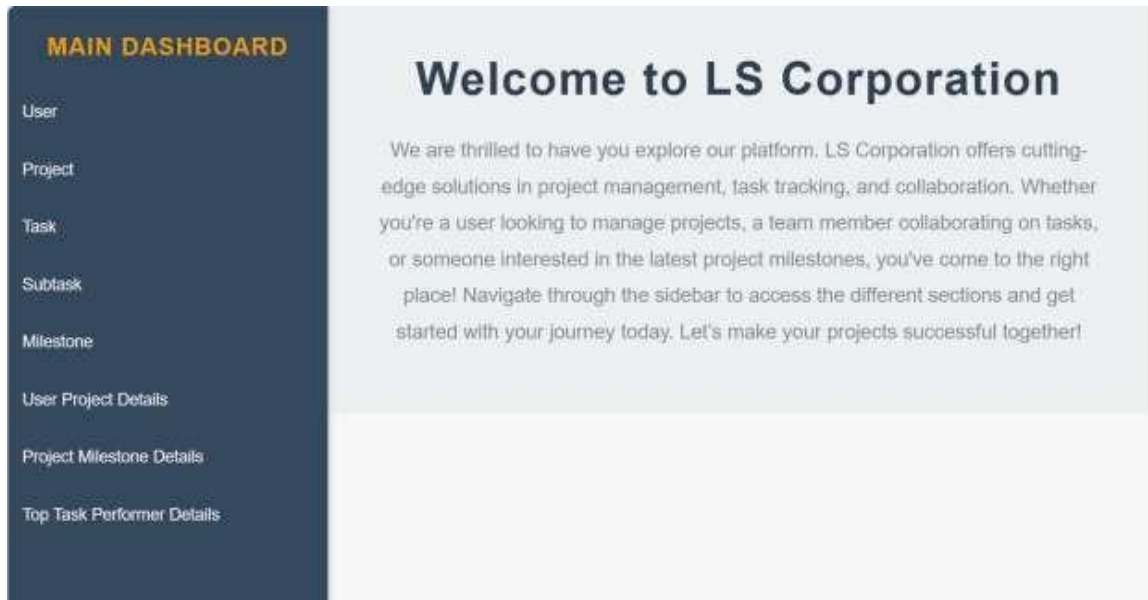


Figure 96 Navigated to Main page

Step 2: The “User” option is clicked to open the User Management form along with User Details table list.

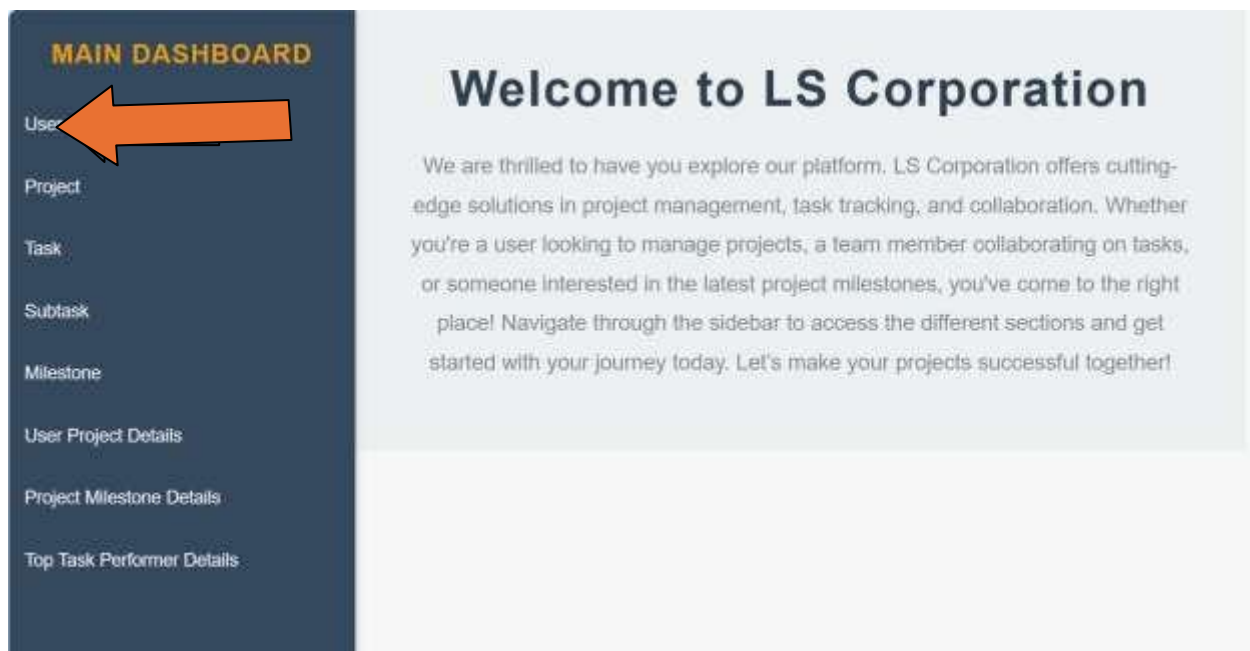


Figure 97 Navigate to 'User' page

Step 3: The “New” button in the User Management form is clicked, which opens empty fields for entering user details.

USER_ID: 21
 USER_NAME: Ningwa
 USER_EMAIL: ningwa@gmail.com
 USER_CONTACT: 9876543211
 USER_ROLE: Designer
[Edit](#) [Delete](#) [New](#)

	USER_ID	USER_NAME	USER_EMAIL	USER_CONTACT	USER_ROLE
Edit Delete	21	Ningwa	ningwa@gmail.com	9876543211	Designer
Edit Delete	1	Alice	alice@gmail.com	9812345678	Developer
Edit Delete	2	Bibisha	bibisha12@gmail.com	9822344678	Tester
Edit Delete	3	Charlie	charlie@gmail.com	9812345675	Project Manager
Edit Delete	4	Sugam	sugam@gmail.com	9812342378	Developer
Edit Delete	5	Sujan	sujan@gmail.com	9892345678	Designer
Edit Delete	7	Bob	bob@gmail.com	9812345654	Designer
Edit Delete	8	Ram	ram@gmail.com	9711345678	Project Manager
Edit Delete	9	Diana	diana@gmail.com	9711345677	Developer
Edit Delete	10	Neha	neha@gmail.com	9711345688	Tester
Edit Delete	11	Numa	numa@gmail.com	9711555678	Developer
Edit Delete	12	Krish	krish@gmail.com	9701345678	Designer
Edit Delete	13	Evan	evan@gmail.com	9711345666	Tester
Edit Delete	14	Aisha	aisha@gmail.com	9811345666	Developer
Edit Delete	15	Binu	binu@gmail.com	9711335666	Designer
Edit Delete	16	Akriti	akriti@gmail.com	9811845666	Developer
Edit Delete	17	Deepson	deepson@gmail.com	9781345666	Designer
Edit Delete	18	Smarika	smarika@gmail.com	9711345606	Tester
Edit Delete	19	Anna	anna@gmail.com	9711346066	Developer
Edit Delete	20	Laxman	laxman@gmail.com	9710045666	Designer

Figure 98 Click on ‘New’ button

USER_ID:
 USER_NAME:
 USER_EMAIL:
 USER_CONTACT:
 USER_ROLE:
[Insert](#) [Cancel](#)

	USER_ID	USER_NAME	USER_EMAIL	USER_CONTACT
Edit Delete	21	Ningwa	ningwa@gmail.com	9876543211
Edit Delete	1	Alice	alice@gmail.com	9812345678
Edit Delete	2	Bibisha	bibisha12@gmail.com	9822344678
Edit Delete	3	Charlie	charlie@gmail.com	9812345675

Figure 99 User Management Form

Step 4: The new user details are entered, and the "Insert" button is clicked, adding the new user to the User Details list.



	USER_ID	USER_NAME	USER_EMAIL	USER_CONTACT	USER_ROLE
Edit Delete	21	Ningwa	ningwa@gmail.com	987654321	Developer
Edit Delete	1	Alice	alice@gmail.com	981234567	Developer

Figure 100 Enter required details and click 'Insert' button

Edit Delete	14	Aisha	aisha@gmail.com	9811345666	Developer
Edit Delete	15	Binu	binu@gmail.com	9711335666	Designer
Edit Delete	16	Akriti	akriti@gmail.com	9811845666	Developer
Edit Delete	17	Deepson	deepson@gmail.com	9781345666	Designer
Edit Delete	18	Smarika	smarika@gmail.com	9711345606	Tester
Edit Delete	19	Anna	anna@gmail.com	9711346066	Developer
Edit Delete	20	Laxman	laxman@gmail.com	9710045666	Designer
Edit Delete	22	Ravi	ravi@gmail.com	9812321234	Developer

Figure 101 New User details inserted and displayed

Step 5: The "Edit" button in the User Details table is clicked, allowing data entry for the selected user. The data is entered and the "Update" button is clicked to save the changes.

USER_ROLE:

[Insert](#) [Cancel](#)

	USER_ID	USER_NAME	USER_EMAIL	USER_CONTACT	USER_ROLE
Edit Delete	21	Ningwa	ningwa@gmail.com	9876543211	Designer
Edit Delete	1	Alice	alice@gmail.com	9812345678	Developer
Update Cancel	2	Bibisha	bibisha12@gmail.com	9822344678	Tester
Edit Delete	3	Charlie	charlie@gmail.com	9812345675	Project Manager
Edit Delete		Sugam	sugam@gmail.com	9812342378	Developer
Edit Delete	5	Sujan	sujan@gmail.com	9892345678	Designer
Edit Delete	6	Anjila	anjila@gmail.com	9712345678	Developer

Figure 102 Update the User data and click 'Update' button

Step 6: The "Delete" button is selected, enabling the deletion of users directly from the User Details table.

	USER_ID	USER_NAME	USER_EMAIL	USER_CONTACT	USER_ROLE
Edit Delete	1	Alice	alice@gmail.com	9812345678	Developer
Edit Delete	2	Bibisha	bibisha@gmail.com	9822345678	Designer
Edit Delete	3	Charlie	charlie@gmail.com	9812345675	Project Manager
Edit Delete	4	Sugam	sugam@gmail.com	9812342378	Developer
Edit Delete	5	Sujan	sujan@gmail.com	9892345678	Designer
Edit Delete	6	Anjila	anjila@gmail.com	9712345678	Developer
Edit Delete		Bob	bob@gmail.com	9812345654	Designer
Edit Delete	8	ram	ram@gmail.com	9711345678	Project Manager
Edit Delete	9	diana	diana@gmail.com	9711345677	Developer
Edit Delete	10	Neha	neha@gmail.com	9711345688	Tester
Edit Delete	11	Numa	numa@gmail.com	9711555678	Developer
Edit Delete	12	Krish	krish@gmail.com	9701345678	Designer
Edit Delete	13	Tom	tom@gmail.com	9711345666	Tester

Figure 103 Click 'Delete' button in User table

	USER_ID	USER_NAME	USER_EMAIL	USER_CONTACT	USER_ROLE
Edit Delete	21	Ningwa	ningwa@gmail.com	9876543211	Designer
Edit Delete	1	Alice	alice@gmail.com	9812345678	Developer
Edit Delete	2	Bibisha	bibisha12@gmail.com	9822344678	Tester
Edit Delete	3	Charlie	charlie@gmail.com	9812345675	Project Manager
Edit Delete	4	Sugam	sugam@gmail.com	9812342378	Developer
Edit Delete	5	Sujan	sujan@gmail.com	9892345678	Designer
Edit Delete	7	Bob	bob@gmail.com	9812345654	Designer
Edit Delete	8	Ram	ram@gmail.com	9711345678	Project Manager
Edit Delete	9	Diana	diana@gmail.com	9711345677	Developer
Edit Delete	10	Neha	neha@gmail.com	9711345688	Tester
Edit Delete	11	Numa	numa@gmail.com	9711555678	Developer
Edit Delete	12	Krish	krish@gmail.com	9701345678	Designer
Edit Delete	13	Evan	evan@gmail.com	9711345666	Tester
Edit Delete	14	Aisha	aisha@gmail.com	9811345666	Developer
Edit Delete	15	Binu	binu@gmail.com	9711335666	Designer
Edit Delete	16	Akriti	akriti@gmail.com	9811845666	Developer
Edit Delete	17	Deepson	deepson@gmail.com	9781345666	Designer
Edit Delete	18	Smarika	smarika@gmail.com	9711345606	Tester
Edit Delete	19	Anna	anna@gmail.com	9711346066	Developer
Edit Delete	20	Laxman	laxman@gmail.com	9710045666	Designer
Edit Delete	22	Ravi	ravi@gmail.com	9812321234	Developer

Figure 104 User Details Table

Step 7: The “Project” option is clicked to open the Project Management form along with Project Details table list.

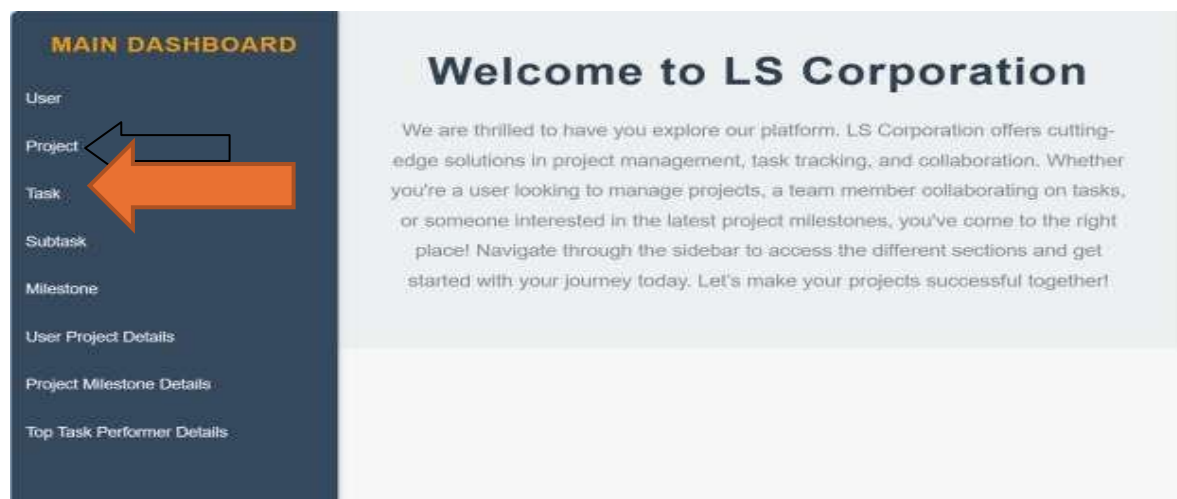


Figure 105 Navigate to Project page

Step 8: The “New” button in the Project Management form is clicked, which opens empty fields for entering project details.

PROJECT_ID: 24 PROJECT_NAME: Betan PROJECT_START_DATE: 1/10/2025 12:00:00 AM PROJECT_DUE_DATE: 12/11/2025 12:00:00 AM PROJECT_STATUS: A USER_USER_ID: 2 Edit Delete New						
	PROJECT_ID	PROJECT_NAME	PROJECT_START_DATE	PROJECT_DUE_DATE	PROJECT_STATUS	USER_USER_ID
Edit Delete	24		1/10/2025 12:00:00 AM	12/11/2025 12:00:00 AM	A	2
Edit Delete	25	Och	9/5/2025 10:00:00 AM	11/11/2025 10:00:00 AM	E	1
Edit Delete	22	Project Gama	1/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	C	1
Edit Delete	23	Zipper	1/15/2025 12:30:00 AM	11/21/2025 12:30:00 AM	A	1
Edit Delete	1	Project Alpha	1/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	A	3
Edit Delete	2	Project B	2/20/2025 12:00:00 AM	11/15/2025 12:00:00 AM	B	7
Edit Delete	3	Project C	3/5/2025 12:00:00 AM	10/20/2025 12:00:00 AM	C	12
Edit Delete	4	Project D	4/10/2024 12:00:00 AM	9/25/2024 12:00:00 AM	D	1
Edit Delete	5	Project E	5/15/2025 12:00:00 AM	8/10/2025 12:00:00 AM	E	20
Edit Delete	7	Project G	7/25/2025 12:00:00 AM	12/5/2025 12:00:00 AM	G	9
Edit Delete	8	Project H	8/15/2023 12:00:00 AM	10/15/2023 12:00:00 AM	H	4
Edit Delete	9	Project I	9/10/2025 12:00:00 AM	11/20/2025 12:00:00 AM	I	18
Edit Delete	10	Project J	10/5/2024 12:00:00 AM	12/20/2024 12:00:00 AM	J	14
Edit Delete	17	Project I	12/10/2023 12:00:00 AM	9/28/2024 12:00:00 AM	I	8

Figure 106 Click the 'New' button

PROJECT_ID:	
PROJECT_NAME:	
PROJECT_START_DATE:	
PROJECT_DUE_DATE:	
PROJECT_STATUS:	
USER_USER_ID:	
Insert Cancel	

	PROJECT_ID	PROJECT_NAME	PROJECT_START_DATE	P
Edit Delete	24	Betaa	1/10/2025 12:00:00 AM	12

Figure 107 Project Management Form

Step 9: The new project details are entered, and the "Insert" button is clicked, adding the new project to the Project Details list.

PROJECT_ID: 26
 PROJECT_NAME: Hectagon
 PROJECT_START_DATE: 09/10/2025 12:00:00 AM
 PROJECT_DUE_DATE: 12/31/2025 12:00:00 AM
 PROJECT_STATUS: C
 USER_USER_ID: 2
[Insert](#) [Cancel](#)

	PROJECT_ID	PROJECT_NAME	PROJECT_START_DATE	PROJECT_DUE_DATE	PROJECT_STATUS	USER_USER_ID
Edit Delete	24	Betaa	1/10/2025 12:00:00 AM	12/11/2025 12:00:00 AM	A	2
Edit Delete	25	Octagon	9/5/2025 10:00:00 AM	11/11/2025 10:00:00 AM	E	1
Edit Delete	22	Project Gama	1/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	C	1
Edit Delete	23	Zipper	1/15/2025 12:30:00 AM	11/21/2025 12:30:00 AM	A	1

Figure 108 Enter required project details and click 'Insert' button

USER_USER_ID: 2
[Edit](#) [Delete](#) [New](#)

	PROJECT_ID	PROJECT_NAME	PROJECT_START_DATE	PROJECT_DUE_DATE	PROJECT_STATUS	USER_USER_ID
Edit Delete	24	Betaa	1/10/2025 12:00:00 AM	12/11/2025 12:00:00 AM	A	2
Edit Delete	25	Octagon	9/5/2025 10:00:00 AM	11/11/2025 10:00:00 AM	E	1
Edit Delete	22	Project Gama	1/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	C	1
Edit Delete	23	Zipper	1/15/2025 12:30:00 AM	11/21/2025 12:30:00 AM	A	1
Edit Delete	26	Hectagon	9/10/2025 12:00:00 AM	12/31/2025 12:00:00 AM	C	2
Edit Delete	1	Project Alpha	1/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	A	3
Edit Delete	2	Project B	2/2/2025 12:00:00 AM	11/15/2025 12:00:00 AM	B	7
Edit Delete	3	Project C	3/5/2025 12:00:00 AM	10/20/2025 12:00:00 AM	C	12
Edit Delete	4	Project D	4/10/2024 12:00:00 AM	9/25/2024 12:00:00 AM	D	1
Edit Delete	5	Project E	5/15/2025 12:00:00 AM	8/10/2025 12:00:00 AM	E	20

Figure 109 New Project details inserted

Step 10: The "Edit" button in the Project Details table is clicked, allowing data entry for the selected project. The data is entered and the "Update" button is clicked to save the changes.

USER_USER_ID:

[Insert](#) [Cancel](#)

	PROJECT_ID	PROJECT_NAME	PROJECT_START_DATE	PROJECT_DUE_DATE	PROJECT_STATUS	USER_USER_ID
Edit Delete	24	Betaa	1/10/2025 12:00:00 AM	12/11/2025 12:00:00 AM	A	2
Edit Delete	25	Octagon	9/5/2025 10:00:00 AM	11/11/2025 10:00:00 AM	E	1
Edit Delete	22	Project Gama	1/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	C	1
Update	23	Zipper	1/15/2025 12:30:00 AM	11/21/2025 12:30:00 AM	A	1
Edit Delete	1	Project Alpha	1/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	A	3
Edit Delete	2	Project B	2/20/2025 12:00:00 AM	11/15/2025 12:00:00 AM	B	7
Edit Delete	3	Project C	3/5/2025 12:00:00 AM	10/20/2025 12:00:00 AM	C	12
Edit Delete	4	Project D	4/10/2024 12:00:00 AM	9/25/2024 12:00:00 AM	D	1
Edit Delete	5	Project E	5/15/2025 12:00:00 AM	8/10/2025 12:00:00 AM	E	20
Edit Delete	6	Project F	6/20/2025 12:00:00 AM	7/30/2025 12:00:00 AM	F	15

Figure 110 Update project details and click on 'Update' button

Step 11: The "Delete" button is selected, enabling the deletion of projects directly from the Project Details table.

[Insert](#) [Cancel](#)

	PROJECT_ID	PROJECT_NAME	PROJECT_START_DATE	PROJECT_DUE_DATE	PROJECT_STATUS	USER_USER_ID
Edit Delete	24	Betaa	1/10/2025 12:00:00 AM	12/11/2025 12:00:00 AM	A	2
Edit Delete	25	Octagon	9/5/2025 10:00:00 AM	11/11/2025 10:00:00 AM	E	1
Edit Delete	22	Project Gama	1/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	C	1
Edit Delete	23	Zipper	1/15/2025 12:30:00 AM	11/21/2025 12:30:00 AM	A	1
Edit Delete	1	Project Alpha	1/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	A	3
Edit Delete	2	Project B	2/20/2025 12:00:00 AM	11/15/2025 12:00:00 AM	B	7
Edit Delete	3	Project C	3/5/2025 12:00:00 AM	10/20/2025 12:00:00 AM	C	12
Edit Delete	4	Project D	4/10/2024 12:00:00 AM	9/25/2024 12:00:00 AM	D	1
Edit Delete	5	Project E	5/15/2025 12:00:00 AM	8/10/2025 12:00:00 AM	E	20
Edit Delete	6	Project F	6/20/2025 12:00:00 AM	7/30/2025 12:00:00 AM	F	15
Edit Delete	7	Project G	7/25/2025 12:00:00 AM	12/5/2025 12:00:00 AM	G	9
Edit Delete	8	Project H	8/15/2023 12:00:00 AM	10/15/2023 12:00:00 AM	H	4

Figure 111 Click 'Delete' button on project table

Edit Delete	1	Project Alpha	1/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	A	3
Edit Delete	2	Project B	2/20/2025 12:00:00 AM	11/15/2025 12:00:00 AM	B	7
Edit Delete	3	Project C	3/5/2025 12:00:00 AM	10/20/2025 12:00:00 AM	C	12
Edit Delete	4	Project D	4/10/2024 12:00:00 AM	9/25/2024 12:00:00 AM	D	1
Edit Delete	5	Project E	5/15/2025 12:00:00 AM	8/10/2025 12:00:00 AM	E	20
Edit Delete	7	Project G	7/25/2025 12:00:00 AM	12/5/2025 12:00:00 AM	G	9
Edit Delete	8	Project H	8/15/2023 12:00:00 AM	10/15/2023 12:00:00 AM	H	4
Edit Delete	9	Project I	9/10/2025 12:00:00 AM	11/20/2025 12:00:00 AM	I	18
Edit Delete	10	Project J	10/5/2024 12:00:00 AM	12/20/2024 12:00:00 AM	J	14
Edit Delete	12	Project L	12/10/2022 12:00:00 AM	2/28/2023 12:00:00 AM	L	8
Edit Delete	13	Project M	1/20/2026 12:00:00 AM	3/25/2026 12:00:00 AM	M	19
Edit Delete	14	Project N	2/15/2026 12:00:00 AM	5/10/2026 12:00:00 AM	N	2
Edit Delete	15	Project O	3/5/2026 12:00:00 AM	6/15/2026 12:00:00 AM	O	11
Edit Delete	16	Project P	4/5/2026 12:00:00 AM	7/10/2026 12:00:00 AM	P	16

Figure 112 Project Details Table

Step 12: The “Task” option is clicked to open the Task Management form along with Task Details table list.

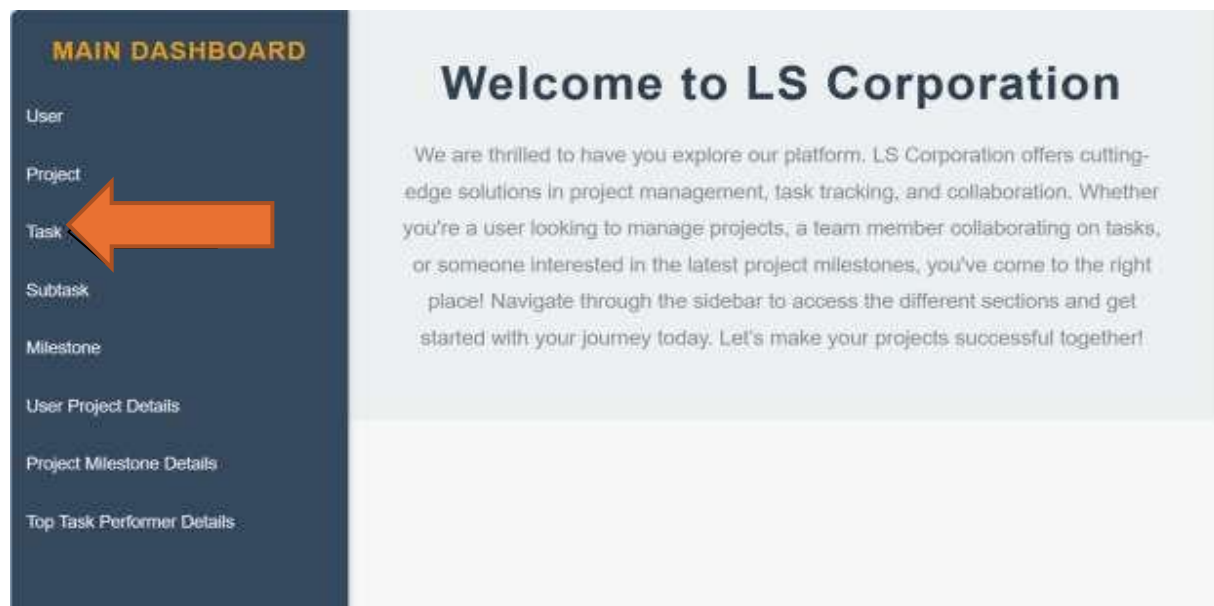


Figure 113 Navigate to Task page

Step 13: The “New” button in the Task Management form is clicked, which opens empty fields for entering task details.

TASK_ID: 21
 TASK_NAME: Development
 START_DATE: 5/10/2025 1:00:00 PM
 DUE_DATE: 7/3/2025 4:30:00 PM
 STATUS: In progress
 PROJECT_PROJECT_ID: 1
[Edit](#) [Delete](#) [New](#)

	TASK_ID	TASK_NAME	START_DATE	DUE_DATE	STATUS	PROJECT_PROJECT_ID
Edit Delete	21	Development	5/10/2025 1:00:00 PM	7/3/2025 4:30:00 PM	In progress	1
Edit Delete	1	Project Scope	1/1/2025 12:00:00 AM	1/5/2025 12:00:00 AM	Completed	1
Edit Delete	2	Project Kickoff	1/6/2025 12:00:00 AM	1/10/2025 12:00:00 AM	In Progress	1
Edit Delete	3	Feasibility Study	1/11/2025 12:00:00 AM	1/15/2025 12:00:00 AM	Pending	1
Edit Delete	4	Project Budgeting	2/1/2025 12:00:00 AM	2/5/2025 12:00:00 AM	Completed	2
Edit Delete	5	Design Documentation	2/6/2025 12:00:00 AM	2/10/2025 12:00:00 AM	In Progress	2
Edit Delete	6	Integration Testing	4/1/2025 12:00:00 AM	4/15/2025 12:00:00 AM	Not Started	1
Edit Delete	7	Acceptance Testing	4/16/2025 12:00:00 AM	4/30/2025 12:00:00 AM	Not Started	1
Edit Delete	8	Deployment Plan	5/1/2025 12:00:00 AM	5/10/2025 12:00:00 AM	Not Started	1
Edit Delete	9	Ground Research	5/11/2025 2:30:00 PM	5/15/2025 11:00:00 PM	Completed	1
Edit Delete	10	Post-Launch	5/16/2025 12:00:00 AM	5/30/2025 12:00:00 AM	Not Started	1
Edit Delete	11	Research	1/1/2025 12:00:00 AM	1/20/2025 12:00:00 AM	Completed	2
Edit Delete	12	Prototype Development	1/21/2025 12:00:00 AM	2/10/2025 12:00:00 AM	In Progress	2
Edit Delete	13	User Feedback	2/11/2025 12:00:00 AM	2/20/2025 12:00:00 AM	Not Started	2

Figure 114 Click on 'New' button to add new task

TASK_ID:
 TASK_NAME:
 START_DATE:
 DUE_DATE:
 STATUS:
 PROJECT_PROJECT_ID:
[Insert](#) [Cancel](#)

	TASK_ID	TASK_NAME	START_DATE	
Edit Delete	21	Development	5/10/2025 1:00:00 PM	7

Figure 115 Task Management Form

Step 14: The new task details are entered, and the "Insert" button is clicked, adding the new user to the Task Details list.

TASK_ID: 22

TASK_NAME: Pre-Launch

START_DATE: 5/10/2025 1:00:00 PM

DUE_DATE: 7/03/2025 4:30:00 PM

STATUS: In progress

PROJECT_PROJECT_ID: 1

[Insert](#) [Cancel](#)

	TASK_ID	TASK_NAME	START_DATE
Edit Delete	21	Development	5/10/2025 1:00:00 PM

Figure 116 Enter required task details and click 'Insert' button

START_DATE: 5/10/2025 1:00:00 PM
DUE_DATE: 7/3/2025 4:30:00 PM
STATUS: In progress
PROJECT_PROJECT_ID: 1
[Edit](#) [Delete](#) [New](#)

	TASK_ID	TASK_NAME	START_DATE	DUE_DATE	STATUS	PROJECT_PROJECT_ID
Edit Delete	21	Development	5/10/2025 1:00:00 PM	7/3/2025 4:30:00 PM	In progress	1
Edit Delete	22	Pre-Launch	5/10/2025 1:00:00 PM	7/3/2025 4:30:00 PM	In progress	1
Edit Delete	1	Project Scope	1/5/2025 12:00:00 AM	1/5/2025 12:00:00 AM	Completed	1
Edit Delete	2	Project Kickoff	1/6/2025 12:00:00 AM	1/10/2025 12:00:00 AM	In Progress	1
Edit Delete	3	Feasibility Study	1/11/2025 12:00:00 AM	1/15/2025 12:00:00 AM	Pending	1
Edit Delete	4	Project Budgeting	2/1/2025 12:00:00 AM	2/5/2025 12:00:00 AM	Completed	2
Edit Delete	5	Design Documentation	2/6/2025 12:00:00 AM	2/10/2025 12:00:00 AM	In Progress	2
Edit Delete	6	Integration Testing	4/1/2025 12:00:00 AM	4/15/2025 12:00:00 AM	Not Started	1

Figure 117 New Task details inserted

Step 15: The "Edit" button in the Task Details table is clicked, allowing data entry for the selected task. The data is entered and the "Update" button is clicked to save the changes.

Insert Cancel						
	TASK_ID	TASK_NAME	START_DATE	DUE_DATE	STATUS	PROJECT_PROJECT_ID
Edit Delete	21	Development	5/10/2025 1:00:00 PM	7/3/2025 4:30:00 PM	In progress	1
Edit Delete	1	Project Scope	1/1/2025 12:00:00 AM	1/5/2025 12:00:00 AM	Completed	1
Edit Delete	2	Project Kickoff	1/6/2025 12:00:00 AM	1/10/2025 12:00:00 AM	In Progress	1
Edit Delete	3	Feasibility Study	1/11/2025 12:00:00 AM	1/15/2025 12:00:00 AM	Pending	1
Edit Delete	4	Project Budgeting	2/1/2025 12:00:00 AM	2/5/2025 12:00:00 AM	Completed	2
Edit Delete	5	Design Documentation	2/6/2025 12:00:00 AM	2/10/2025 12:00:00 AM	In Progress	2
Edit Delete	6	Integration Testing	4/1/2025 12:00:00 AM	4/15/2025 12:00:00 AM	Not Started	1
Edit Delete	7	Acceptance Testing	4/16/2025 12:00:00 AM	4/30/2025 12:00:00 AM	Not Started	1
Edit Delete	8	Deployment Plan	5/1/2025 12:00:00 AM	5/10/2025 12:00:00 AM	Not Started	1
Update		Ground Research	5/11/2025 2:30:00 PM	5/15/2025 11:00:00 PM	Completed	1
Edit Delete		Launch	5/16/2025 12:00:00 AM	5/30/2025 12:00:00 AM	Not Started	1
Edit Delete	11	Research	1/1/2025 12:00:00 AM	1/20/2025 12:00:00 AM	Completed	2
Edit Delete	12	Prototype Development	1/21/2025 12:00:00 AM	2/10/2025 12:00:00 AM	In Progress	2

Figure 118 Update the task data and click on 'Update' button

Step 16: The "Delete" button is selected, enabling the deletion of tasks directly from the Task Details table.

Edit Delete	13	User Feedback	2/11/2025 12:00:00 AM	2/20/2025 12:00:00 AM	Not Started	2
Edit Delete	14	Final Testing	2/21/2025 12:00:00 AM	3/1/2025 12:00:00 AM	Not Started	2
Edit Delete	15	Planning	3/2/2025 12:00:00 AM	3/15/2025 12:00:00 AM	Not Started	2
Edit Delete	16	Marketing	3/16/2025 12:00:00 AM	4/1/2025 12:00:00 AM	Not Started	2
Edit Delete	17	Feedback	4/2/2025 12:00:00 AM	4/15/2025 12:00:00 AM	Not Started	2
Edit Delete	18	Documentation	4/16/2025 12:00:00 AM	4/30/2025 12:00:00 AM	Not Started	2

Figure 119 Click 'Delete' button on task details table

Edit Delete	1	Project Scope	1/1/2025 12:00:00 AM	1/5/2025 12:00:00 AM	Completed	1
Edit Delete	2	Project Kickoff	1/6/2025 12:00:00 AM	1/10/2025 12:00:00 AM	In Progress	1
Edit Delete	3	Feasibility Study	1/11/2025 12:00:00 AM	1/15/2025 12:00:00 AM	Pending	1
Edit Delete	4	Project Budgeting	2/1/2025 12:00:00 AM	2/5/2025 12:00:00 AM	Completed	2
Edit Delete	5	Design Documentation	2/6/2025 12:00:00 AM	2/10/2025 12:00:00 AM	In Progress	2
Edit Delete	6	Integration Testing	4/1/2025 12:00:00 AM	4/15/2025 12:00:00 AM	Not Started	1
Edit Delete	7	Acceptance Testing	4/16/2025 12:00:00 AM	4/30/2025 12:00:00 AM	Not Started	1
Edit Delete	8	Deployment Plan	5/1/2025 12:00:00 AM	5/10/2025 12:00:00 AM	Not Started	1
Edit Delete	9	Ground Research	5/11/2025 2:30:00 PM	5/15/2025 11:00:00 PM	Completed	1
Edit Delete	10	Post-Launch	5/16/2025 12:00:00 AM	5/30/2025 12:00:00 AM	Not Started	1
Edit Delete	11	Research	1/1/2025 12:00:00 AM	1/20/2025 12:00:00 AM	Completed	2
Edit Delete	12	Prototype Development	1/21/2025 12:00:00 AM	2/10/2025 12:00:00 AM	In Progress	2
Edit Delete	13	User Feedback	2/11/2025 12:00:00 AM	2/20/2025 12:00:00 AM	Not Started	2
Edit Delete	14	Final Testing	2/21/2025 12:00:00 AM	3/1/2025 12:00:00 AM	Not Started	2
Edit Delete	16	Marketing	3/16/2025 12:00:00 AM	4/1/2025 12:00:00 AM	Not Started	2
Edit Delete	17	Feedback	4/2/2025 12:00:00 AM	4/15/2025 12:00:00 AM	Not Started	2
Edit Delete	18	Documentation	4/16/2025 12:00:00 AM	4/30/2025 12:00:00 AM	Not Started	2

Figure 120 Task Details Table

Step 17: The “Subtask” option is clicked to open the Subtask Management form along with Subtask Details table list.

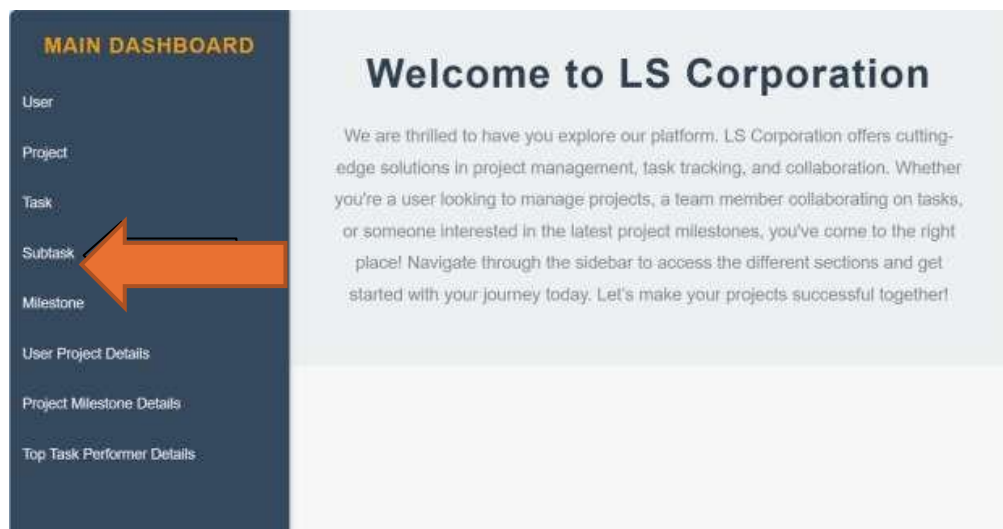


Figure 121 Navigate to Subtask page

Step 18: The “New” button in the Subtask Management form is clicked, which opens empty fields for entering subtask details.

SUBTASK_ID: 21
SUBTASK_NAME: AI research
SUBTASK_STATUS: Completed
TASK_DETAILS_TASK_ID: 1
[Edit](#) [Delete](#) [New](#)

	SUBTASK_ID	SUBTASK_NAME	SUBTASK_STATUS	TASK_DETAILS_TASK_ID
Edit Delete	21	AI research	Completed	1
Edit Delete	1	Design UI	In Progress	1
Edit Delete	2	Develop API	Not Started	2
Edit Delete	3	Create Database Schema	Completed	2
Edit Delete	4	Integrate Unit Test	In progress	3
Edit Delete	5	Conduct Integration Tests	In Progress	3
Edit Delete	6	Prepare Deployment Plan	Not Started	4
Edit Delete	7	Gather User Feedback	Not Started	4
Edit Delete	8	Finalize Documentation	In Progress	5
Edit Delete	9	Conduct User Training	Not Started	6
Edit Delete	11	Set Up Monitoring	In Progress	8
Edit Delete	12	Post-Launch Review Meeting	Not Started	9
Edit Delete	13	Follow Up on Feedback	Not Started	10
Edit Delete	14	Plan Next Phase	Not Started	11
Edit Delete	15	Conduct Stakeholder Meeting	In Progress	12
Edit Delete	16	Review Project Outcomes	Not Started	13
Edit Delete	17	Identify Improvements	Not Started	14
Edit Delete	19	Set Up Post-Mortem Analysis	Not Started	16
Edit Delete	20	Prepare Final Report	Not Started	17

Figure 122 Click on 'New' button to insert new subtask

SUBTASK_ID:
 SUBTASK_NAME:
 SUBTASK_STATUS:
 TASK_DETAILS_TASK_ID:
[Insert](#) [Cancel](#)

	SUBTASK_ID	SUBTASK_NAME	SUBTA
Edit Delete	21	AI research	Comple

Figure 123 Subtask Management Form

Step 19: The new subtask details are entered, and the "Insert" button is clicked, adding the new subtask to the Subtask Details list.

SUBTASK_ID: 22
 SUBTASK_NAME: Financial Research
 SUBTASK_STATUS: In progress
 TASK_DETAILS_TASK_ID: 1
[Insert](#) [Cancel](#)

	SUBTASK_ID	SUBTASK_NAME	SUB
Edit Delete	21	AI research	Com

Figure 124 Enter required subtask details and click 'Insert' button

SUBTASK_NAME: AI research
 SUBTASK_STATUS: Completed
 TASK_DETAILS_TASK_ID: 1
[Edit](#) [Delete](#) [New](#)

	SUBTASK_ID	SUBTASK_NAME	SUBTASK_STATUS	TASK_DETAILS_TASK_ID
Edit Delete	21	AI research	Completed	1
Edit Delete	22	Financial Research	In progress	1
Edit Delete	1	Design UI	In progress	1
Edit Delete	2	Develop API	Not started	2
Edit Delete	3	Create Database Schema	Completed	2
Edit Delete	4	Integrate Unit Test	In progress	3

Figure 125 New subtask details inserted

Step 20: The "Edit" button in the Subtask Details table is clicked, allowing data entry for the selected subtask. The data is entered and the "Update" button is clicked to save the changes.

	SUBTASK_ID	SUBTASK_NAME	SUBTASK_STATUS	TASK_DETAILS_TASK_ID
Edit Delete	21	AI research	Completed	1
Edit Delete	1	Design UI	In Progress	1
Edit Delete	2	Develop API	Not Started	2
Edit Delete	3	Create Database Schema	Completed	2
Update Cancel	4	Integrate Unit Test	In progress	3
Edit Delete	5	Conduct Integration Tests	In Progress	3
Edit Delete		Prepare Deployment Plan	Not Started	4
Edit Delete	7	Gather User Feedback	Not Started	4
Edit Delete	8	Finalize Documentation	In Progress	5

Figure 126 Update the subtask data and click 'Update' button

Step 21: The "Delete" button is selected, enabling the deletion of subtasks directly from the Subtask Details table.

TASK_DETAILS_TASK_ID:				
Insert Cancel				
	SUBTASK_ID	SUBTASK_NAME	SUBTASK_STATUS	TASK_DETAILS_TASK_ID
Edit Delete	21	AI research	Completed	1
Edit Delete	1	Design UI	In Progress	1
Edit Delete	2	Develop API	Not Started	2
Edit Delete	3	Create Database Schema	Completed	2
Edit Delete	4	Integrate Unit Test	In progress	3
Edit Delete	5	Conduct Integration Tests	In Progress	3
Edit Delete	6	Prepare Deployment Plan	Not Started	4
Edit Delete	7	Gather User Feedback	Not Started	4
Edit Delete	8	Finalize Documentation	In Progress	5
Edit Delete		Conduct User Training	Not Started	6
Edit Delete	10	Create Marketing Materials	Not Started	7
Edit Delete	11	Set Up Monitoring	In Progress	8

Figure 127 Click 'Delete' button on subtask table

Edit Delete	1	Design UI	In Progress	1
Edit Delete	2	Develop API	Not Started	2
Edit Delete	3	Create Database Schema	Completed	2
Edit Delete	4	Integrate Unit Test	In progress	3
Edit Delete	5	Conduct Integration Tests	In Progress	3
Edit Delete	6	Prepare Deployment Plan	Not Started	4
Edit Delete	7	Gather User Feedback	Not Started	4
Edit Delete	8	Finalize Documentation	In Progress	5
Edit Delete	9	Conduct User Training	Not Started	6
Edit Delete	11	Set Up Monitoring	In Progress	8
Edit Delete	12	Post-Launch Review Meeting	Not Started	9
Edit Delete	13	Follow Up on Feedback	Not Started	10
Edit Delete	14	Plan Next Phase	Not Started	11
Edit Delete	15	Conduct Stakeholder Meeting	In Progress	12
Edit Delete	16	Review Project Outcomes	Not Started	13
Edit Delete	17	Identify Improvements	Not Started	14
Edit Delete	19	Set Up Post-Mortem Analysis	Not Started	16

Figure 128 Subtask Details

Step 22: The “Milestone” option is clicked to open the Milestone Management form along with Milestone Details table list.

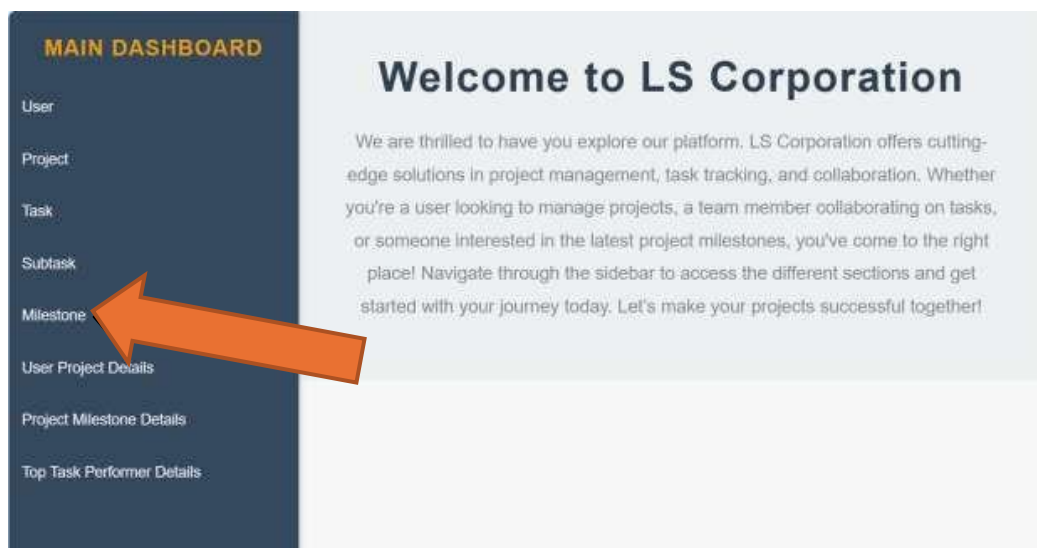


Figure 129 Navigate to Milestone page

Step 23: The “New” button in the Milestone Management form is clicked, which opens empty fields for entering milestone details.

MILESTONE_ID: 17
 MILESTONE_NAME: Meeting
 MILESTONE_PERIOD: 3/15/2025 12:20:00 PM
 PROJECT_PROJECT_ID: 1
[Edit](#) [Delete](#) [New](#)

	MILESTONE_ID	MILESTONE_NAME	MILESTONE_PERIOD	PROJECT_PROJECT_ID
Edit Delete	17	Meeting	3/15/2025 12:20:00 PM	1
Edit Delete	16	Testing	1/19/2025 12:30:00 AM	1
Edit Delete	1	Initiation	1/15/2025 12:00:00 AM	1
Edit Delete	2	Planning	2/15/2025 12:00:00 AM	1
Edit Delete	3	Execution	3/15/2025 12:00:00 AM	1
Edit Delete	4	Monitoring	4/15/2025 12:00:00 AM	1
Edit Delete	6	Requirements Gathering	1/20/2025 12:00:00 AM	2
Edit Delete	7	Design Phase	2/20/2025 12:00:00 AM	2
Edit Delete	9	Testing Phase	4/20/2025 12:00:00 AM	2
Edit Delete	10	Final Deployment	7/27/2025 4:30:00 PM	2
Edit Delete	11	Requirements Review	1/25/2025 12:00:00 AM	3
Edit Delete	12	Prototype	2/25/2025 12:00:00 AM	3
Edit Delete	13	Final Review	3/25/2025 12:00:00 AM	3
Edit Delete	14	User Acceptance Testing	4/25/2025 12:00:00 AM	3
Edit Delete	15	Go Live	5/25/2025 12:00:00 AM	3

Figure 130 Click on 'New' button

MILESTONE_ID:
 MILESTONE_NAME:
 MILESTONE_PERIOD:
 PROJECT_PROJECT_ID:
[Insert](#) [Cancel](#)

	MILESTONE_ID	MILESTONE_NAME	MILESTONE_PERIOD
Edit Delete	17	Meeting	3/15/2025 12:20:00 PM

Figure 131 Milestone Management Form

Step 24: The new milestone details are entered, and the "Insert" button is clicked, adding the new milestone to the Milestone Details list.

MILESTONE_ID: 18
 MILESTONE_NAME: Demonstration
 MILESTONE_PERIOD: 4/25/2025 12:40:00 PM
 PROJECT_PROJECT_ID: 1
[Insert](#) [Cancel](#)

	MILESTONE_ID	MILESTONE_NAME	MILESTONE_PERIOD	PROJECT_PROJECT_ID
Edit Delete	17	Meeting	3/15/2025 12:20:00 PM	1

Figure 132 Enter require milestone details and click on 'Insert' button

MILESTONE_ID: 17
 MILESTONE_NAME: Meeting
 MILESTONE_PERIOD: 3/15/2025 12:20:00 PM
 PROJECT_PROJECT_ID: 1
[Edit](#) [Delete](#) [New](#)

	MILESTONE_ID	MILESTONE_NAME	MILESTONE_PERIOD	PROJECT_PROJECT_ID
Edit Delete	17	Meeting	3/15/2025 12:20:00 PM	1
Edit Delete	16	Testing	1/19/2025 12:30:00 AM	1
Edit Delete	18	Demonstration	4/25/2025 12:40:00 PM	1
Edit Delete	1	Initiation	1/15/2025 12:00:00 AM	1
Edit Delete	2	Planning	2/15/2025 12:00:00 AM	1
Edit Delete	3	Execution	3/15/2025 12:00:00 AM	1
Edit Delete	4	Monitoring	4/15/2025 12:00:00 AM	1

Figure 133 New milestone details inserted

Step 25: The "Edit" button in the Milestone Details table is clicked, allowing data entry for the selected milestone. The data is entered and the "Update" button is clicked to save the changes.

Insert Cancel		MILESTONE_ID	MILESTONE_NAME	MILESTONE_PERIOD	PROJECT PROJECT_ID
Edit Delete		17	Meeting	3/15/2025 12:20:00 PM	1
Edit Delete		16	Testing	1/19/2025 12:30:00 AM	1
Edit Delete		1	Initiation	1/15/2025 12:00:00 AM	1
Edit Delete		2	Planning	2/15/2025 12:00:00 AM	1
Edit Delete		3	Execution	3/15/2025 12:00:00 AM	1
Edit Delete		4	Monitoring	4/15/2025 12:00:00 AM	1
Edit Delete		5	Closure	5/15/2025 12:00:00 AM	1
Edit Delete		6	Requirements Gathering	1/20/2025 12:00:00 AM	2
Edit Delete		7	Design Phase	2/20/2025 12:00:00 AM	2
Edit Delete		9	Testing Phase	4/20/2025 12:00:00 AM	2
Update Cancel		10	Final Deployment	7/27/2025 04:30:00 PM	2
Edit Delete		11	Requirements Review	1/25/2025 12:00:00 AM	3
Edit Delete		12	Prototype	2/25/2025 12:00:00 AM	3
Edit Delete		13	Final Review	3/25/2025 12:00:00 AM	3
Edit Delete		14	User Acceptance Testing	4/25/2025 12:00:00 AM	3
Edit Delete		15	Go Live	5/25/2025 12:00:00 AM	3

Figure 134 Update the required milestone data and click on 'Update' button

Step 26: The "Delete" button is selected, enabling the deletion of milestones directly from the Milestone Details table.

		MILESTONE_ID	MILESTONE_NAME	MILESTONE_PERIOD	PROJECT PROJECT_ID
Edit Delete		17	Meeting	3/15/2025 12:20:00 PM	1
Edit Delete		16	Testing	1/19/2025 12:30:00 AM	1
Edit Delete		1	Initiation	1/15/2025 12:00:00 AM	1
Edit Delete		2	Planning	2/15/2025 12:00:00 AM	1
Edit Delete		3	Execution	3/15/2025 12:00:00 AM	1
Edit Delete		4	Monitoring	4/15/2025 12:00:00 AM	1
Edit Delete		5	Closure	5/15/2025 12:00:00 AM	1
Edit Delete		6	Requirements Gathering	1/20/2025 12:00:00 AM	2
Edit Delete		7	Design Phase	2/20/2025 12:00:00 AM	2

Figure 135 Click 'Delete' button on Milestone details table

	EDIT	DELETE	MILESTONE_ID	MILESTONE_NAME	MILESTONE_PERIOD	PROJECT	PROJECT_ID
	Edit	Delete	17	Meeting	3/15/2025 12:20:00 PM	1	
	Edit	Delete	16	Testing	1/19/2025 12:30:00 AM	1	
	Edit	Delete	1	Initiation	1/15/2025 12:00:00 AM	1	
	Edit	Delete	2	Planning	2/15/2025 12:00:00 AM	1	
	Edit	Delete	3	Execution	3/15/2025 12:00:00 AM	1	
	Edit	Delete	4	Monitoring	4/15/2025 12:00:00 AM	1	
	Edit	Delete	6	Requirements Gathering	1/20/2025 12:00:00 AM	2	
	Edit	Delete	7	Design Phase	2/20/2025 12:00:00 AM	2	
	Edit	Delete	9	Testing Phase	4/20/2025 12:00:00 AM	2	
	Edit	Delete	10	Final Deployment	7/27/2025 4:30:00 PM	2	
	Edit	Delete	11	Requirements Review	1/25/2025 12:00:00 AM	3	

Figure 136 Milestone Details Table

Step 27: The “User Project Details” option is clicked to open the User details along with their project involvement details.

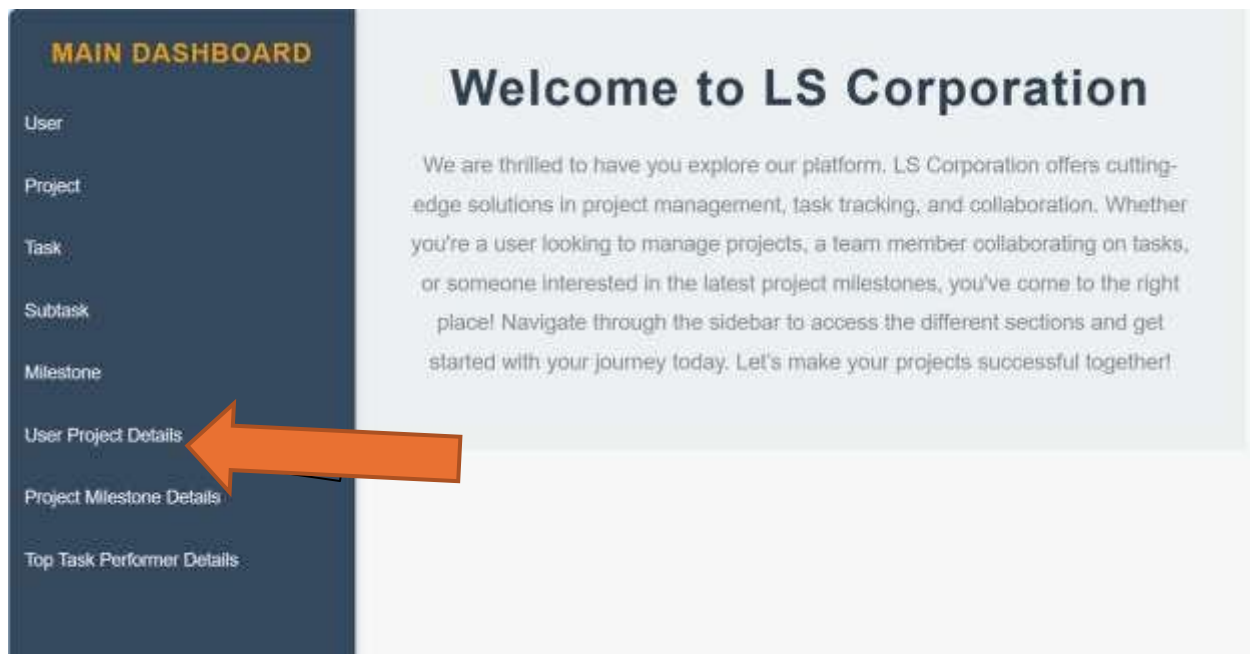


Figure 137 Navigate to User Project Details Table

Step 28: Select a user from the dropdown list on the left side, which displays comprehensive details of all the projects linked to the selected user.



NAME	PROJECT_START_DATE	PROJECT_DUE_DATE	PROJECT_STATUS	USER_NAME	USER_EMAIL	USER_CONTACT	USER_ROLE
Alice	1/10/2025 12:00:00 AM	12/11/2025 12:00:00 AM	A	Bibsha	bibsha12@gmail.com	9822344678	Tester
Bibsha	9/10/2025 12:00:00 AM	12/31/2025 12:00:00 AM	C	Bibsha	bibsha12@gmail.com	9822344678	Tester
Charlie	10/26 12:00:00 AM	5/10/2026 12:00:00 AM	N	Bibsha	bibsha12@gmail.com	9822344678	Tester

Figure 138 Choosing desired username from dropdown button



PROJECT_NAME	PROJECT_START_DATE	PROJECT_DUE_DATE	PROJECT_STATUS	USER_NAME	USER_EMAIL	USER_CONTACT	USER_ROLE
Octagon	9/5/2025 10:00:00 AM	11/11/2025 10:00:00 AM	E	Alice	alice@gmail.com	9812345678	Developer
Project Gama	1/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	C	Alice	alice@gmail.com	9812345678	Developer
Zipper	1/15/2025 12:30:00 AM	11/21/2025 12:30:00 AM	A	Alice	alice@gmail.com	9812345678	Developer
Project D	4/10/2024 12:00:00 AM	9/25/2024 12:00:00 AM	D	Alice	alice@gmail.com	9812345678	Developer
Project U	1/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	H	Alice	alice@gmail.com	9812345678	Developer

Figure 139 User Project Details

Step 29: The “Project Milestone Details” option is clicked to open the Project details with the selected project milestone details.

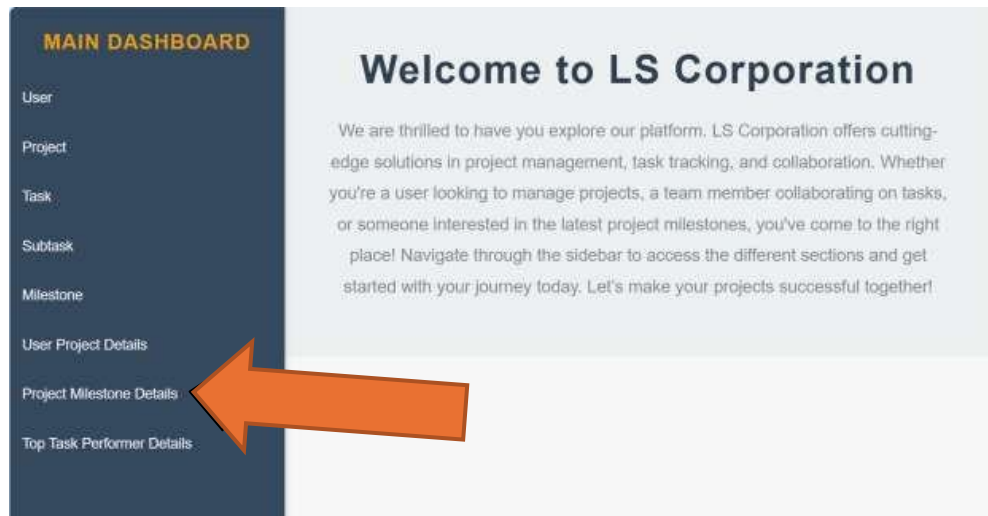


Figure 140 Navigate to Project Milestone Details

Step 30: Select a project from the dropdown list on the left side, which displays comprehensive details of all the milestones linked to the selected project.

ID	MILESTONE_NAME	MILESTONE_PERIOD	PROJECT_PROJECT_ID	PROJECT_NAME	PROJECT_START_DATE	PROJECT_DUE_DATE	PROJECT_STATUS
1	Requirements Gathering	1/20/2025 12:00:00 AM	2	Project B	2/20/2025 12:00:00 AM	11/15/2025 12:00:00 AM	B
2	Design Phase	2/20/2025 12:00:00 AM	2	Project B	2/20/2025 12:00:00 AM	11/15/2025 12:00:00 AM	B
3	Testing Phase	4/20/2025 12:00:00 AM	2	Project B	3/20/2025 12:00:00 AM	11/15/2025 12:00:00 AM	B
4	Final Deployment	7/27/2025 4:30:00 PM	2	Project B	2/20/2025 12:00:00 AM	11/15/2025 12:00:00 AM	B

Figure 141 Choosing desired project from dropdown button

Project Alpha							
MILESTONE_ID	MILESTONE_NAME	MILESTONE_PERIOD	PROJECT_PROJECT_ID	PROJECT_NAME	PROJECT_START_DATE	PROJECT_DUE_DATE	PROJECT_STATUS
17	Meeting	3/15/2025 12:29:00 PM	1	Project Alpha	3/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	A
16	Testing	1/19/2025 12:30:00 AM	1	Project Alpha	1/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	A
18	Demonstration	4/25/2025 12:40:00 PM	1	Project Alpha	1/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	A
1	Initiation	1/15/2025 12:00:00 AM	1	Project Alpha	1/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	A
2	Planning	2/15/2025 12:00:00 AM	1	Project Alpha	1/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	A
3	Execution	3/15/2025 12:00:00 AM	1	Project Alpha	1/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	A
4	Monitoring	4/15/2025 12:00:00 AM	1	Project Alpha	1/1/2025 12:00:00 AM	12/31/2025 12:00:00 AM	A

Figure 142 Project Milestone Details

Step 31: The “Top Task Performer Details” option is clicked to open the Project details with the selected project milestone details.

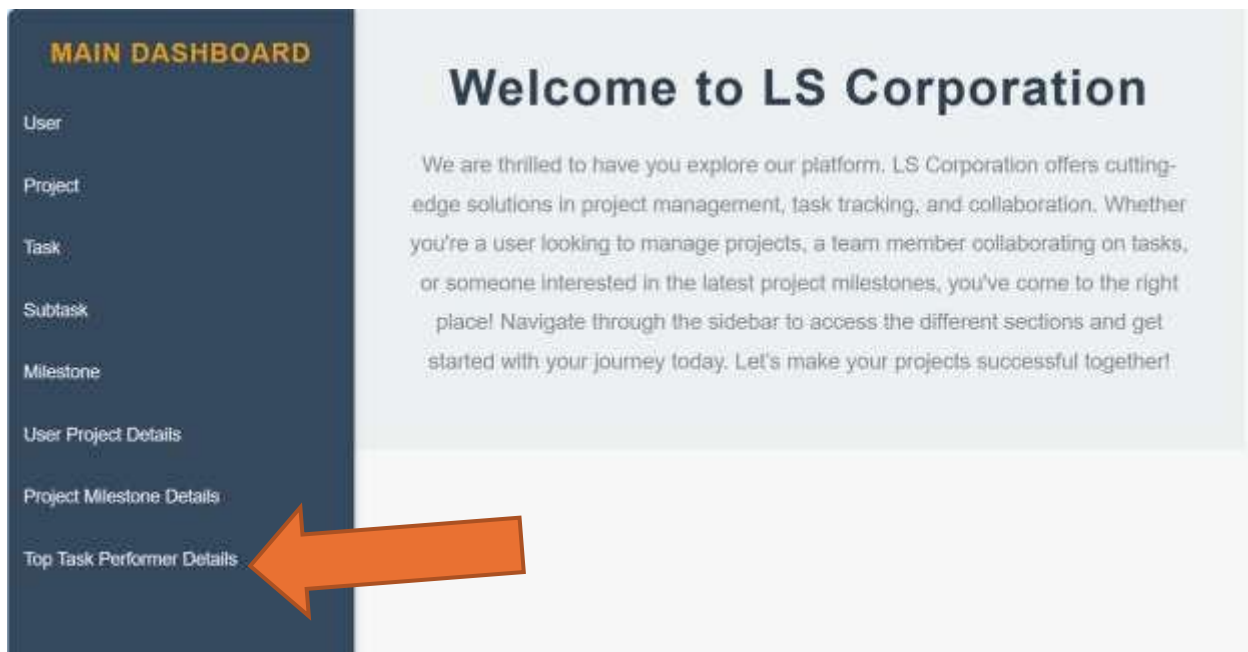
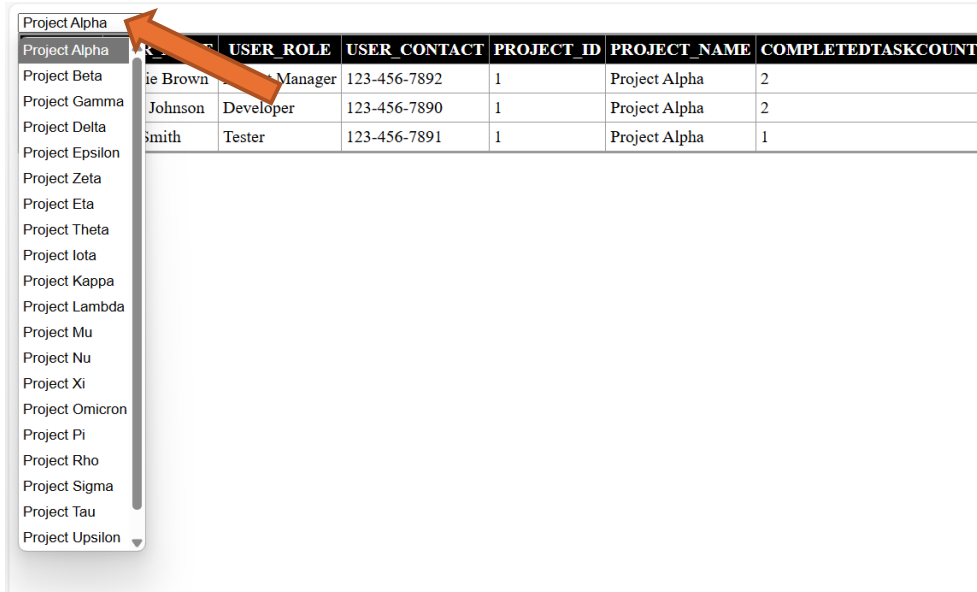


Figure 143 Navigate to Top Task Performer Details

Step 32: Select a project from the dropdown list on the left side, which displays comprehensive details of all the projects with its top performer details.



The screenshot displays a web application interface. On the left, a vertical dropdown menu is open, showing a list of project names from 'Project Alpha' at the top to 'Project Upsilon' at the bottom. An orange arrow points to the 'Project Alpha' option. To the right of the dropdown is a table with the following columns: USER_ID, USER_ROLE, USER_CONTACT, PROJECT_ID, PROJECT_NAME, and COMPLETEDTASKCOUNT. The table contains three rows of data, all associated with 'Project Alpha'.

USER_ID	USER_ROLE	USER_CONTACT	PROJECT_ID	PROJECT_NAME	COMPLETEDTASKCOUNT
1	Manager	123-456-7892	1	Project Alpha	2
2	Developer	123-456-7890	1	Project Alpha	2
3	Tester	123-456-7891	1	Project Alpha	1

Figure 144 Choose desired project name from dropdown option

11 Further Discussion

The coursework highly focuses on enriching the experience regarding the database management along with web application development. Throughout the coursework, I have engaged with different tools and techniques which are essential for data and web development scenario.

I involved in creating Entity-Relationship Diagrams (ERD) using of Oracle SQL Developer Data Modular tool. This allowed me to properly visualize and design database schema through identifying relationships between various entities in database. I further learned the importance of normalization in building database schema to eliminate data redundancy and ensure data integrity. With the involvement in understanding and performing the different normal forms like 1NF, 2NF, and 3NF, helped in properly structure database. SQL Developer tool was used for creating database tables while SQL queries were written for data manipulation. CRUD operations were performed to enrich the experience on handling data retrieve and management. Through the database schema, a web-based application is developed using ASP.NET. I learned development of user-friendly interface along with implementation of back-end logic to create functional web applications. I engaged in performing testing of the application through CRUD operations to ensure the functioning as expected for quality assurance purpose.

Overall, the coursework has helped me to enrich me with theoretical knowledge long with practical skills crucial for future career in data and web development. The combined form of designing database schema, SQL proficiency, web application development and testing has prepared me to tackle any real-world problems in this field.

12 Reference

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